Optimasi Pemuatan Kendaraan pada Geladak Kapal Menggunakan Algoritma *Hill-Climbing*, *Simulated Annealing*, dan *Genetic Algorithm*

IF3070 Dasar Inteligensi Artifisial

Kelompok 23:

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I DESKRIPSI PERSOALAN



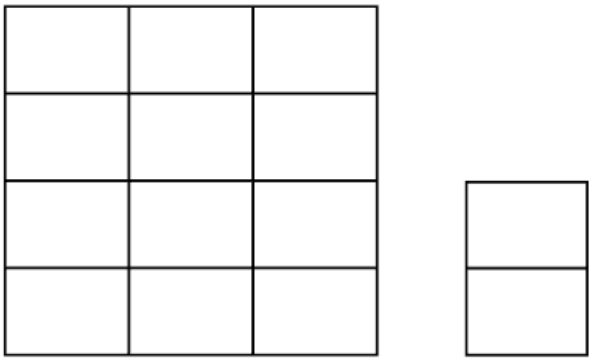
Gambar 1.1 *(Sumber: Gambar Kapal)*

Anda adalah seorang *Forward Deployed Engineer* (FDE) di PT ~~Owen~~ Intim Line yang mengoperasikan kapal kargo pengangkut kendaraan. Perusahaan baru saja memperoleh kontrak pengiriman dari beberapa dealer kendaraan sekaligus. Pada setiap jadwal keberangkatan, perusahaan menerima daftar kendaraan dengan dimensi, berat, tarif pengiriman, dan deadline yang berbeda-beda.

Selama ini, penempatan kendaraan pada geladak kapal dilakukan secara manual. Akibatnya, ruang geladak sering tidak termanfaatkan dengan baik dan keputusan pemilihan letak kendaraan belum tentu menghasilkan keuntungan yang optimal.

Tim operasional meminta Anda membangun sistem pendukung keputusan untuk menentukan kendaraan yang akan diangkut serta posisi dan orientasi setiap kendaraan pada geladak. Sistem tersebut harus dapat mengevaluasi berbagai kemungkinan konfigurasi, mempertimbangkan fungsi-fungsi objektif yang ditentukan, dan menghasilkan solusi penataan yang optimal.

Tugas ini bertujuan untuk membuat konfigurasi penempatan kendaraan paling optimal pada sebuah geladak kapal kargo. Setiap kendaraan dan geladak direpresentasikan sebagai grid dua dimensi berbentuk persegi atau persegi panjang. Solusi dapat memilih kendaraan mana yang diangkut dan menentukan posisi serta orientasinya di dalam geladak, dengan tetap memenuhi seluruh batasan.



Kapal Kendaraan

Tabel 1.1 Daftar Entitas dan Atribut

Entitas	Atribut	Contoh	Keterangan
Vehicle	id	"Owen"	*Identifier*
	Dimensions	{w:3, l:4}	Ukuran sebuah kendaraan, terdiri atas *width* dan *length* (satuan grid)
	Orientation	Horizontal	*Horizontal* atau *Vertical*
	ShippingFee	60	Tarif/keuntungan pengiriman kendaraan (satuan juta rupiah)
	Weight	5	Berat kendaraan (satuan unit)
	ETA	3	Estimated Time of Arrival, perkiraan waktu kedatangan kendaraan (dalam sisa hari)
Ship	Dimensions	{w:10, l:20}	Ukuran geladak pada kapal, terdiri atas *width* dan *length* (satuan grid)
	MaxCapacity	500	Kapasitas angkut maksimum suatu kapal (satuan unit)

Ketentuan Representasi

Berikut prosedur bagaimana dalam merepresentasikan permasalahan ini:

- Kendaraan dan geladak kapal direpresentasikan sebagai grid dua dimensi berbentuk persegi atau persegi panjang.

- Kendaraan dapat berbeda ukuran satu sama lain dan tidak semua harus dimasukkan ke dalam peti kemas pada kapal sehingga suatu kendaraan dapat berada di luar **(seluruh bagian kendaraan di luar kapal)**
- Koordinat berupa bilangan bulat. Cara representasi koordinat dibebaskan.
- Struktur data yang digunakan untuk representasi dibebaskan. Misalnya menggunakan *key-value* seperti JSON.
- Inisialisasi (***initial state***) random.
- Langkah neighbor (***move***) yang diperbolehkan tiap iterasi:
    1. Menukar dua kendaraan, baik di dalam maupun di luar kapal
    2. Memindahkan sebuah kendaraan ke suatu koordinat yang berbeda
    3. Mengubah orientasi kendaraan
- Representasi solusi (***final state***) adalah pemetaan semua paket yang ada dan juga pemetaan semua koordinat dalam peti kemas pada kapal. Secara detail yang harus dapat ditunjukkan oleh state dan visualisasi adalah:
    1. Bagaimana keadaan semua kendaraan yang di dalam maupun yang di luar
    2. Bagaimana keadaan geladak pada kapal
    3. *Value* pada *state* tersebut

Objective Function & Constraints

Agar permasalahan ini tidak menjadi kompleks, berikut arahan yang diberikan

- **Objective function**
  Permasalahan ini bertujuan untuk memaksimalkan *ShippingFee* pada kapal.
- **Kendaraan tidak boleh saling bertumpuk (*no overlapping*)**
  Tidak boleh ada satu sel grid geladak yang ditempati oleh lebih dari satu kendaraan.
- **Kendaraan harus berada di dalam batas geladak.**

Seluruh sel grid yang ditempati suatu kendaraan harus berada di dalam dimensi geladak kapal.

- **Kapal tidak boleh mengangkut melebihi kapasitasnya.**

II PEMBAHASAN

Bab ini membahas penyelesaian permasalahan optimasi penempatan kendaraan pada geladak kapal kargo PT Intim Line menggunakan pendekatan *local search algorithms* yang meliputi *Hill-Climbing*, *Simulated Annealing*, dan *Genetic Algorithm*. Implementasi sistem dibangun menggunakan bahasa pemrograman Python dengan arsitektur kode yang modular di dalam direktori src/. Struktur modul dirancang untuk memisahkan tanggung jawab antara representasi *state* dan validasi *constraints*, evaluasi *objective function*, operator *move neighbor*, pemuatan data uji JSON, hingga visualisasi geladak kapal dan grafik eksperimen.

2.1 Inisiasi *File* dan Fondasi Sistem

Sebelum menjalankan algoritma pencarian lokal, diperlukan fondasi sistem yang bertugas memodelkan entitas permasalahan ke dalam struktur data pemrograman, memvalidasi aturan penempatan pada grid dua dimensi, serta menyediakan antarmuka keluar-masuk data.

2.1.1 Representasi *Entitas* dan *State*

File `src/models.py` berfungsi sebagai *core engine* yang mendefinisikan struktur data kendaraan, geladak kapal, status penempatan kendaraan, serta representasi solusi lengkap. Mengacu pada konsep *local search*, setiap *state* menyimpan konfigurasi utuh dari seluruh kendaraan baik yang berada di dalam geladak kapal maupun yang tidak terangkut di luar kapal.

A. Inisiasi Class `Vehicle`, `Ship`, dan `VehiclePlacement`

```python
@dataclass(frozen=True)
class Vehicle:
    id: str
    w: int
    l: int
    shipping_fee: int
    weight: int
    eta: int
```

```
    def get_effective_dimensions(self, orientation: str) -> Tuple[int,
int]:
        if orientation == "Horizontal":
            return (self.w, self.l)
        elif orientation == "Vertical":
            return (self.l, self.w)
        raise ValueError(f"Orientasi tidak valid: {orientation}")


@dataclass(frozen=True)
class Ship:
    w: int
    l: int
    max_capacity: int
    id: str = "Ship-1"


@dataclass
class VehiclePlacement:
    vehicle_id: str
    in_ship: bool
    x: int = -1
    y: int = -1
    orientation: str = "Horizontal"
    ship_id: Optional[str] = None

    def copy(self) -> 'VehiclePlacement':
          return VehiclePlacement(self.vehicle_id, self.in_ship, self.x,
self.y, self.orientation, self.ship_id)
```

Penjelasan atribut dan metode pada ketiga class dasar tersebut adalah sebagai berikut:

1. **`Class Vehicle`**: Merepresentasikan entitas kendaraan dari dealer yang akan dikirim melalui kapal kargo. Kelas ini memiliki atribut:
    - **`id: str`** → *Identifier* untuk setiap kendaraan.
    - **`w: int`** dan **`l: int`** → Ukuran dimensi kendaraan dalam satuan sel grid dua dimensi.
    - **`shipping_fee: int`** → Tarif pengiriman kendaraan dalam satuan juta rupiah yang menjadi komponen utama untuk dimaksimalkan.

- **`weight: int`** → Berat kendaraan dalam satuan unit beban.
- **`eta: int`** → *Estimated Time of Arrival*, yaitu perkiraan waktu kedatangan atau tenggat waktu kendaraan dalam satuan sisa hari.
- **`get_effective_dimensions(orientation: str) → Tuple[int, int]:`** Mengembalikan ukuran dimensi efektif kendaraan di atas grid berdasarkan orientasinya. Jika orientasi bernilai "Horizontal", ukuran efektifnya adalah (w, l), sedangkan jika bernilai "Vertical", ukuran efektifnya menjadi (l, w).

2. **`Class Ship:`** Merepresentasikan entitas geladak kapal kargo dengan atribut:
   - **`id: str`** → *Identifier* kapal untuk mendukung skenario kapal tunggal maupun multi-kapal.
   - **`w: int`** dan **`l: int`** → Ukuran dimensi lebar dan panjang geladak kapal dalam satuan sel *grid*.
   - **`max_capacity: int`** → Kapasitas beban angkut maksimum kapal dalam satuan unit berat.
3. **`Class VehiclePlacement`**: Menyimpan informasi pemetaan posisi dan orientasi dari satu kendaraan pada suatu state, dengan atribut:
   - **`vehicle_id: str`** → ID kendaraan yang dipetakan.
   - **`in_ship: bool`** → Penanda status apakah kendaraan dimasukkan ke dalam geladak kapal (True) atau berada sepenuhnya di luar kapal (False).
   - **`x: int`** dan **`y: int`** → Koordinat bilangan bulat yang merepresentasikan titik pojok kiri-atas kendaraan pada grid geladak kapal (bernilai -1 apabila kendaraan berada di luar kapal).
   - **`orientation: str`** → Orientasi penempatan kendaraan, yaitu "Horizontal" atau "Vertical".

B. Inisiasi Class `CargoState` dan Pemetaan Grid 2D

```
def get_grid(self, ship_id: Optional[str] = None) -> List[List[Optional[str]]]:
        ship = self._resolve_ship(ship_id)

        grid = [[None for _ in range(ship.w)] for _ in range(ship.l)]

        for vid, p in self.placements.items():
```

```
            if p.in_ship and (p.ship_id or self.ship.id) == ship.id:
                v = self.vehicles[vid]
                eff_w, eff_l = v.get_effective_dimensions(p.orientation)
                for dy in range(eff_l):
                    for dx in range(eff_w):
                        gx, gy = p.x + dx, p.y + dy
                        if 0 <= gx < ship.w and 0 <= gy < ship.l:
                            grid[gy][gx] = vid
        return grid


    def get_total_weight(self, ship_id: Optional[str] = None) -> int:
                return sum(self.vehicles[vid].weight for vid, p in
self.placements.items()
                    if p.in_ship and (ship_id is None or (p.ship_id or
self.ship.id) == ship_id))


    def get_total_shipping_fee(self) -> int:
            return sum(self.vehicles[vid].shipping_fee for vid, p in
self.placements.items() if p.in_ship)
```

`Class CargoState` membungkus objek `Ship`, kamus data `vehicles`, serta pemetaan posisi seluruh kendaraan dalam `placements` untuk membentuk satu *state* lengkap. Fungsi-fungsi dasar pada `CargoState` meliputi:

- **`get_grid()`** → Membangun matriks dua dimensi berukuran `ship.l x ship.w` yang memetakan setiap koordinat sel geladak ke ID kendaraan yang menempatinya, atau None jika sel tersebut kosong.
- **`get_total_weight()`** → Menghitung akumulasi weight dari seluruh kendaraan yang berstatus `in_ship == True`.
- **`get_total_shipping_fee()`** → Menghitung akumulasi `shipping_fee` dari seluruh kendaraan yang berhasil diangkut di dalam geladak kapal.

C. Validasi *Constraints* pada `CargoState`

Penempatan kendaraan pada geladak kapal wajib mematuhi tiga batasan mutlak, yakni kendaraan tidak boleh saling bertumpuk, seluruh bagian kendaraan harus berada di dalam batas dimensi geladak, dan total berat muatan tidak boleh melampaui kapasitas maksimal kapal.

```
def count_violations(self) -> Tuple[int, int, int]:
        grid_counts = {
            ship.id: [[0 for _ in range(ship.w)] for _ in range(ship.l)]
            for ship in self.ships
        }
        ship_weights = {ship.id: 0 for ship in self.ships}
        out_of_bounds, overlap = 0, 0

        for vid, p in self.placements.items():
            if not p.in_ship: continue
            v = self.vehicles[vid]
            ship_id = p.ship_id or self.ship.id
            ship = self._resolve_ship(ship_id)
            grid_count = grid_counts[ship.id]
            ship_weights[ship.id] += v.weight
            eff_w, eff_l = v.get_effective_dimensions(p.orientation)

              if p.x < 0 or p.y < 0 or (p.x + eff_w) > ship.w or (p.y +
eff_l) > ship.l:
                for dy in range(eff_l):
                    for dx in range(eff_w):
                        gx, gy = p.x + dx, p.y + dy
                            if gx < 0 or gx >= ship.w or gy < 0 or gy >=
ship.l:
                            out_of_bounds += 1
                        else:
                            grid_count[gy][gx] += 1
                            if grid_count[gy][gx] > 1: overlap += 1
            else:
                for dy in range(eff_l):
                    for dx in range(eff_w):
                        gx, gy = p.x + dx, p.y + dy
                        grid_count[gy][gx] += 1
                        if grid_count[gy][gx] > 1: overlap += 1
```

```
        excess_weight = sum(
            max(0, ship_weights[ship.id] - ship.max_capacity)
            for ship in self.ships
        )
        return out_of_bounds, overlap, excess_weight

    def is_valid(self) -> bool:
        oob, ovl, ex_w = self.count_violations()
        return (oob == 0) and (ovl == 0) and (ex_w == 0)

     def can_place_vehicle(self, vid: str, x: int, y: int, orientation:
str,
                          ignore_vids: Optional[set] = None,
                          ship_id: Optional[str] = None) -> bool:
         ignore_vids = {vid} if ignore_vids is None else set(ignore_vids)
| {vid}
        v = self.vehicles[vid]
        current = self.placements.get(vid)
          target_ship = self._resolve_ship(ship_id or (current.ship_id if
current and current.in_ship else None))
        eff_w, eff_l = v.get_effective_dimensions(orientation)

         if x < 0 or y < 0 or (x + eff_w) > target_ship.w or (y + eff_l) >
target_ship.l: return False

        current_weight = sum(
                     self.vehicles[other_id].weight for other_id, p in
self.placements.items()
            if p.in_ship and other_id not in ignore_vids
            and (p.ship_id or self.ship.id) == target_ship.id
        )
          if current_weight + v.weight > target_ship.max_capacity: return
False

        for other_id, p in self.placements.items():
            if (not p.in_ship or other_id in ignore_vids
                       or (p.ship_id or self.ship.id) != target_ship.id):
continue
            other_v = self.vehicles[other_id]
            ow, ol = other_v.get_effective_dimensions(p.orientation)
```

```
            no_overlap = (x + eff_w <= p.x) or (p.x + ow <= x) or (y +
eff_l <= p.y) or (p.y + ol <= y)
            if not no_overlap: return False
        return True
```

Penjelasan ketiga fungsi *constraint* diatas adalah sebagai berikut:

1. Fungsi **`count_violations()`** dan **`is_valid()`**
   Memeriksa seluruh kendaraan yang berada di dalam kapal dan menghitung tiga jenis pelanggaran: jumlah sel kendaraan yang keluar dari batas koordinat geladak, jumlah sel geladak yang ditempati oleh lebih dari satu kendaraan, serta selisih kelebihan beban di atas kapasitas maksimum kapal. Suatu state dinyatakan valid apabila ketiga nilai pelanggaran tersebut sama dengan nol.
2. Fungsi **`can_place_vehicle(vid, x, y, orientation, ignore_vids)`**
   Melakukan pengecekan kelayakan penempatan satu kendaraan secara cepat tanpa harus membangun ulang seluruh matriks grid. Fungsi ini memastikan persegi panjang kendaraan tidak keluar batas geladak, penambahan beratnya tidak melampaui `max_capacity`, dan area persegi panjangnya tidak beririsan dengan kendaraan lain yang sedang berada di dalam kapal.

D. *Generate Random Initial State*

```
def generate_random_initial_state(cls, ship: Ship | List[Ship], vehicles:
Dict[str, Vehicle], use_bonus_objective: bool = False) -> 'CargoState':

        ships = list(ship) if isinstance(ship, (list, tuple)) else [ship]

        if len(ships) > 1:

                placements = {vid: VehiclePlacement(vid, False) for vid in
vehicles}

            state = cls(ships, vehicles, placements, use_bonus_objective)

            vehicle_ids = list(vehicles)

            random.shuffle(vehicle_ids)

            for vid in vehicle_ids:

                if random.random() >= 0.85:
```

```
                    continue
                options = []
                for target in ships:
                    for orientation in ("Horizontal", "Vertical"):
                                                        width, length =
vehicles[vid].get_effective_dimensions(orientation)
                        if width <= target.w and length <= target.l:
                            options.append((target, orientation))
                random.shuffle(options)
                for target, orientation in options:
                                                        width, length =
vehicles[vid].get_effective_dimensions(orientation)
                    coords = [(x, y) for x in range(target.w - width + 1)
                              for y in range(target.l - length + 1)]
                    random.shuffle(coords)
                    for x, y in coords[:15]:
                                    if state.can_place_vehicle(vid, x, y,
orientation, ship_id=target.id):
                            state.placements[vid] = VehiclePlacement(
                                vid, True, x, y, orientation, target.id
                            )
                            break
                    if state.placements[vid].in_ship:
                        break
            return state
              placements = {vid: VehiclePlacement(vid, False, -1, -1,
"Horizontal") for vid in vehicles}
```

```
        state = cls(ships[0], vehicles, placements, use_bonus_objective)
        vehicle_ids = list(vehicles.keys())
        random.shuffle(vehicle_ids)


        for vid in vehicle_ids:
            if random.random() < 0.85:
                orientation = random.choice(["Horizontal", "Vertical"])
                v = vehicles[vid]
                eff_w, eff_l = v.get_effective_dimensions(orientation)
                if eff_w <= ship.w and eff_l <= ship.l:
                      candidate_coords = [(x, y) for x in range(ship.w -
eff_w + 1) for y in range(ship.l - eff_l + 1)]
                    random.shuffle(candidate_coords)
                    for cx, cy in candidate_coords[:15]:
                                if state.can_place_vehicle(vid, cx, cy,
orientation):
                             state.placements[vid] = VehiclePlacement(vid,
True, cx, cy, orientation)
                            break
        state.invalidate_cache()
        return state
```

Fungsi `generate_random_initial_state()` pertama-tama menginisialisasi seluruh kendaraan di luar kapal (`in_ship = False`). Selanjutnya, daftar ID kendaraan diacak menggunakan `random.shuffle()`, lalu setiap kendaraan dicoba untuk dimasukkan ke dalam geladak kapal dengan orientasi acak "Horizontal" atau "Vertical" dan koordinat (x, y) acak yang memenuhi pemeriksaan `can_place_vehicle()`. Apabila tidak ditemukan posisi yang muat atau kapasitas telah penuh, kendaraan tersebut tetap berada di luar kapal sehingga *initial state* yang terbentuk dijamin acak sekaligus valid.

2.1.2 Perancangan *Objective Function*

Pada permasalahan penataan kendaraan PT Intim Line, tujuan utama optimasi adalah memaksimalkan total tarif pengiriman (`ShippingFee`) dari kendaraan yang diangkut di dalam geladak kapal dengan tetap mematuhi seluruh batasan dimensi, *overlap*, dan kapasitas angkut.

A. Formulasi *Objective Function*

Secara matematis, untuk suatu state S, himpunan kendaraan yang berada di dalam kapal $V_{in}(S)$, dan himpunan pelanggaran batasan, fungsi objektif $f(S)$ didefinisikan sebagai fungsi maksimasi:

$$f(S) = \sum_{v \in V_{in}(S)} ShippingFee(v) - P(S)$$

dengan fungsi penalti $P(S)$ bernilai:

$$P(S) = 1000 \cdot C_{oob}(S) + 1000 \cdot C_{overlap}(S) + 500 \cdot C_{excess_weight}(S)$$

di mana $C_{oob}(S)$ adalah banyaknya sel kendaraan di luar batas geladak, $C_{overlap}(S)$ adalah banyaknya sel yang saling bertumpuk, dan $C_{excess_weight}(S)$ adalah kelebihan berat muatan di atas `MaxCapacity`. Apabila state S valid, maka $P(S) = 0$ sehingga nilai objektif murni merepresentasikan total `ShippingFee` dalam satuan juta rupiah. Sebaliknya, penalti bernilai besar diterapkan untuk mengeliminasi solusi tidak valid apabila terdapat operator genetik yang menghasilkan keturunan yang melanggar batasan.

B. Formulasi *Objective Function* Alternatif (Bonus)

Spesifikasi bonus pertama, yaitu merancang fungsi objektif alternatif tanpa menambah variabel baru pada persoalan, program juga menyediakan mode evaluasi berbasis urgensi waktu kedatangan (ETA) dan efisiensi pemanfaatan ruang geladak:

$$f_{bonus}(S) = \sum_{v \in V_{in}(S)} ShippingFee(v) \cdot \left(1 + \frac{0.1}{\max(1, ETA(v))}\right) - P(S)$$

Melalui formulasi ini, kendaraan dengan tenggat ETA yang lebih singkat memberikan kontribusi skor objektif ekstra tanpa mengubah variabel input yang telah didefinisikan.

Nilai *state* hasil perhitungan `evaluate()` kemudian disimpan sementara pada atribut `_cached_value` agar tidak dihitung berulang-ulang pada *state*

yang tidak berubah, serta diekspor bersama pemetaan kendaraan ke dalam struktur *key-value* melalui metode `to_dict()`.

2.1.3 Implementasi Operator Perubahan *Neighbour*

Eksplorasi ruang solusi dilakukan dengan berpindah dari *current state* menuju *neighbor state* melalui operasi modifikasi kecil atau *move* dalam algoritma *local search*. Terdapat tiga langkah *neighbor move* yang diperbolehkan pada setiap iterasi:

1. Menukar dua kendaraan, baik keduanya berada di dalam kapal maupun salah satu di dalam dan satu lagi di luar kapal.
2. Memindahkan sebuah kendaraan ke suatu koordinat (x, y) yang berbeda di geladak, termasuk memasukkan kendaraan dari luar ke koordinat kosong di dalam kapal atau mengeluarkan kendaraan dari kapal.
3. Mengubah orientasi kendaraan dari Horizontal menjadi Vertical atau sebaliknya.

Pembangkitan *neighbor* dibagi menjadi dua mekanisme sesuai kebutuhan algoritma:

- Fungsi `get_random_neighbor()`, *Random Successor*
  Fungsi ini digunakan oleh algoritma *Simulated Annealing* dan *Stochastic Hill-Climbing* yang membutuhkan satu suksesor acak pada setiap iterasi. Fungsi memilih secara acak salah satu dari tiga jenis move (`1: Swap, 2: Move, 3: Rotate`), lalu memilih kendaraan dan koordinat target secara acak yang memenuhi validasi `can_place_vehicle()`.
- Fungsi `get_all_neighbors()`, *All Successors*
  Fungsi ini digunakan oleh algoritma *Steepest-Ascent Hill-Climbing*, *Hill-Climbing with Sideways Move*, dan *Random-Restart Hill-Climbing* untuk membangkitkan himpunan kandidat suksesor dari ketiga operasi *move* sekaligus, sehingga algoritma dapat memilih *highest-valued successor* pada setiap langkah iterasi.

2.1.4 Pemuatan Dataset dan Hasil

File `src/dataset.py` berfungsi untuk mengelola data masukan permasalahan berupa daftar kendaraan dan spesifikasi kapal serta menyimpan hasil eksekusi ke dalam format JSON di folder `output/`. Fungsi-fungsi utama yang terdapat pada `src/dataset.py` meliputi:

1. `get_default_problem()` → Menghasilkan studi kasus standar pengiriman kendaraan PT Intim Line dengan geladak kapal berukuran 10 × 20 sel grid dan kapasitas maksimum 500 unit, beserta daftar 18 kendaraan dengan variasi dimensi, tarif pengiriman, berat, dan ETA.
2. `load_problem_from_json(filepath)` → Membaca file konfigurasi .json eksternal dan mengonversinya menjadi objek Ship (atau daftar Ship untuk skenario multi-kapal) dan kamus objek Vehicle.
3. `save_problem_to_json(filepath, ...)` → Menyimpan spesifikasi persoalan maupun representasi state hasil pencarian ke dalam file .json secara terformat.

2.1.5 Visualisasi Geladak Kapal dan Grafik Eksperimen

Initial state, *final state* (keadaan geladak kapal, kendaraan di dalam dan di luar kapal, serta *value* pada *state* tersebut), dan grafik hasil eksperimen setiap algoritma divisualisasikan di dalam file `src/visualizer.py` menggunakan pustaka `matplotlib`.

2.1.6 Orkestrasi Program

File `src/main.py` dan `src/run_experiments.py` digunakan untuk mengintegrasikan seluruh modul dan mengotomatisasi pengujian sesuai cakupan spesifikasi. Peran kedua modul orkestrasi ini adalah:

- Menjalankan setiap algoritma *Hill-Climbing* dan *Simulated Annealing* sebanyak masing-masing 3 kali percobaan dari *initial state* yang konsisten pada tiap percobaan agar perbandingan antar-algoritma berlangsung adil.
- Menjalankan eksperimen *Genetic Algorithm* dengan menjadikan jumlah populasi sebagai variabel kontrol yang diuji pada 3 variasi banyak iterasi, masing-masing 3 kali *run* serta menjadikan banyak iterasi sebagai variabel kontrol yang diuji pada 3 variasi jumlah populasi, masing-masing 3 kali *run*.
- Mencatat durasi waktu proses pencarian menggunakan *modul time*, mengekspor rekapitulasi metrik ke dalam file .json dan .csv pada folder `output/experiments/`, serta memanggil `visualizer.py` untuk menghasilkan seluruh *plot state* dan grafik konvergensi secara otomatis.

2.2 *Hill Climbing*

Hill-Climbing merupakan algoritma *local search* yang memperbaiki solusi secara bertahap dengan berpindah dari *state* saat ini ke *neighbor* yang memiliki nilai fungsi objektif lebih baik. Pencarian dilakukan hingga tidak ditemukan *neighbor* yang memberikan perbaikan atau hingga batas iterasi yang ditentukan tercapai.

Pada permasalahan pemuatan kendaraan ke dalam geladak kapal, setiap state direpresentasikan oleh objek `CargoState` yang menyimpan informasi kapal, kendaraan, dan posisi penempatannya. Fungsi objektif yang digunakan bertujuan memaksimalkan *total shipping fee* dengan tetap memenuhi batasan dimensi geladak, tidak terjadi *overlap*, dan tidak melebihi kapasitas berat kapal. Implementasi *Hill-Climbing* menggunakan modul *helper* untuk menyediakan fungsi pembantu dan struktur hasil yang dapat digunakan oleh setiap variannya.

Hill Climbing Helper

Fungsi: to_dict()

```
def to_dict(self) -> dict[str, Any]:
        return {
            "initial_state": self.initial_state.to_dict(),
            "final_state": self.final_state.to_dict(),
            "initial_objective": float(self.initial_state.evaluate()),
            "final_objective": float(self.final_state.evaluate()),
            "objective_history": self.objective_history,
            "iterations": self.iterations,
            "accepted_moves": self.accepted_moves,
            "duration_seconds": self.duration_seconds,
            "termination_reason": self.termination_reason,
            "metadata": self.metadata.copy(),
        }
```

Fungsi `to_dict()` mengubah objek `HillClimbingResult` menjadi *dictionary* agar hasil algoritma dapat dicatat dan disimpan oleh *runner* eksperimen. Kondisi awal dan akhir dikonversi melalui `to_dict()` milik masing-masing *state*, sedangkan nilai objektifnya dihitung menggunakan `evaluate()` dan dikonversi ke tipe `float`. Informasi lain yang disertakan meliputi riwayat

objektif, jumlah iterasi, perpindahan yang diterima, durasi eksekusi, dan alasan penghentian. `metadata.copy()` digunakan untuk menyertakan salinan dictionary metadata, sehingga informasi khusus setiap varian dapat disimpan tanpa menghilangkan struktur hasil umum.

Fungsi: is_strictly_better()

```
def is_strictly_better(
    candidate_value: float,
    current_value: float,
) -> bool:
    return candidate_value > current_value + OBJECTIVE_EPSILON
```

Fungsi `is_strictly_better()` menentukan apakah nilai objektif kandidat benar-benar lebih tinggi daripada nilai objektif saat ini. Perbandingan dilakukan dengan menambahkan `OBJECTIVE_EPSILON` sebesar (10^{-9}) pada nilai saat ini. Fungsi hanya mengembalikan `True` apabila kandidat melampaui nilai tersebut. Toleransi ini mencegah perbedaan numerik yang sangat kecil dianggap sebagai peningkatan. Pada Steepest Ascent, fungsi ini menjadi dasar keputusan untuk menerima perpindahan dan menghentikan pencarian ketika tidak ditemukan peningkatan objektif.

Fungsi: is_equal_objective()

```
def is_equal_objective(
    first_value: float,
    second_value: float,
) -> bool:
    return isclose(
        first_value,
        second_value,
        rel_tol=0.0,
        abs_tol=OBJECTIVE_EPSILON,
    )
```

Fungsi `is_equal_objective()` memeriksa apakah dua nilai objektif dapat dianggap setara dengan toleransi absolut `OBJECTIVE_EPSILON`. Parameter `rel_tol=0.0` memastikan perbandingan tidak menggunakan toleransi relatif. Pada Steepest Ascent, fungsi ini digunakan untuk mengumpulkan

kandidat-kandidat yang memiliki nilai objektif setara dengan nilai terbaik. Jika terdapat lebih dari satu kandidat terbaik, algoritma dapat memilih salah satunya secara acak tanpa mengorbankan kualitas objektif yang dipilih.

Fungsi: get_valid_neighbors()

```
def get_valid_neighbors(state: Any) -> List[Any]:
    return [
        neighbor
        for neighbor in state.get_all_neighbors()
        if neighbor.is_valid()
    ]
```

Fungsi `get_valid_neighbors()` menghasilkan daftar *neighbor* valid dari suatu *state*. Fungsi memanggil `state.get_all_neighbors()` untuk memperoleh kandidat, kemudian menyaringnya menggunakan `neighbor.is_valid()`. Hanya kandidat yang memenuhi batasan masalah yang dimasukkan ke dalam daftar hasil. Pemisahan ini memastikan pemilihan solusi terbaik dilakukan terhadap kandidat yang valid, bukan terhadap seluruh kandidat tanpa pemeriksaan. Fungsi ini digunakan oleh Steepest Ascent untuk memperoleh lingkungan solusi yang akan dievaluasi pada setiap iterasi.

Fungsi: get_random_valid_neighbor()

```
def get_random_valid_neighbor(
    state: Any,
    max_attempts: int = 100,
) -> Optional[Any]:
      """Sample a random valid neighbor, returning None if sampling
fails."""
    if max_attempts < 1:
        raise ValueError("max_attempts harus minimal 1.")

    for _ in range(max_attempts):
        neighbor = state.get_random_neighbor()

        if neighbor is not None and neighbor.is_valid():
            return neighbor
```

```
    return None
```

Fungsi `get_random_valid_neighbor()` mencari satu *neighbor* valid melalui sampling acak. Parameter `max_attempts` membatasi jumlah percobaan dan harus bernilai minimal satu, jika tidak, fungsi menghasilkan `ValueError`.

Pada setiap percobaan, fungsi mengambil kandidat melalui `state.get_random_neighbor()`. Kandidat langsung dikembalikan apabila tidak bernilai `None` dan memenuhi seluruh batasan. Jika tidak ada kandidat valid setelah jumlah percobaan maksimum, fungsi mengembalikan `None`. Mekanisme ini membatasi proses sampling agar tidak berlangsung tanpa akhir ketika kandidat valid sulit ditemukan. Berbeda dari `get_valid_neighbors()`, fungsi ini hanya mencari satu kandidat secara acak. Karena itu, fungsi ini digunakan oleh Stochastic Hill-Climbing dan bukan mekanisme utama pemilihan tetangga pada Steepest Ascent.

2.2.1 *Steepest Ascent*

Steepest Ascent Hill-Climbing merupakan variasi *Hill-Climbing* yang mengevaluasi seluruh *neighbor state* valid dari kondisi saat ini, kemudian memilih tetangga dengan nilai fungsi objektif tertinggi. Perpindahan dilakukan apabila tetangga terpilih memberikan peningkatan nilai fungsi objektif. Jika tidak ditemukan tetangga yang lebih baik, pencarian dihentikan karena algoritma telah mencapai *local optimum*.

Fungsi: steepest_ascent_hill_climbing()

```
def steepest_ascent_hill_climbing(
    initial_state: Any,
    max_iterations: int = 1000,
) -> HillClimbingResult:
    if max_iterations < 0:
        raise ValueError("max_iterations tidak boleh negatif.")

    start_time = perf_counter()

    initial = initial_state.copy()

    if not initial.is_valid():
```

```
            raise ValueError("Initial state harus memenuhi seluruh
constraint.")

    current = initial.copy()
    objective_history = [float(current.evaluate())]

    iterations = 0
    accepted_moves = 0
    termination_reason = "max_iterations"

    while iterations < max_iterations:
        iterations += 1
        current_value = float(current.evaluate())

        valid_neighbors = get_valid_neighbors(current)

        if not valid_neighbors:
            objective_history.append(current_value)
            termination_reason = "no_valid_successors"
            break

        best_value = max(
            float(neighbor.evaluate())
            for neighbor in valid_neighbors
        )

        if not is_strictly_better(best_value, current_value):
            objective_history.append(current_value)
            termination_reason = "local_optimum"
            break

        best_neighbors = [
            neighbor
            for neighbor in valid_neighbors
            if is_equal_objective(
                float(neighbor.evaluate()),
                best_value,
            )
        ]
```

```
            current = choice(best_neighbors)
            accepted_moves += 1
            objective_history.append(float(current.evaluate()))

        duration = perf_counter() - start_time

        return HillClimbingResult(
            initial_state=initial,
            final_state=current.copy(),
            objective_history=objective_history,
            iterations=iterations,
            accepted_moves=accepted_moves,
            duration_seconds=duration,
            termination_reason=termination_reason,
        )
```

Fungsi `steepest_ascent_hill_climbing()` menjalankan pencarian dengan selalu berpindah ke tetangga bernilai objektif tertinggi pada tiap iterasinya. Proses dimulai dengan menyalin dan memvalidasi state awal, mencatat waktu mulai, serta menginisialisasi riwayat objektif dan alasan penghentian bawaan ("max_iterations"). Pada setiap iterasi, fungsi membangkitkan seluruh tetangga valid melalui `get_valid_neighbors()` jika kosong, pencarian terhenti dengan alasan "no_valid_successors". Jika tersedia, algoritma mencari kandidat dengan nilai tertinggi (`best_value`) dan membandingkannya dengan state saat ini menggunakan `is_strictly_better()`. Apabila tidak ada peningkatan nilai, pencarian berhenti di "local_optimum", namun jika lebih baik, fungsi akan mengumpulkan semua kandidat bernilai setara (`is_equal_objective()`), memilih salah satunya secara acak, memperbarui state saat ini, dan mencatat pergerakan tersebut. Perulangan ini terus berlanjut hingga mencapai kondisi henti atau batas iterasi, dan diakhiri dengan perhitungan durasi eksekusi serta pengembalian objek `HillClimbingResult` yang merangkum seluruh metrik eksperimen secara konsisten.

2.2.2 *Sideways*

Hill-Climbing with Sideways Move merupakan varian Hill-Climbing yang memungkinkan perpindahan ke *neighbor* dengan nilai fungsi objektif yang

sama ketika tidak ditemukan perpindahan yang lebih baik. Mekanisme ini memberikan kesempatan untuk mengeksplorasi solusi lain pada *plateau*, yaitu bagian ruang pencarian yang memiliki nilai objektif setara. Dalam implementasi ini, perpindahan yang meningkatkan objektif tetap diprioritaskan, sedangkan perpindahan *sideways* dibatasi agar pencarian tidak berputar tanpa kemajuan.

Fungsi: _state_signature()

```
def _state_signature(state: Any) -> Tuple:
    return tuple(
        (
            vehicle_id,
            placement.in_ship,
            placement.x if placement.in_ship else None,
            placement.y if placement.in_ship else None,
            placement.orientation if placement.in_ship else None,
        )
        for vehicle_id, placement in sorted(state.placements.items())
    )
```

Fungsi `_state_signature()` mengubah konfigurasi penempatan kendaraan menjadi *tuple* yang diurutkan berdasarkan `vehicle_id` agar menghasilkan tanda tangan state yang konsisten. Tanda tangan ini mencatat detail status penempatan, koordinat, dan orientasi untuk kendaraan yang berada di dalam kapal, namun secara sengaja mengabaikan atribut tersebut (diganti dengan `None`) untuk kendaraan di luar kapal karena tidak memengaruhi susunan muatan di geladak. Keluaran dari fungsi ini kemudian dimanfaatkan oleh struktur `visited` untuk melacak riwayat pencarian dan mencegah algoritma mengevaluasi konfigurasi yang persis sama secara berulang.

Fungsi: hill_climbing_with_sideways_move()

```
def hill_climbing_with_sideways_move(
    initial_state: Any,
    max_iterations: int = 1000,
    max_sideways_moves: int = 100,
) -> HillClimbingResult:
    if max_iterations < 0:
```

```
        raise ValueError("max_iterations tidak boleh negatif.")

    if max_sideways_moves < 0:
        raise ValueError("max_sideways_moves tidak boleh negatif.")

    start_time = perf_counter()

    initial = initial_state.copy()

    if not initial.is_valid():
            raise ValueError("Initial state harus memenuhi seluruh
constraint.")

    current = initial.copy()
    objective_history = [float(current.evaluate())]

    iterations = 0
    accepted_moves = 0
    total_sideways_moves = 0
    consecutive_sideways_moves = 0
    sideways_move_iterations: List[int] = []

    visited = {_state_signature(current)}
    termination_reason = "max_iterations"

    while iterations < max_iterations:
        iterations += 1
        current_value = float(current.evaluate())

        valid_neighbors = get_valid_neighbors(current)

        if not valid_neighbors:
            objective_history.append(current_value)
            termination_reason = "no_valid_successors"
            break

        # Ignore previously visited configurations to avoid cycling.
        unique_neighbors = {}
        current_signature = _state_signature(current)
```

```
        for neighbor in valid_neighbors:
            signature = _state_signature(neighbor)

            if signature == current_signature or signature in visited:
                continue

            unique_neighbors.setdefault(signature, neighbor)

        candidates = list(unique_neighbors.values())

        # 1. Always prioritize an improving move when one exists.
        improving_neighbors = [
            neighbor
            for neighbor in candidates
            if is_strictly_better(
                float(neighbor.evaluate()),
                current_value,
            )
        ]

        if improving_neighbors:
            best_value = max(
                float(neighbor.evaluate())
                for neighbor in improving_neighbors
            )

            best_neighbors = [
                neighbor
                for neighbor in improving_neighbors
                if is_equal_objective(
                    float(neighbor.evaluate()),
                    best_value,
                )
            ]

            current = choice(best_neighbors)
            visited.add(_state_signature(current))
```

```
            accepted_moves += 1
            consecutive_sideways_moves = 0
            objective_history.append(float(current.evaluate()))
            continue

        # 2. If no improving move exists, consider sideways moves.
        equal_neighbors = [
            neighbor
            for neighbor in candidates
            if is_equal_objective(
                float(neighbor.evaluate()),
                current_value,
            )
        ]

        if not equal_neighbors:
            objective_history.append(current_value)
            termination_reason = "local_optimum"
            break

        if consecutive_sideways_moves >= max_sideways_moves:
            objective_history.append(current_value)
            termination_reason = "sideways_limit_reached"
            break

        current = choice(equal_neighbors)
        visited.add(_state_signature(current))

        accepted_moves += 1
        total_sideways_moves += 1
        consecutive_sideways_moves += 1
        sideways_move_iterations.append(iterations)

        objective_history.append(float(current.evaluate()))

    duration = perf_counter() - start_time

    return HillClimbingResult(
        initial_state=initial,
```

```
        final_state=current.copy(),
        objective_history=objective_history,
        iterations=iterations,
        accepted_moves=accepted_moves,
        duration_seconds=duration,
        termination_reason=termination_reason,
        metadata={
            "variant": "sideways",
            "max_sideways_moves": max_sideways_moves,
            "total_sideways_moves": total_sideways_moves,
            "consecutive_sideways_moves_at_end": (
                consecutive_sideways_moves
            ),
            "sideways_move_iterations": sideways_move_iterations,
        },
    )
```

Fungsi `hill_climbing_with_sideways_move()` menjalankan pencarian yang mengizinkan perpindahan ke *neighbor* bernilai objektif setara demi menghindari jebakan optimum lokal, dengan batasan `max_sideways_moves` berturut-turut. Setelah memvalidasi *initial state* dan menginisialisasi metrik pencatatan (termasuk penyimpanan *signature* state ke himpunan `visited` untuk mencegah kunjungan ulang atau siklus), algoritma membangkitkan dan menyaring kandidat *neighbor* unik yang belum dikunjungi. Fungsi ini selalu memprioritaskan perpindahan yang meningkatkan objektif, jika kandidat yang lebih baik ditemukan, yang bernilai tertinggi akan dipilih dan penghitung `consecutive_sideways_moves` akan direset ke nol. Namun, bila tidak ada peningkatan, pencarian beralih ke kandidat setara secara acak sembari menaikkan penghitung sideways.

Eksekusi akan berhenti jika tidak ada kandidat valid (`no_valid_successors`), tidak ada opsi yang lebih baik maupun setara (`local_optimum`), atau batas sideways beruntun telah tercapai (`sideways_limit_reached`). Pada akhirnya, fungsi mengembalikan objek `HillClimbingResult` yang merangkum seluruh metrik pencarian, dilengkapi dengan *dictionary* metadata khusus (seperti total sideways, sisa sideways beruntun, dan riwayat iterasinya) untuk membedakan pergerakan peningkatan objektif dari pergerakan mendatar pada tahap analisis.

2.2.3 *Stochastic*

Stochastic Hill-Climbing merupakan varian Hill-Climbing yang memilih satu *neighbor* valid secara acak pada setiap iterasi, bukan mengevaluasi seluruh tetangga untuk menentukan kandidat terbaik seperti Steepest Ascent. Kandidat yang terpilih hanya diterima apabila menghasilkan peningkatan nilai fungsi objektif. Jika tidak, algoritma mempertahankan *state* saat ini dan melanjutkan pencarian hingga batas iterasi tercapai atau proses sampling gagal menemukan *neighbor* valid.

Fungsi: stochastic_hill_climbing()

```
def stochastic_hill_climbing(
    initial_state: Any,
    max_iterations: int = 1000,
    max_neighbor_attempts: int = 100,
) -> HillClimbingResult:
    if max_iterations < 0:
        raise ValueError("max_iterations tidak boleh negatif.")

    if max_neighbor_attempts < 1:
        raise ValueError("max_neighbor_attempts harus minimal 1.")

    start_time = perf_counter()

    initial = initial_state.copy()

    if not initial.is_valid():
            raise ValueError("Initial state harus memenuhi seluruh
constraint.")

    current = initial.copy()
    objective_history = [float(current.evaluate())]

    iterations = 0
    accepted_moves = 0
    rejected_moves = 0
    sampling_failures = 0
    termination_reason = "max_iterations"
```

```
while iterations < max_iterations:
    iterations += 1
    current_value = float(current.evaluate())

    neighbor = get_random_valid_neighbor(
        current,
        max_attempts=max_neighbor_attempts,
    )

    if neighbor is None:
        sampling_failures += 1
        objective_history.append(current_value)
        termination_reason = "neighbor_sampling_failed"
        break

    neighbor_value = float(neighbor.evaluate())

    if is_strictly_better(neighbor_value, current_value):
        current = neighbor.copy()
        accepted_moves += 1
    else:
        rejected_moves += 1
    objective_history.append(float(current.evaluate()))

duration = perf_counter() - start_time

return HillClimbingResult(
    initial_state=initial,
    final_state=current.copy(),
    objective_history=objective_history,
    iterations=iterations,
    accepted_moves=accepted_moves,
    duration_seconds=duration,
    termination_reason=termination_reason,
    metadata={
        "variant": "stochastic",
        "max_iterations": max_iterations,
        "max_neighbor_attempts": max_neighbor_attempts,
```

```
            "rejected_moves": rejected_moves,
            "neighbor_sampling_failures": sampling_failures,
        },
    )
```

Fungsi `stochastic_hill_climbing()` menjalankan pencarian efisien dengan hanya mengevaluasi satu tetangga valid secara acak pada setiap iterasinya, sehingga membebaskan algoritma dari kewajiban membangkitkan seluruh *successor* (seperti pada *Steepest Ascent*). Setelah memvalidasi state awal dan menginisialisasi metrik pelacakan, fungsi berulang kali memanggil `get_random_valid_neighbor()` dengan batas percobaan `max_neighbor_attempts`. Jika sampling gagal menemukan kandidat, pencarian langsung dihentikan dengan status `neighbor_sampling_failed`. Namun, jika kandidat ditemukan, nilainya akan dievaluasi menggunakan `is_strictly_better()`, kandidat yang lebih baik akan diterima sebagai state baru, sedangkan yang lebih buruk atau setara akan ditolak (`rejected_moves` bertambah) tanpa menghentikan pencarian.

Karena pencarian tetap berlanjut meskipun ada penolakan, nilai state saat ini dipertahankan, memastikan pencatatan riwayat objektif tidak pernah menurun dan iterasi selanjutnya tetap berpeluang menemukan kandidat yang lebih baik. Eksekusi berakhir ketika mencapai batas `max_iterations` atau gagal sampling, lalu mengembalikan `HillClimbingResult` beserta metadata khusus—seperti jumlah penolakan dan kegagalan sampling—untuk mempermudah analisis karakteristik pencarian stokastik.

2.2.4 *Random Restart*

Random Restart Hill-Climbing merupakan varian Hill-Climbing yang menjalankan pencarian lokal berulang kali dengan *initial state* yang dibangkitkan secara acak pada setiap *restart*. Pada implementasi ini, setiap pencarian menggunakan algoritma Steepest Ascent Hill-Climbing. Setelah seluruh *restart* selesai, algoritma memilih *final state* dengan nilai fungsi objektif tertinggi dari seluruh pencarian, sehingga hasilnya tidak bergantung pada satu kondisi awal saja.

Fungsi: random_restart_hill_climbing()

```python
def random_restart_hill_climbing(
    ship: Ship,
    vehicles: Dict[str, Vehicle],
    max_restarts: int = 10,
    max_iterations_per_restart: int = 1000,
    use_bonus_objective: bool = False,
) -> HillClimbingResult:
    if max_restarts < 1:
        raise ValueError("max_restarts harus minimal 1.")

    if max_iterations_per_restart < 1:
        raise ValueError(
            "max_iterations_per_restart harus minimal 1."
        )

    start_time = perf_counter()
    restart_results = []

    for restart_number in range(1, max_restarts + 1):
        initial_state = CargoState.generate_random_initial_state(
            ship=ship,
            vehicles=vehicles,
            use_bonus_objective=use_bonus_objective,
        )

        if not initial_state.is_valid():
            raise ValueError(
                f"Initial state pada restart {restart_number} "
                "tidak memenuhi seluruh constraint."
            )

        result = steepest_ascent_hill_climbing(
            initial_state=initial_state,
            max_iterations=max_iterations_per_restart,
        )

        if not result.final_state.is_valid():
            raise RuntimeError(
                f"Final state pada restart {restart_number} "
```

```
                "tidak valid."
            )

        restart_results.append(result)

    best_restart_index = max(
        range(len(restart_results)),
        key=lambda index: (
            restart_results[index].final_state.evaluate()
        ),
    )

    best_result = restart_results[best_restart_index]

    total_iterations = sum(
        result.iterations for result in restart_results
    )

    total_accepted_moves = sum(
        result.accepted_moves for result in restart_results
    )

    best_seen = float(restart_results[0].objective_history[0])

    aggregate_history = [best_seen]
    history_iterations = [0]
    cumulative_iterations = 0

    for restart_index, result in enumerate(restart_results):
        if restart_index > 0:
            initial_value = float(result.initial_state.evaluate())
            best_seen = max(best_seen, initial_value)

            aggregate_history.append(best_seen)
            history_iterations.append(cumulative_iterations)

        for value in result.objective_history[1:]:
            cumulative_iterations += 1
            best_seen = max(best_seen, float(value))
```

```
            aggregate_history.append(best_seen)
            history_iterations.append(cumulative_iterations)

    restart_summaries = []

    for restart_number, result in enumerate(
        restart_results,
        start=1,
    ):
        restart_summaries.append({
            "restart_number": restart_number,
            "initial_state": result.initial_state.to_dict(),
            "final_state": result.final_state.to_dict(),
            "initial_objective": float(
                result.initial_state.evaluate()
            ),
            "final_objective": float(
                result.final_state.evaluate()
            ),
            "objective_history": result.objective_history,
            "iterations": result.iterations,
            "accepted_moves": result.accepted_moves,
            "duration_seconds": result.duration_seconds,
            "termination_reason": result.termination_reason,
        })

    duration = perf_counter() - start_time

    return HillClimbingResult(
        initial_state=restart_results[0].initial_state.copy(),
        final_state=best_result.final_state.copy(),
        objective_history=aggregate_history,
        iterations=total_iterations,
        accepted_moves=total_accepted_moves,
        duration_seconds=duration,
        termination_reason="max_restarts_reached",
        metadata={
            "variant": "random_restart",
```

```
            "max_restarts": max_restarts,
            "restarts_completed": len(restart_results),
            "max_iterations_per_restart": (
                max_iterations_per_restart
            ),
            "iterations_per_restart": [
                result.iterations for result in restart_results
            ],
            "best_restart_number": best_restart_index + 1,
            "best_objective_per_restart": [
                float(result.final_state.evaluate())
                for result in restart_results
            ],
            "restart_results": restart_summaries,
            "total_iterations": total_iterations,
            "objective_history_iterations": history_iterations,
        },
    )
```

Fungsi `random_restart_hill_climbing()` menjalankan serangkaian pencarian independen (`max_restarts`) dengan batas iterasinya masing-masing (`max_iterations_per_restart`). Setelah memvalidasi parameter dan mencatat waktu mulai, algoritma melakukan perulangan di mana setiap iterasi membangkitkan `initial_state` acak yang valid, lalu mengeksekusinya menggunakan `steepest_ascent_hill_climbing()`. State awal dan state akhir dari setiap pencarian akan divalidasi secara ketat; apabila lolos, hasilnya disimpan ke dalam `restart_results`. Pendekatan ini murni memanfaatkan pengulangan titik awal pencarian untuk menghindari *local optimum*, tanpa mengubah mekanisme internal pemilihan tetangganya.

Setelah seluruh *restart* diselesaikan, fungsi membandingkan nilai objektif akhir dari setiap pencarian dan memilih hasil tertinggi (`best_result`) menggunakan ekspresi `max()`. Algoritma kemudian mengagregasi total beban komputasi dengan menjumlahkan seluruh iterasi dan `accepted_moves`. Untuk pelacakan progres, fungsi membangun `aggregate_history` yang merekam nilai objektif tertinggi global (`best_seen`) sepanjang rangkaian *restart*. Riwayat ini dipetakan terhadap sumbu `cumulative_iterations` yang sengaja mengizinkan angka iterasi berulang pada batas antar-pencarian, sehingga transisi antara evaluasi akhir

pencarian sebelumnya dan kondisi awal pencarian berikutnya terekam dengan akurat.

Sebagai luaran, algoritma menyusun `restart_summaries` yang merinci metrik spesifik setiap pencarian (seperti alasan henti internal dan durasi). Fungsi kemudian mengembalikan `HillClimbingResult` yang merangkum *initial state* pencarian pertama, *final state* terbaik keseluruhan, metrik agregat, dan status henti `"max_restarts_reached"`. Objek ini dilengkapi *dictionary metadata* komprehensif—termasuk indeks *restart* terbaik dan riwayat objektif agregat—untuk mendukung analisis mendalam terhadap evolusi solusi dan karakteristik setiap pencarian lokal.

2.3 *Simulated Annealing*

Algoritma *Simulated Annealing* adalah metode pencarian lokal yang menggabungkan efisiensi algoritma *Hill-Climbing* dengan sifat kelengkapan dari *Purely Random Walk*. Algoritma *Hill-Climbing* standar tidak pernah melakukan langkah menurun, sehingga rentan terjebak pada *local maximum*. Untuk mengatasi kelemahan tersebut, *Simulated Annealing* mengadopsi analogi proses *annealing* pada bidang metalurgi, yaitu memanaskan logam atau kaca hingga suhu tinggi lalu mendinginkannya secara perlahan guna menghilangkan tegangan internal dan mencapai konfigurasi energi yang paling stabil.

Dalam konteks optimasi penataan kendaraan pada geladak kapal kargo PT Intim Line, *Simulated Annealing* bekerja sebagai versi modifikasi dari *Stochastic Hill-Climbing* yang mengizinkan algoritma berpindah ke *neighbor state* yang memiliki nilai fungsi objektif lebih buruk dengan probabilitas tertentu yang terus menurun seiring berjalannya waktu.

A. Konsep *Cooling Schedule* dan Probabilitas

Pada setiap iterasi ke-t, algoritma memetakan waktu iterasi menjadi nilai temperatur T melalui fungsi *cooling schedule* eksponensial:

$$T(t) = T_0 \cdot \alpha^t$$

dengan T_0 adalah suhu awal dan α adalah laju pendinginan dengan rentang $0 < \alpha < 1$. Selanjutnya, algoritma membangkitkan tepat satu suksesor acak dari *current state* melalui metode `get_random_neighbor()` dan menghitung selisih nilai fungsi objektif ΔE:

$$\Delta E = next.VALUE - current.VALUE$$

Mekanisme pengambilan keputusan perpindahan *state* mengikuti aturan berikut:

1. Tetangga Lebih Baik ($\Delta E > 0$): Apabila *neighbor state* menghasilkan total keuntungan pengiriman (`ShippingFee`) yang lebih tinggi dibandingkan *current state*, algoritma selalu menerima langkah tersebut (`current <- next`) dan mencatat nilai probabilitas penerimaan sebesar 1.0.
2. Tetangga Lebih Buruk atau Sama ($\Delta E \leq 0$): Apabila *neighbor state* tidak memberikan perbaikan nilai, algoritma tetap memiliki peluang untuk berpindah ke *state* tersebut dengan probabilitas penerimaan sesuai distribusi Boltzmann:

   $$P(\Delta E, T) = e^{\frac{\Delta E}{T}}$$

   Nilai $P(\Delta E, T)$ dibandingkan terhadap suatu bilangan acak seragam `random(0, 1)`. Jika `random(0, 1) < P`, maka langkah penurunan diterima. Ketika temperatur T masih tinggi di awal pencarian, nilai $e^{\frac{\Delta E}{T}}$ mendekati 1, sehingga algoritma aktif melakukan eksplorasi *random walk* untuk meloloskan diri dari *local optima*. Ketika T mendekati 0 di akhir iterasi, nilai $e^{\frac{\Delta E}{T}}$ mendekati 0, sehingga algoritma bersifat semakin selektif menyerupai *Stochastic Hill-Climbing*.

B. Mekanisme Deteksi Frekuensi *Stuck* di *Local Optima*

Kondisi tertahan di *local optima* dilacak menggunakan penghitung penolakan beruntun, `consecutive_rejections`. Apabila algoritma gagal berpindah ke *state* baru, baik karena tidak menemukan tetangga dengan $\Delta E > 0$ maupun karena langkah buruk ditolak oleh probabilitas $e^{\frac{\Delta E}{T}}$) selama lebih dari ambang batas konsekutif, 20 iterasi berturut-turut, maka variabel `stuck_frequency` akan ditambah 1 dan penghitung di-reset.

C. Deskripsi Kelas dan *Source Code* `simulated_annealing.py`

Implementasi algoritma *Simulated Annealing* dibungkus di dalam kelas `SimulatedAnnealing` pada file `src/simulated_annealing.py`. Parameter konstruktor yang digunakan meliputi:

- **`initial_temperature: float = 1000.0`** → Suhu awal (T_0) untuk memulai proses pencarian.
- **`cooling_rate: float = 0.99`** → Faktor pengali penurunan suhu (α) pada setiap iterasi.
- **`min_temperature: float = 1e-3`** → Ambang batas suhu minimum (T_{min}) yang menghentikan pencarian ketika suhu mendekati nol.
- **`max_iterations: int = 5000`** → Batas maksimum iterasi pencarian.

Class: SimulatedAnnealing

```
class SimulatedAnnealing:
    def __init__(
        self,
        initial_temperature: float = 1000.0,
        cooling_rate: float = 0.99,
        min_temperature: float = 1e-3,
        max_iterations: int = 5000
    ):
        self.initial_temperature = initial_temperature
        self.cooling_rate = cooling_rate
        self.min_temperature = min_temperature
        self.max_iterations = max_iterations

    def schedule(self, t: int) -> float:
        return self.initial_temperature * (self.cooling_rate ** t)

      def run(self, initial_state: CargoState) -> Tuple[CargoState,
Dict[str, Any]]:
        current_state = initial_state
        current_value = current_state.evaluate()

        history_obj: List[float] = [current_value]
        history_prob: List[float] = []

        stuck_frequency = 0
        consecutive_rejections = 0
```

```
        start_time = time.time()

        for t in range(self.max_iterations):
            temperature = self.schedule(t)
            if temperature < self.min_temperature:
                break

            next_state = current_state.get_random_neighbor()
            if next_state is None:
                break
            next_value = next_state.evaluate()
            delta_e = next_value - current_value

            if delta_e > 0:
                current_state = next_state
                current_value = next_value
                history_prob.append(1.0)
                consecutive_rejections = 0
            else:
                exponent = delta_e / temperature
                if exponent < -700:
                    acceptance_probability = 0.0
                else:
                    acceptance_probability = math.exp(exponent)

                history_prob.append(acceptance_probability)

                if random.random() < acceptance_probability:
                    current_state = next_state
                    current_value = next_value
                    consecutive_rejections = 0
                else:
                    consecutive_rejections += 1
                    if consecutive_rejections > 20:
                        stuck_frequency += 1
                        consecutive_rejections = 0

            history_obj.append(current_value)
```

```
    end_time = time.time()

    metrics = {
        "initial_value": initial_state.evaluate(),
        "final_value": current_value,
        "duration_seconds": end_time - start_time,
        "iterations_ran": len(history_obj) - 1,
        "history_obj": history_obj,
        "history_prob": history_prob,
        "stuck_frequency": stuck_frequency,
        "is_valid": current_state.is_valid()
    }

    return current_state, metrics
```

Implementasi metode `run()` menggunakan pengamanan numerik `if exponent < -700: acceptance_probability = 0.0` untuk mencegah kesalahan *floating-point underflow/overflow* pada fungsi `math.exp()` ketika suhu T sudah sangat kecil. Setelah perulangan berakhir, baik karena T < Tmin maupun mencapai `max_iterations`, fungsi mengembalikan objek `current_state` sebagai solusi akhir beserta *dictionary metrics* yang memuat nilai objektif awal dan akhir, durasi eksekusi dalam detik, daftar riwayat nilai objektif (`history_obj`), riwayat nilai $e^{\frac{\Delta E}{T}}$ (`history_prob`), dan frekuensi stuck (`stuck_frequency`).

2.4 Genetic Algorithm

Algoritma ini bertugas mencari penempatan kendaraan paling optimal dengan cara memodelkan setiap konfigurasi *state* sebagai satu individu. Algoritma ini mengevaluasi banyak *state* sekaligus di dalam satu populasi. *State* dengan total *shipping fee* tinggi dan minim pelanggaran *constraint* akan dipertahankan. Posisi kendaraan dari *state* unggulan ini kemudian saling disilangkan melalui proses *crossover*, serta dimodifikasi sedikit posisinya lewat mutasi. Siklus generasi ini diulang terus menerus sampai batas iterasi habis untuk mendapatkan *state* akhir bernilai paling tinggi.

Fungsi: _initialize_population

```
def _initialize_population(self) -> List[CargoState]:
        population: List[CargoState] = []
        for _ in range(self.pop_size):
            state = CargoState.generate_random_initial_state(
                ship=self.ship,
                vehicles=self.vehicles,
                use_bonus_objective=self.use_bonus_objective
            )
            population.append(state)

        return population
```

Fungsi ini bertugas membuat populasi awal yang berisi sekumpulan *state* acak. Jumlah state yang diinisialisasi disesuaikan dengan nilai variabel `pop_size`. Pembuatannya cukup memanggil fungsi `generate_random_initial_state` dari kelas `CargoState` berulang kali ke dalam sebuah list, sehingga kita langsung mendapatkan populasi awal dengan variasi letak kendaraan yang beragam.

Fungsi: _calculate_fitness()

```
def _calculate_fitness(self, state: CargoState) -> float:
        oob, ovl, ex_w = state.count_violations()
        base_fee = state.get_total_shipping_fee()

        if oob == 0 and ovl == 0 and ex_w == 0:
            if not self.use_bonus_objective:
                return float(base_fee)
            else:
                bonus_score = 0.0
                for vid, p in state.placements.items():
                    if p.in_ship:
                        v = self.vehicles[vid]
                          urgency_bonus = v.shipping_fee * (1.0 / max(1,
v.eta)) * 0.1
                        bonus_score += v.shipping_fee + urgency_bonus
                return round(bonus_score, 4)
```

```
        else:
            penalty = (oob * 500.0) + (ovl * 500.0) + (ex_w * 50.0)
            return float(base_fee) - penalty
```

Fungsi ini menilai seberapa bagus sebuah *state*. Jika penempatan kendaraan valid, nilai *fitness* murni diambil dari *total shipping fee*. Kita juga punya opsi menambah skor bonus dari fungsi objektif berdasarkan urgensi waktu kedatangan. Namun jika *state* tersebut melanggar *constraint*, sistem akan memberikan penalti bertingkat. Contohnya setiap satu sel kendaraan yang menabrak batas *grid* atau *out of bounds* akan memotong 500 poin. Semakin parah tingkat *overlap* atau kelebihan beratnya, nilai *fitness* akan dipotong semakin besar hingga angkanya bisa menjadi negatif.

Fungsi: _roulette_wheel_selection()

```
def _roulette_wheel_selection(self, population: List[CargoState]) ->
CargoState:
      fitnesses = [self._calculate_fitness(ind) for ind in population]
      min_fit = min(fitnesses)

      shifted_fitnesses = [f - min_fit + 1.0 for f in fitnesses]

      selected = random.choices(population, weights=shifted_fitnesses,
k=1)[0]

      return selected.copy()
```

Fungsi ini memilih individu secara acak untuk disilangkan nantinya. *State* yang punya *fitness* besar mendapat peluang kepilih yang jauh lebih tinggi. Karena evaluasi *fitness* bisa menghasilkan angka minus akibat penalti, fungsi ini harus menggeser seluruh skor ke atas terlebih dahulu. Caranya dengan mencari nilai *fitness* paling minus, lalu semua skor ditambahkan nilai tersebut plus satu. Hasilnya state dengan nilai terburuk sekalipun berubah menjadi positif satu, sehingga fungsi *random* bawaan Python bisa bekerja normal tanpa merusak perhitungan peluang.

Fungsi: _crossover()

```
def _crossover(self, parent1: CargoState, parent2: CargoState) ->
Tuple[CargoState, CargoState]:
        if random.random() > self.crossover_rate:
            return parent1.copy(), parent2.copy()

        child1 = parent1.copy()
        child2 = parent2.copy()

        for vid in self.vehicles.keys():
            if random.random() < 0.5:
                child1.placements[vid] = parent2.placements[vid].copy()
                child2.placements[vid] = parent1.placements[vid].copy()

        child1.invalidate_cache()
        child2.invalidate_cache()

        return child1, child2
```

Fungsi ini bertugas menyilangkan dua *state parent* untuk menghasilkan dua *state child* baru. Di awal kode terdapat pengecekan probabilitas berdasarkan variabel `crossover_rate`. Jika angka acak yang keluar melebihi nilai tersebut, kedua *parent* akan di-*return* utuh tanpa perubahan. Jika lolos masuk proses persilangan, sistem akan melakukan iterasi ke seluruh ID kendaraan. Pada setiap kendaraan ada peluang 50% letak dan posisinya saling ditukar antara *child* pertama dan *child* kedua. Sebelum di-*return*, fungsi `invalidate_cache` dipanggil untuk menghapus riwayat nilai kedua *child* supaya skor *fitness* mereka bisa dihitung ulang nanti.

Fungsi: _mutate()

```
def _mutate(self, state: CargoState) -> CargoState:
        if random.random() < self.mutation_rate:
            mutated_state = state.get_random_neighbor(max_attempts=15)

            if mutated_state is not None:
                return mutated_state

        return state
```

Fungsi ini memberikan sedikit modifikasi posisi secara acak pada *state* hasil persilangan. Jika syarat probabilitas dari `mutation_rate` terpenuhi, fungsi akan memanggil `get_random_neighbor` bawaan kelas `CargoState` untuk menggeser letak atau orientasi salah satu kendaraannya. Proses pencarian tetangga acak ini sengaja kita batasi lewat parameter `max_attempts` sebanyak 15x saja agar sistem tidak terjebak komputasi lama akibat mencari posisi kosong. Jika mutasi berhasil dan tidak memberikan *return* bernilai `None`, *state* yang baru akan dikembalikan. Sebaliknya jika gagal, fungsi sekadar mengembalikan *state* aslinya.

Fungsi: run()

```
def run(self) -> Tuple[CargoState, List[float], List[float]]:
        population = self._initialize_population()
        best_state = max(population, key=self._calculate_fitness)
        self.initial_best_state = best_state.copy()

        for generation in range(self.max_generations):
                fitnesses = [self._calculate_fitness(ind) for ind in
population]
            gen_max_fit = max(fitnesses)
            gen_avg_fit = sum(fitnesses) / self.pop_size

            self.history_max_fitness.append(gen_max_fit)
            self.history_avg_fitness.append(gen_avg_fit)

            current_best = max(population, key=self._calculate_fitness)
                        if self._calculate_fitness(current_best) >
self._calculate_fitness(best_state):
                best_state = current_best.copy()

                                    sorted_pop = sorted(population,
key=self._calculate_fitness, reverse=True)
                                new_population = [sorted_pop[0].copy(),
sorted_pop[1].copy()]

            while len(new_population) < self.pop_size:
                p1 = self._roulette_wheel_selection(population)
                p2 = self._roulette_wheel_selection(population)
```

```
                c1, c2 = self._crossover(p1, p2)

                c1 = self._mutate(c1)
                c2 = self._mutate(c2)

                new_population.append(c1)
                if len(new_population) < self.pop_size:
                    new_population.append(c2)

            population = new_population

            if (generation + 1) % 10 == 0:
                print(
                    f"  Generasi {generation + 1}/{self.max_generations}
| "
                    f"fitness maks: {gen_max_fit:.2f} | "
                    f"fitness rata-rata: {gen_avg_fit:.2f}"
                )

    return best_state, self.history_max_fitness, self.history_avg_fitness
```

Fungsi ini adalah eksekutor yang mengeksekusi semua komponen sebelumnya. Alurnya dimulai dari pembuatan populasi awal lalu masuk ke loop evolusi sebanyak target batas generasi. Pada setiap generasinya, fungsi akan mengevaluasi *fitness* semua individu lalu mendata skor tertinggi dan rata-ratanya ke dalam sebuah list untuk bahan pembuatan plot grafik. Dua individu dengan skor *fitness* paling tinggi akan langsung diamankan ke generasi berikutnya melalui mekanisme elitism.[1] Sisa slot populasi lalu dipenuhi dengan *state* baru hasil seleksi, persilangan, dan mutasi. Setelah seluruh iterasi generasi selesai, fungsi ini memberikan hasil akhir berupa *state* dengan penempatan paling optimal beserta dua riwayat evaluasinya.

2.5 Hasil & Analisis Eksperimen

2.5.1 *Steepest Ascent*

[1] Stuart Russell dan Peter Norvig, Artificial Intelligence: A Modern Approach

Percobaan 1		

Initial	<pre>"initial_state": {
	"ship": {
	"id": "IntimLine-Deck-01",
	"w": 10,
	"l": 20,
	"max_capacity": 500
	},
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	"l": 4,
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	"weight": 50,
	"eta": 3
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	"V02": {
	"id": "V02",
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	"l": 5,
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"weight": 95,
"eta": 1
},
"V05": {
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"w": 2,
"l": 3,
"shipping_fee": 35,
"weight": 25,
"eta": 5
},
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},
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"y": -1,
"orientation": "Horizontal"
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"V06": {
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"in_ship": true,
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"weight": 80,
"eta": 3
},
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"w": 2,
"l": 5,
"shipping_fee": 55,
"weight": 40,
"eta": 2
},
"V09": {
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            "eta": 3
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"V12",
            "w": 4,
            "l": 4,

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60,
            "eta": 2
          },
          "V13": {
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"V13",
            "w": 2,
            "l": 4,

"shipping_fee": 48,
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            "eta": 5
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          "V14": {
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true,
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"orientation":
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false,
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"orientation":
"Horizontal"
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"orientation":
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            "l": 5,

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60,
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"V15",
            "w": 2,
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"shipping_fee": 28,
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"V16",
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            "l": 4,

"shipping_fee": 62,
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"V17",
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            "l": 5,

"shipping_fee": 105,
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"orientation":
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          },
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false,
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            "y": -1,

"orientation":
"Horizontal"
          },
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true,
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"orientation":
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          },
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true,
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"V18",
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            "l": 3,

"shipping_fee": 38,
            "weight":
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            "eta": 2
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<pre>
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        "vehicles": {
          "V01": {
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"V01",
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            "l": 4,

"shipping_fee": 60,
            "weight":
50,
            "eta": 3
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"orientation":
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"V03": {
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"w": 2,
"l": 4,
"shipping_fee": 45,
"weight": 35,
"eta": 4
},
"V04": {
"id": "V04",
"w": 4,
"l": 6,
"shipping_fee": 130,
"weight": 95,
"eta": 1
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"V05": {
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"l": 3,
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},
"V04": {
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"V05": {
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"shipping_fee": 65,
"weight": 45,
"eta": 2
},
"V07": {
"id": "V07",
"w": 4,
"l": 5,

"shipping_fee": 110,
"weight": 80,
"eta": 3
},
"V08": {
"id": "V08",
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"l": 5,

"shipping_fee": 55,
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"V09": {

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            "w": 2,
            "l": 3,

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            "w": 3,
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"orientation":
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            "weight": 35,
            "eta": 5
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          "V14": {
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            "w": 3,
            "l": 5,
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            "weight": 60,
            "eta": 1
          },
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          "V16": {
            "id":
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            "in_ship": true,
            "x": 3,
            "y": 13,
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            "orientation": "Horizontal"
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"V17",
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"shipping_fee": 105,
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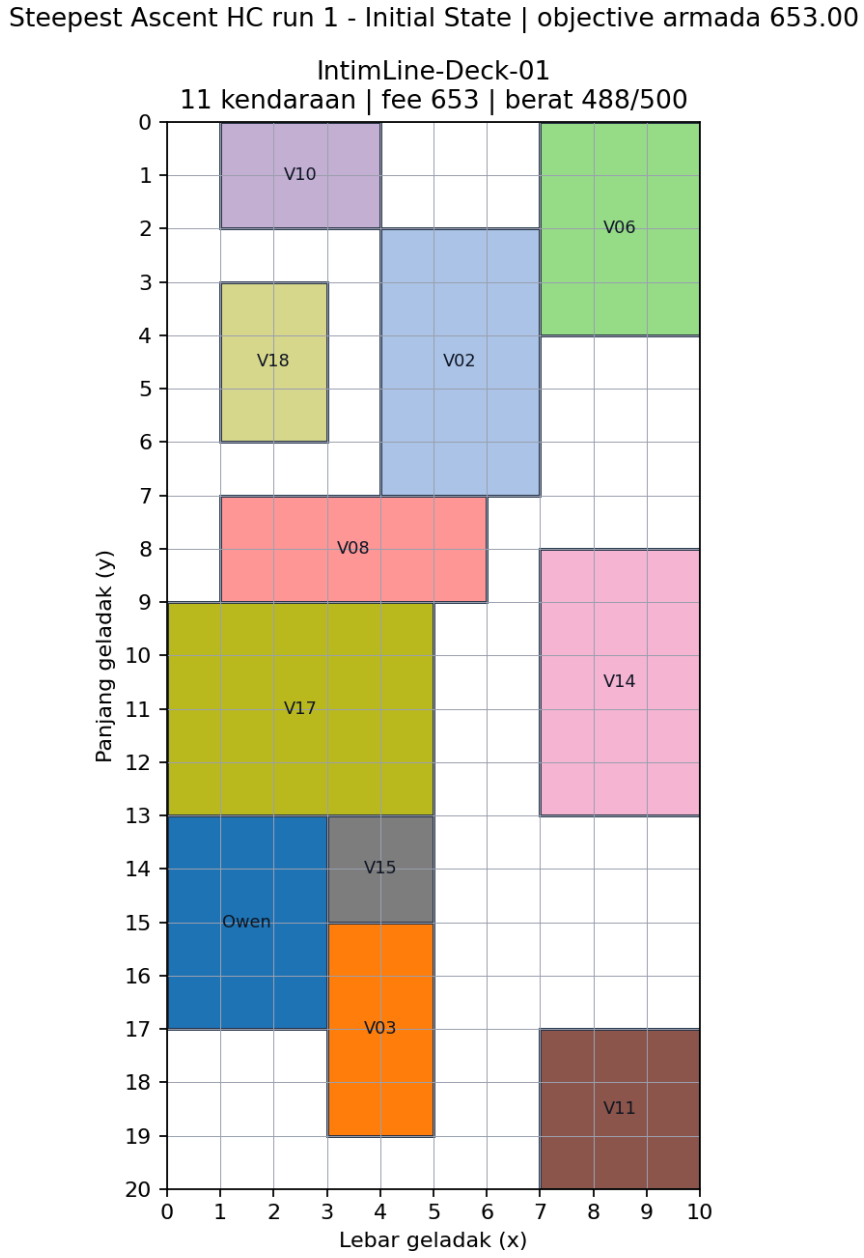
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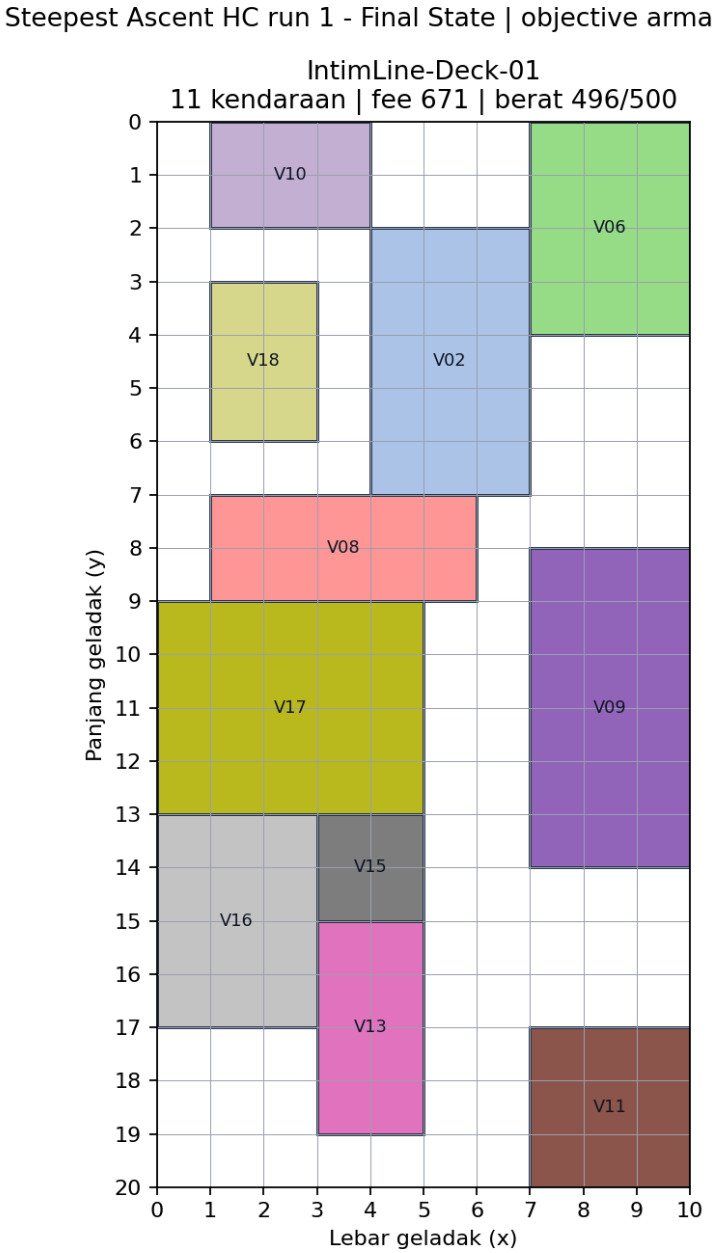
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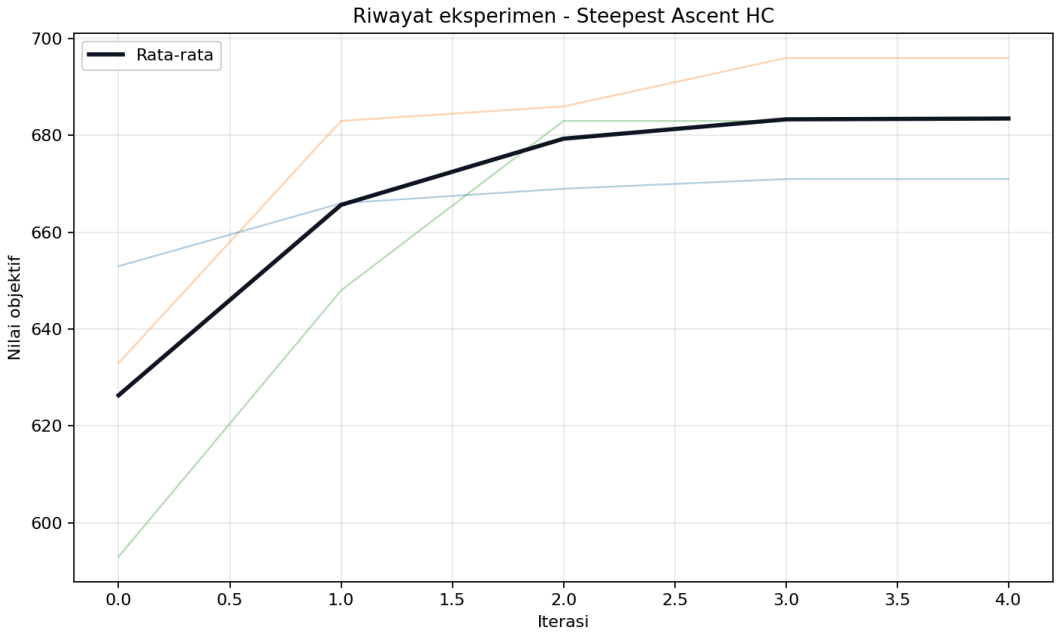
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<td colspan="2">Objective Function Score: 671</td>
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<tr>
<td colspan="3">Visualisasi Initial State:</td>
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Visualisasi Final State:


Steepest Ascent HC run 1 - Final State | objective armada 671.00
IntimLine-Deck-01
11 kendaraan | fee 671 | berat 496/500
V10
V06
V18
V02
V08
V17
V09
V15
V16
V13
V11
Panjang geladak (y)
Lebar geladak (x)

Plot:



Highest Objective Function Score: 671

Iterations: 4

Duration: 0,1848

Percobaan 2		
Initial	"initial_state": { "ship": { "id": "IntimLine-Deck-01", "w": 10, "l": 20, "max_capacity": 500 }, "vehicles": { "V01": { "id": "V01", "w": 3, "l": 4,	"placements": { "V01": { "vehicle_id": "V01", "in_ship": true, "x": 5, "y": 16, "orientation": "Horizontal" }, "V02": { "vehicle_id": "V02",

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            "l": 5,

"shipping_fee": 85,
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            "eta": 2
          },
          "V03": {
            "id": "V03",
            "w": 2,
            "l": 4,

"shipping_fee": 45,
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            "eta": 4
          },
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            "x": -1,
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"orientation": "Horizontal"
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"vehicle_id": "V04",
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            "x": -1,
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"orientation": "Horizontal"
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"l": 4,
"shipping_fee": 65,
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"eta": 2
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"V07": {
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"l": 5,
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"in_ship": false,
"x": -1,
"y": -1,
"orientation": "Horizontal"
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"V08": {
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"vehicle_id": "V09",
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"V14",
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"V17",
            "w": 4,
            "l": 5,

"shipping_fee": 105,
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            "eta": 4
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"orientation":
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"vehicle_id": "V18",
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"orientation":
"Vertical"
          }
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	Objective Function Score: 633	
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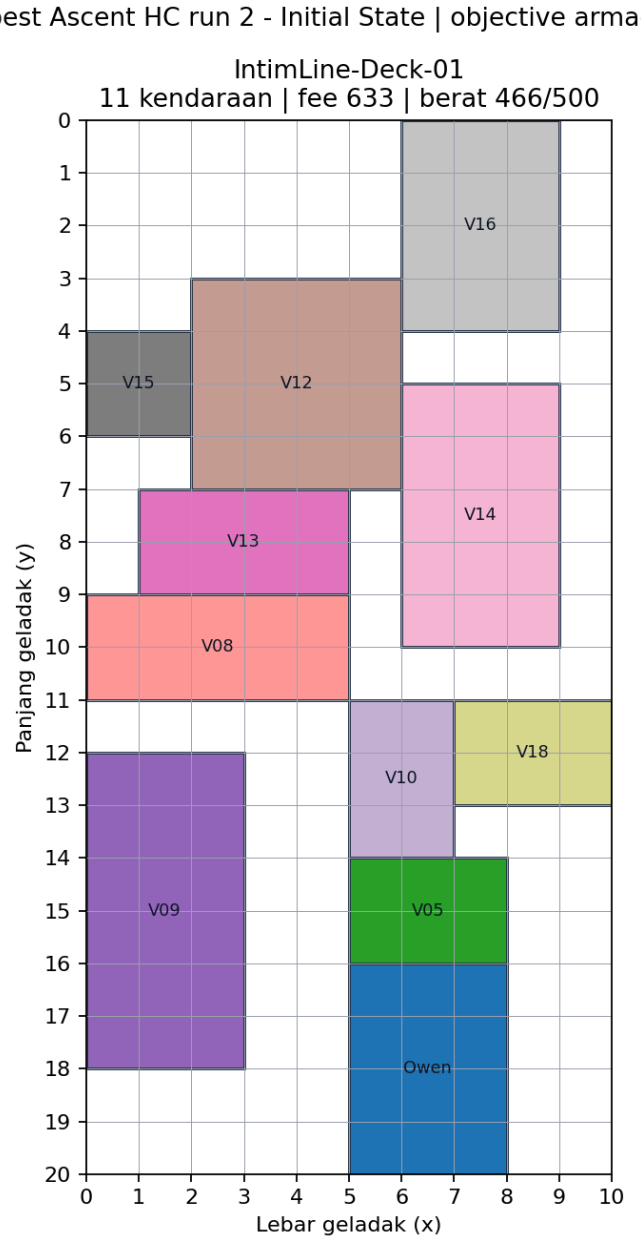
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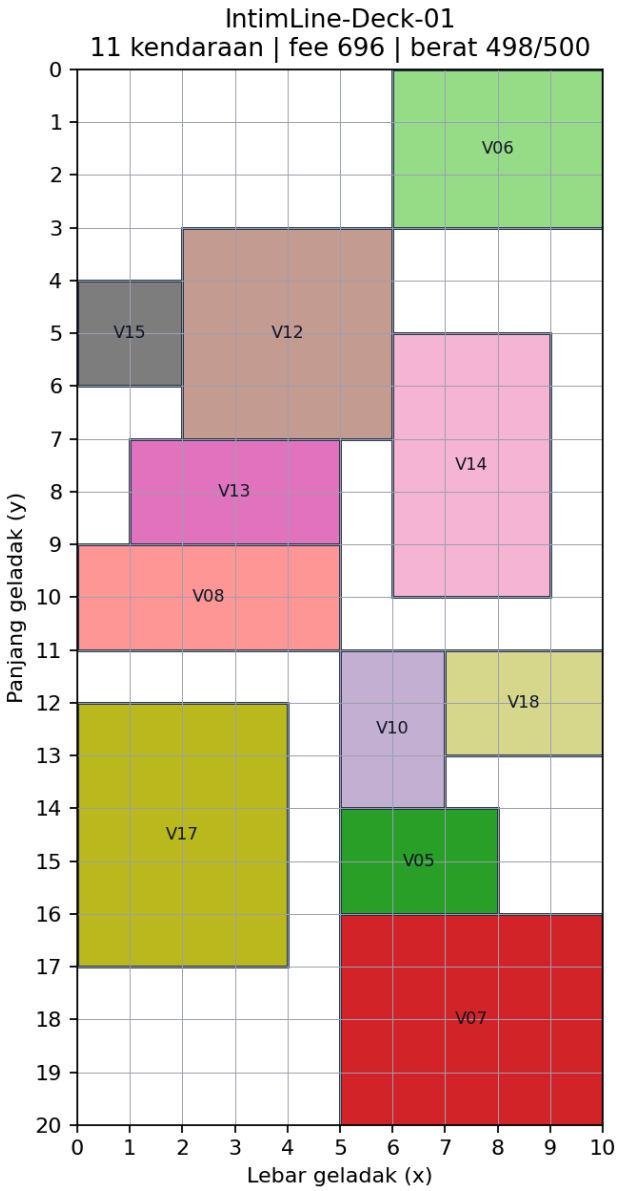
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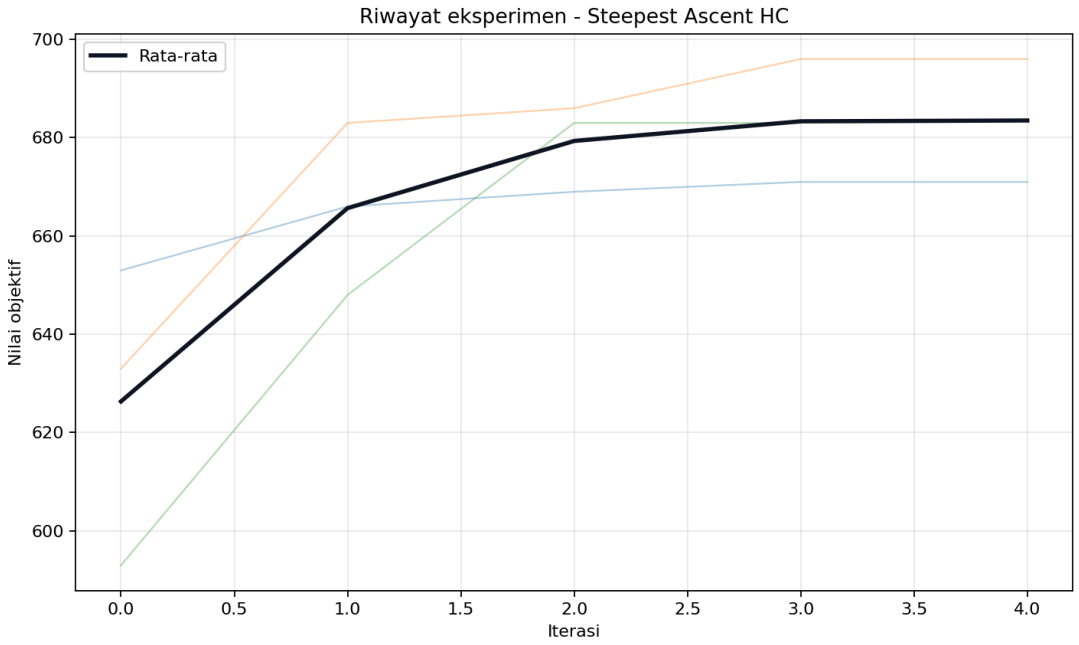
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<td colspan="2">Objective Function Score: 696</td>
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<td colspan="3">Visualisasi Initial State:</td>
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Visualisasi Final State:


Steepest Ascent HC run 2 - Final State | objective armada 696.00
IntimLine-Deck-01
11 kendaraan | fee 696 | berat 498/500
Panjang geladak (y)
Lebar geladak (x)
V06
V15
V12
V14
V13
V08
V18
V10
V17
V05
V07

Plot:



Highest Objective Function Score: 696

Iterations: 4

Duration: 0,1468

Percobaan 3		
Initial	"initial_state": { "ship": { "id": "IntimLine-Deck-01", "w": 10, "l": 20, "max_capacity": 500 }, "vehicles": { "V01": { "id": "V01", "w": 3,	"placements": { "V01": { "vehicle_id": "V01", "in_ship": true, "x": 1, "y": 3, "orientation": "Horizontal" }, "V02": {

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            "l": 5,
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            "eta": 2
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            "w": 2,
            "l": 4,
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            "weight":
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	Objective Function Score: 593	
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          "V16": {

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            "in_ship":
true,
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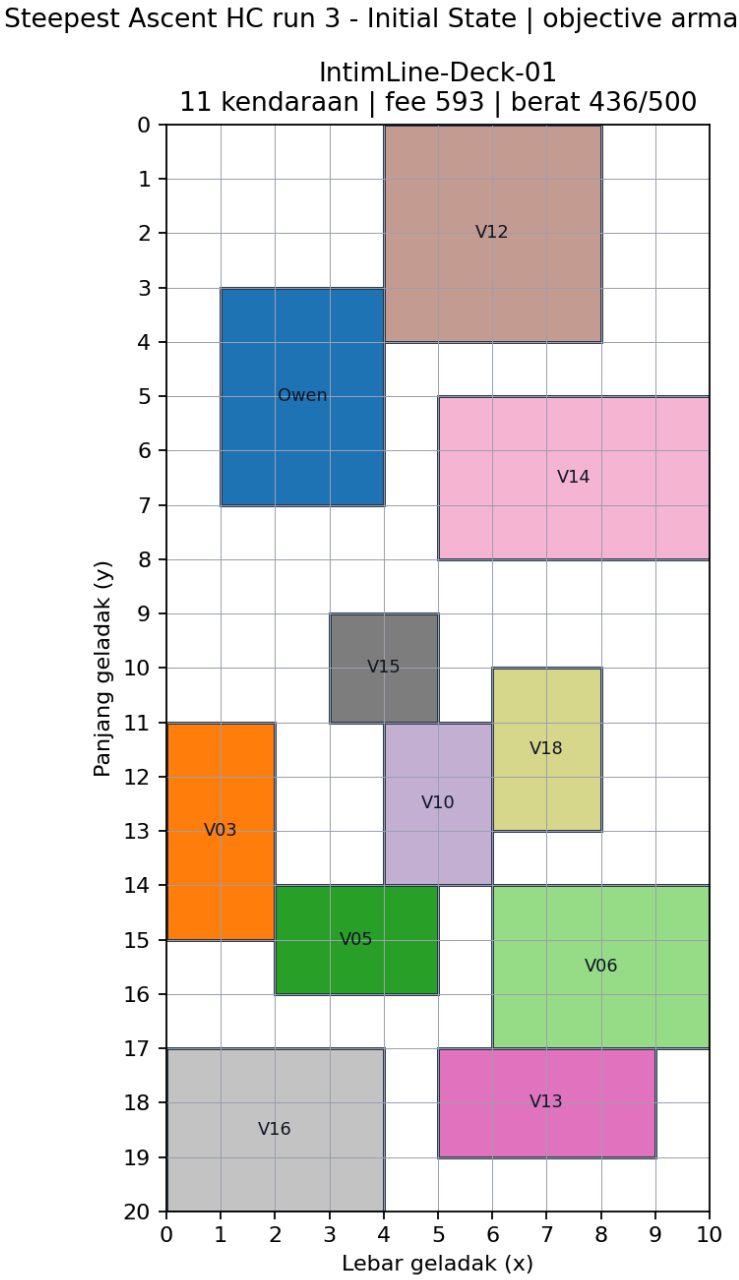
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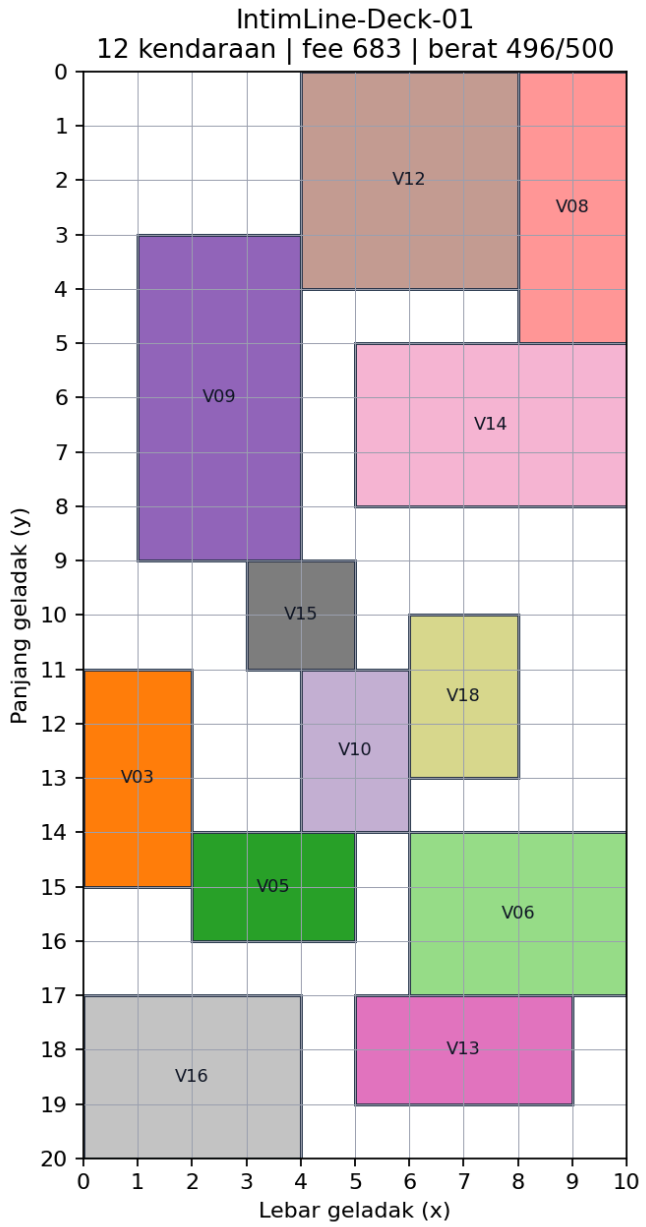
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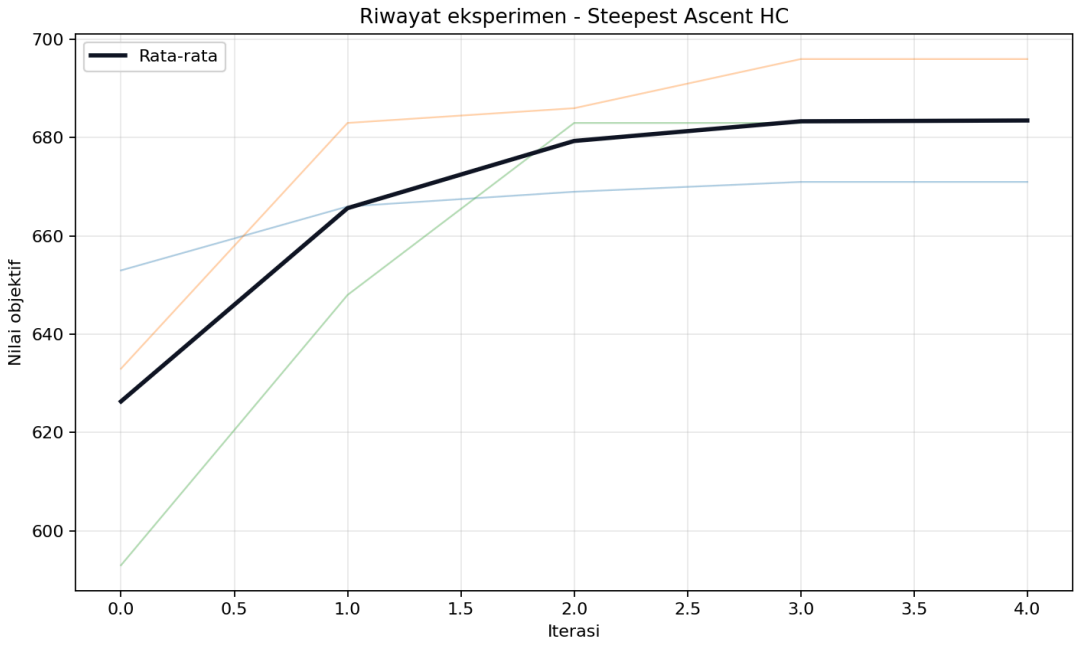
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<td colspan="2">Objective Function Score: 683</td>
</tr>
<tr>
<td colspan="3">Visualisasi Initial State:</td>
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</table>



Visualisasi Final State:


Steepest Ascent HC run 3 - Final State | objective armada 683.00
IntimLine-Deck-01
12 kendaraan | fee 683 | berat 496/500
Panjang geladak (y)
Lebar geladak (x)
V12
V08
V09
V14
V15
V18
V10
V03
V05
V06
V16
V13

Plot:



Highest Objective Function Score: 683
Iterations: 3
Duration: 0,1237

2.5.2 *Sideways Move*

Percobaan 1		
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            "eta": 3
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"V02",
            "w": 3,
            "l": 5,

"shipping_fee": 85,
            "weight":
65,
            "eta": 2
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"V03",
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            "l": 4,

"shipping_fee": 45,
            "weight":
35,
            "eta": 4
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"V04",
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            "weight":
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            "y": -1,

"orientation":
"Horizontal"
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true,
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"orientation":
"Horizontal"
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false,
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"shipping_fee": 35,
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            "eta": 5
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70,
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"V10",
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true,
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"orientation":
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"orientation":
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"V16",
            "w": 3,
            "l": 4,
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            "eta": 2
          },
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"V17",
            "w": 4,
            "l": 5,
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      "vehicles": {
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"shipping_fee": 60,
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            "l": 6,
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            "weight": 95,
            "eta": 1
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          "V05": {
            "id": "V05",
            "w": 2,
            "l": 3,
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            "eta": 5
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          "V06": {
            "vehicle_id": "V06",
            "in_ship": true,
            "x": 1,
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            "orientation": "Horizontal"
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```
            "eta": 2
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          "V07": {
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"V07",
            "w": 4,
            "l": 5,

"shipping_fee": 110,
            "weight":
80,
            "eta": 3
          },
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"V08",
            "w": 2,
            "l": 5,

"shipping_fee": 55,
            "weight":
40,
            "eta": 2
          },
          "V09": {
            "id":
"V09",
            "w": 3,
            "l": 6,

"shipping_fee": 95,
            "weight":
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            "eta": 1
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            "id":
"V10",
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```

"orientation":
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            "in_ship": true,
            "x": 1,
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            "orientation": "Vertical"
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          "V13": {
            "vehicle_id": "V13",
            "in_ship": true,
            "x": 2,
            "y": 0,
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            "eta": 1
          },
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"V15",
            "w": 2,
            "l": 2,

"shipping_fee": 28,
            "weight":
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            "eta": 3
          },
          "V16": {
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"V16",
            "w": 3,
            "l": 4,

"shipping_fee": 62,
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48,
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"V17",
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            "l": 5,
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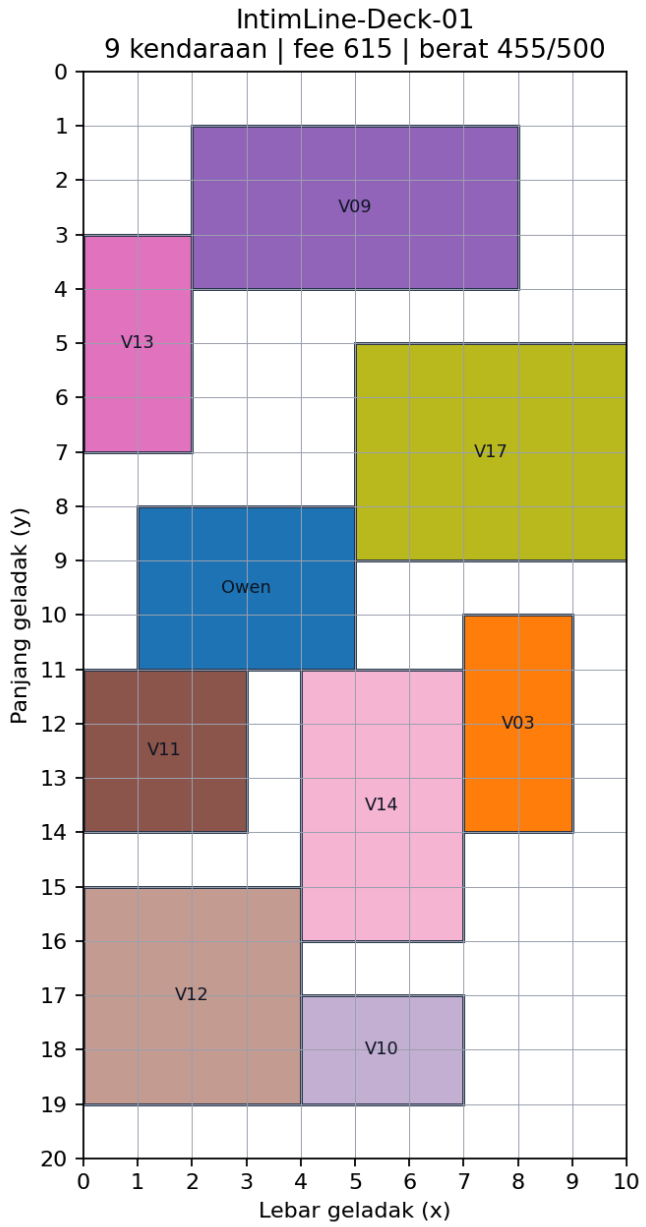
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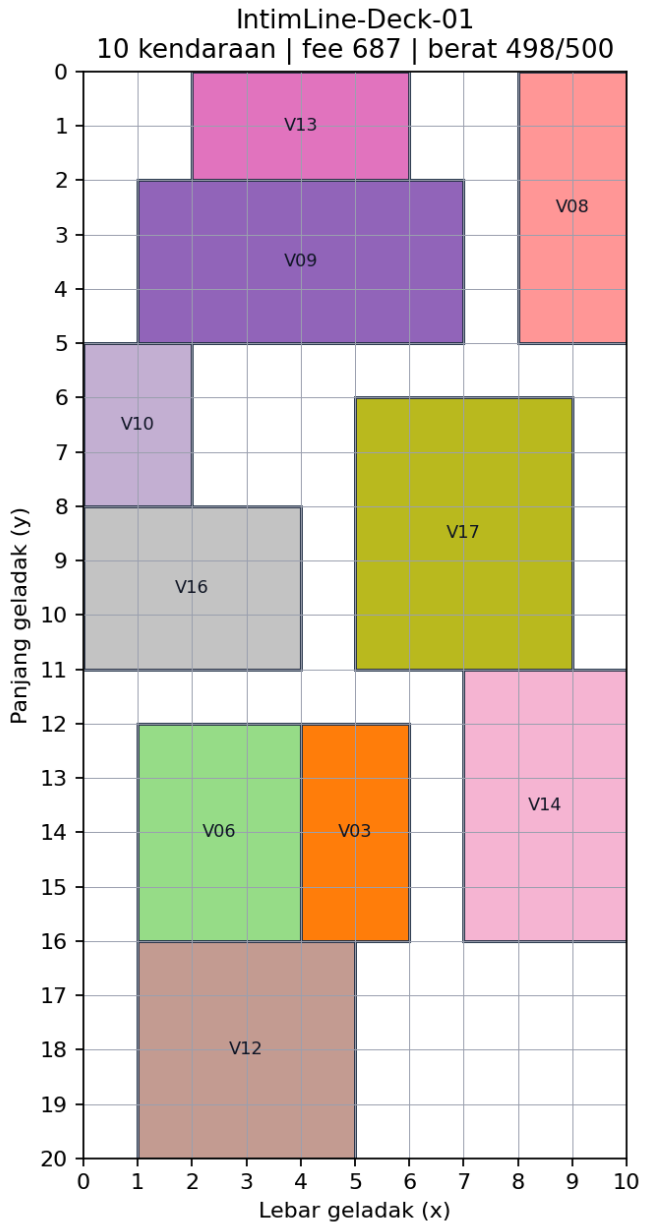
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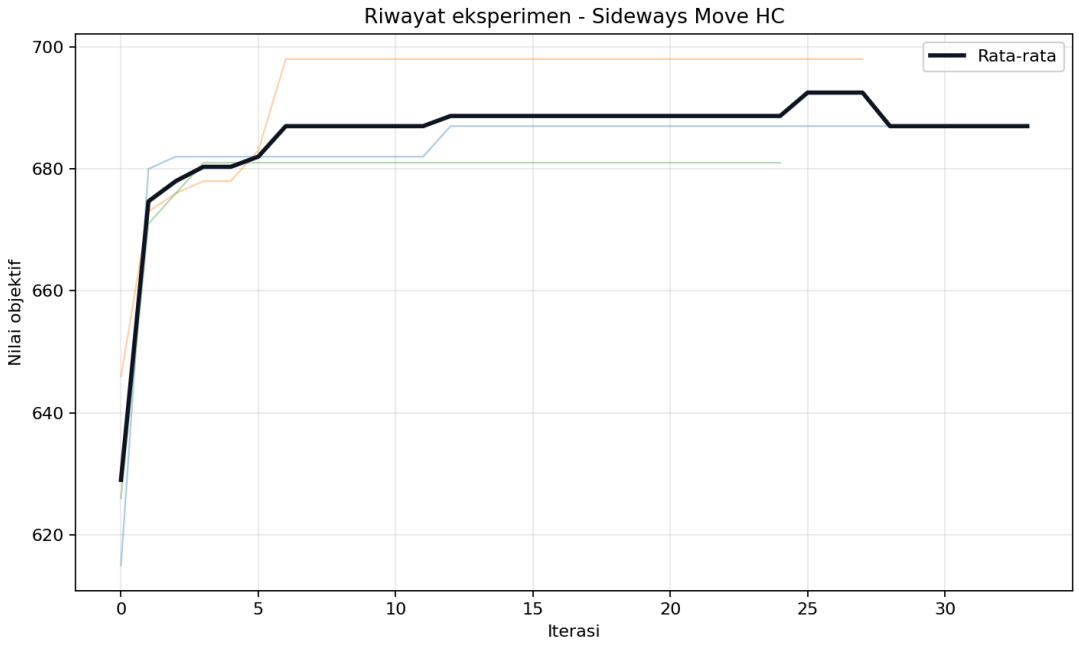
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<td colspan="2">Objective Function Score: 687</td>
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<tr>
<td colspan="3">Visualisasi Initial State:</td>
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</table>



Visualisasi Final State:


Sideways Move HC run 1 - Final State | objective armada 687.00
IntimLine-Deck-01
10 kendaraan | fee 687 | berat 498/500
V13
V08
V09
V10
V17
V16
V14
V06
V03
V12
Panjang geladak (y)
Lebar geladak (x)
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20

Plot:



Highest Objective Function Score: 687
Iterations: 33
Duration: 0.8324

Percobaan 2		
Initial	"initial_state": { "ship": { "id": "IntimLine-Deck-01", "w": 10, "l": 20, "max_capacity": 500 }, "vehicles": { "V01": { "id": "V01", "w": 3,	"placements": { "V01": { "vehicle_id": "V01", "in_ship": false, "x": -1, "y": -1, "orientation": "Horizontal" }, "V02": {

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"w": 3,
"l": 5,
"shipping_fee": 85,
"weight": 65,
"eta": 2
},
"V03": {
"id": "V03",
"w": 2,
"l": 4,
"shipping_fee": 45,
"weight": 35,
"eta": 4
},
"V04": {
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"l": 6,
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"y": 12,
"orientation": "Horizontal"
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"in_ship": true,
"x": 3,
"y": 1,
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          "eta": 5
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"V06",
          "w": 3,
          "l": 4,

"shipping_fee": 65,
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          "eta": 2
        },
        "V07": {
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"V07",
          "w": 4,
          "l": 5,

"shipping_fee": 110,
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"weight": 70,
"eta": 1
},
"V10": {
"id": "V10",
"w": 2,
"l": 3,
"shipping_fee": 40,
"weight": 30,
"eta": 4
},
"V11": {
"id": "V11",
"w": 3,
"l": 3,
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"weight": 40,
"eta": 3
},
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"x": -1,
"y": -1,
"orientation": "Horizontal"
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"V11": {
"vehicle_id": "V11",
"in_ship": true,
"x": 0,
"y": 11,
"orientation": "Vertical"
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            "weight":
60,
            "eta": 2
          },
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"V13",
            "w": 2,
            "l": 4,

"shipping_fee": 48,
            "weight":
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            "eta": 5
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"orientation":
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"weight": 48,
"eta": 2
},
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"eta": 4
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"y": 7,

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	Objective Function Score: 646	
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"shipping_fee": 130,
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            "eta": 1
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"shipping_fee": 35,
            "weight": 25,
            "eta": 5
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            "eta": 2
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"orientation": "Vertical"
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            "y": -1,

"orientation": "Horizontal"
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"orientation":
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            "eta": 3
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true,
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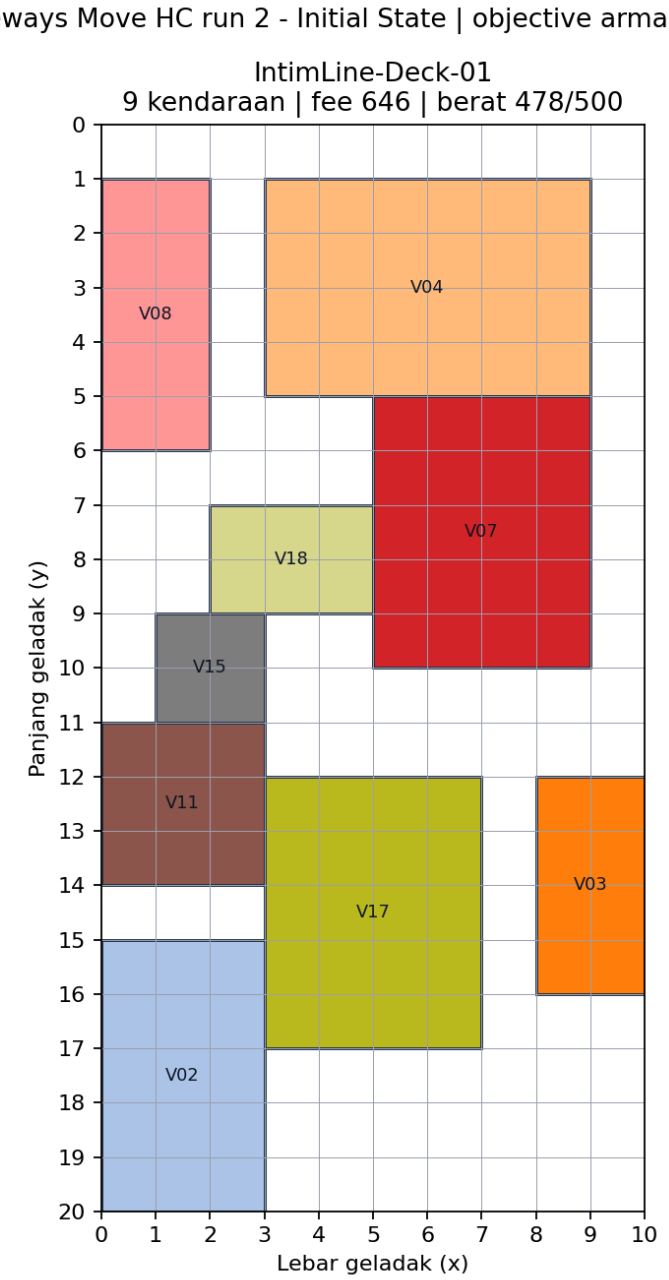
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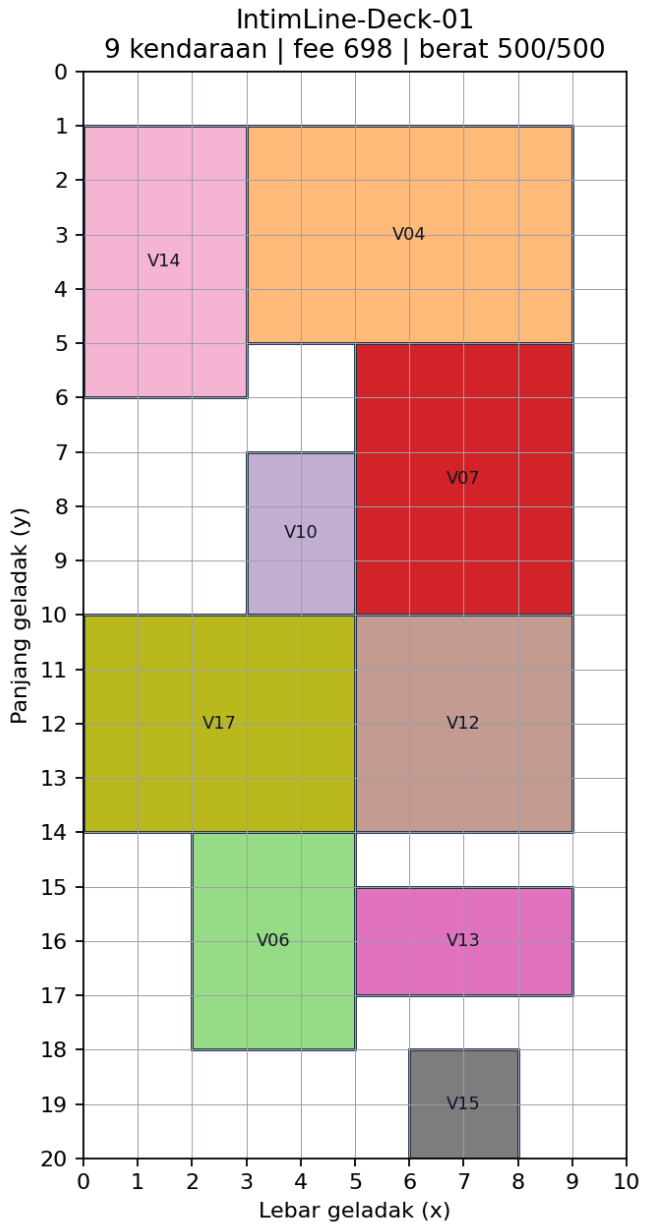
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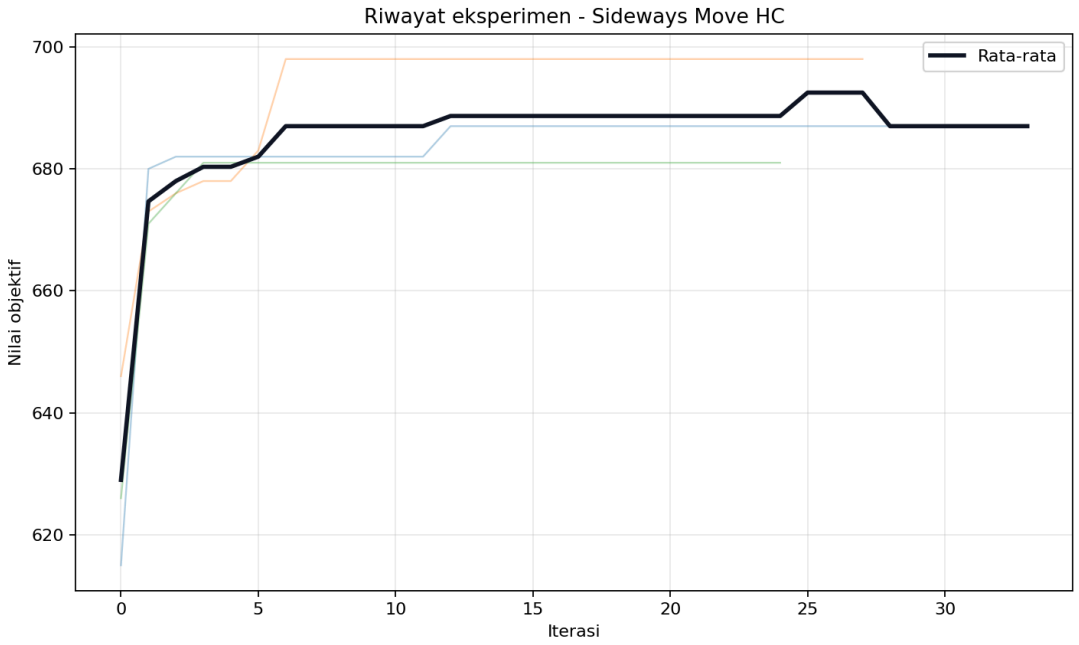
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<tr><td colspan="3">Visualisasi Initial State:</td></tr>
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Visualisasi Final State:


Sideways Move HC run 2 - Final State | objective armada 698.00
IntimLine-Deck-01
9 kendaraan | fee 698 | berat 500/500
V14
V04
V07
V10
V17
V12
V06
V13
V15
Panjang geladak (y)
Lebar geladak (x)
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20

Plot:



Highest Objective Function Score: 698

Iterations: 27

Duration: 0.6811

Percobaan 3		
Initial	"initial_state": { "ship": { "id": "IntimLine-Deck-01", "w": 10, "l": 20, "max_capacity": 500 }, "vehicles": { "V01": { "id": "V01", "w": 3,	"placements": { "V01": { "vehicle_id": "V01", "in_ship": true, "x": 5, "y": 8, "orientation": "Vertical" }, "V02": {

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            "w": 3,
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            "w": 2,
            "l": 4,

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35,
            "eta": 4
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"orientation":
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            "eta": 1
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            "eta": 5
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```

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"orientation":
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	Objective Function Score: 626	
Final	"final_state": { "ship": { "id": "IntimLine-Deck-01", "w": 10, "l": 20, "max_capacity": 500 }, "vehicles": { "V01": { "id": "V01", "w": 3, "l": 4, "shipping_fee": 60, "weight": 50, "eta": 3 }, "V02": { "id": "V02", "w": 3, "l": 5, "shipping_fee": 85, "weight": 65, "eta": 2 }, "V03": { "id": "V03", "w": 2, "l": 4,	"placements": { "V01": { "vehicle_id": "V01", "in_ship": false, "x": -1, "y": -1, "orientation": "Vertical" }, "V02": { "vehicle_id": "V02", "in_ship": true, "x": 1, "y": 17, "orientation": "Vertical" }, "V03": { "vehicle_id": "V03", "in_ship": false, "x": -1, "y": -1, "orientation": "Horizontal" }, "V04": { "vehicle_id": "V04",

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"weight": 95,
"eta": 1
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"w": 2,
"l": 3,
"shipping_fee": 35,
"weight": 25,
"eta": 5
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            "eta": 1
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            "w": 2,
            "l": 2,

"shipping_fee": 28,
            "weight":
20,
            "eta": 3
          },
          "V16": {
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"V16",
            "w": 3,
            "l": 4,

"shipping_fee": 62,
            "weight":
48,
            "eta": 2
          },
          "V17": {
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"V17",
            "w": 4,
            "l": 5,

"shipping_fee": 105,
```

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          "V15": {

"vehicle_id": "V15",
            "in_ship":
true,
            "x": 8,
            "y": 18,

"orientation":
"Horizontal"
          },
          "V16": {

"vehicle_id": "V16",
            "in_ship":
false,
            "x": -1,
            "y": -1,

"orientation":
"Horizontal"
          },
          "V17": {

"vehicle_id": "V17",
            "in_ship":
true,
            "x": 0,
            "y": 12,

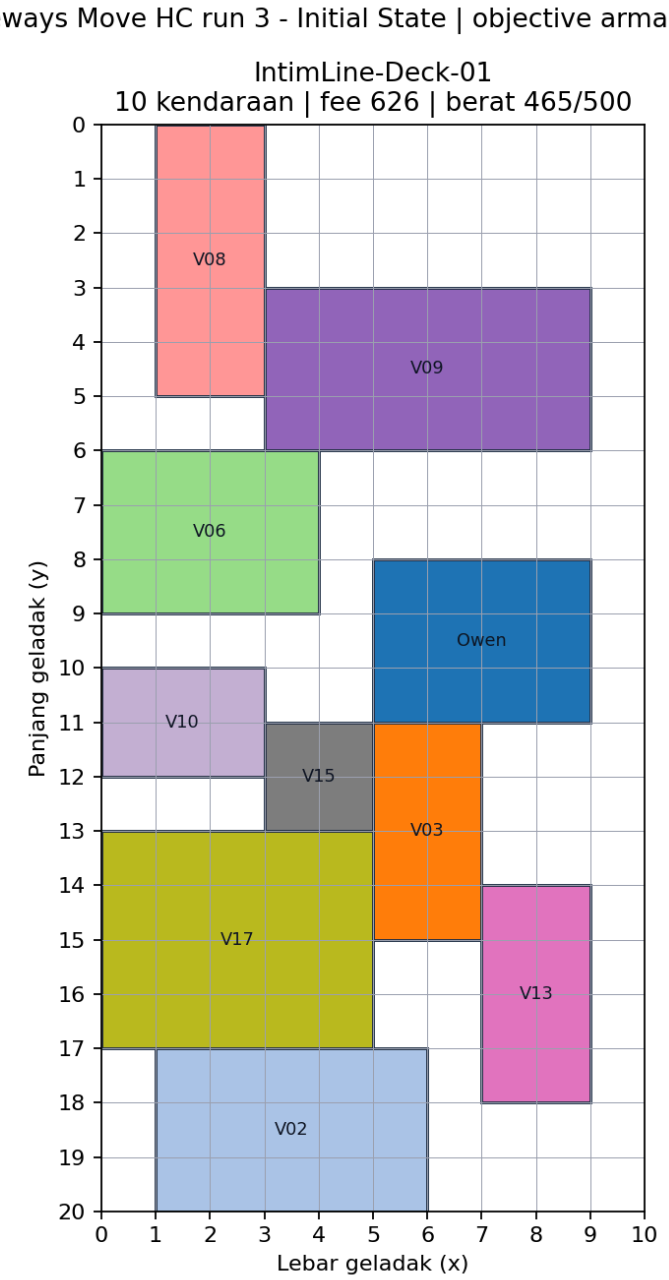
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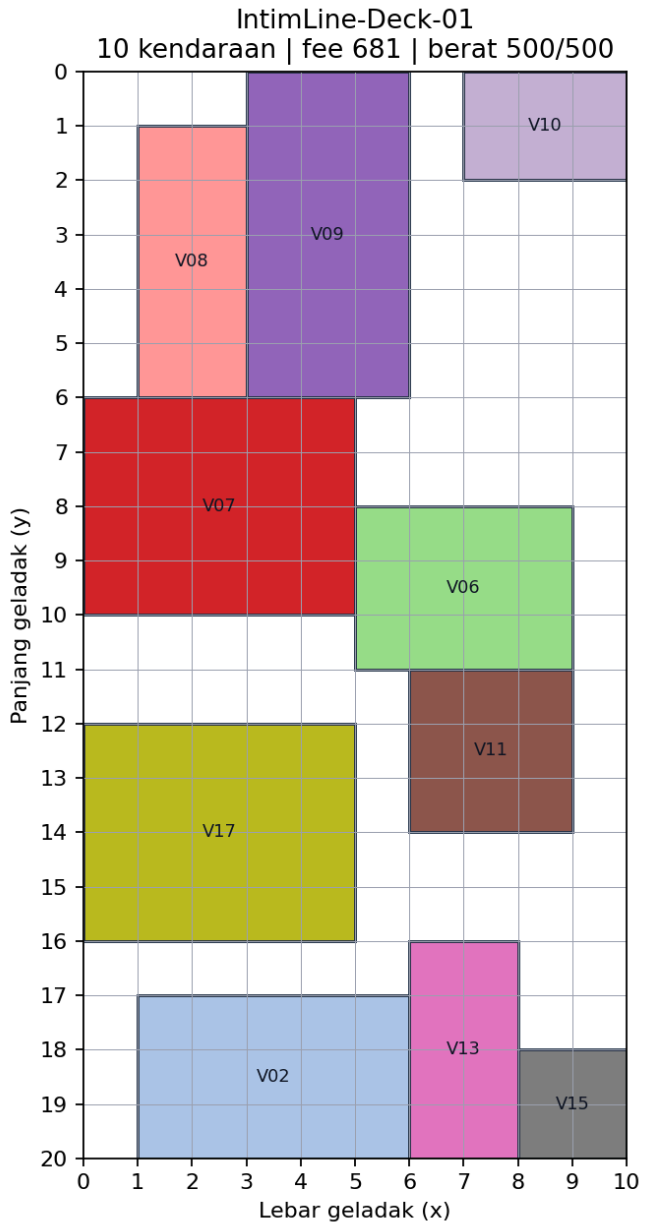
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"V18",
            "w": 2,
            "l": 3,

"shipping_fee": 38,
            "weight":
28,
            "eta": 2
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            "x": -1,
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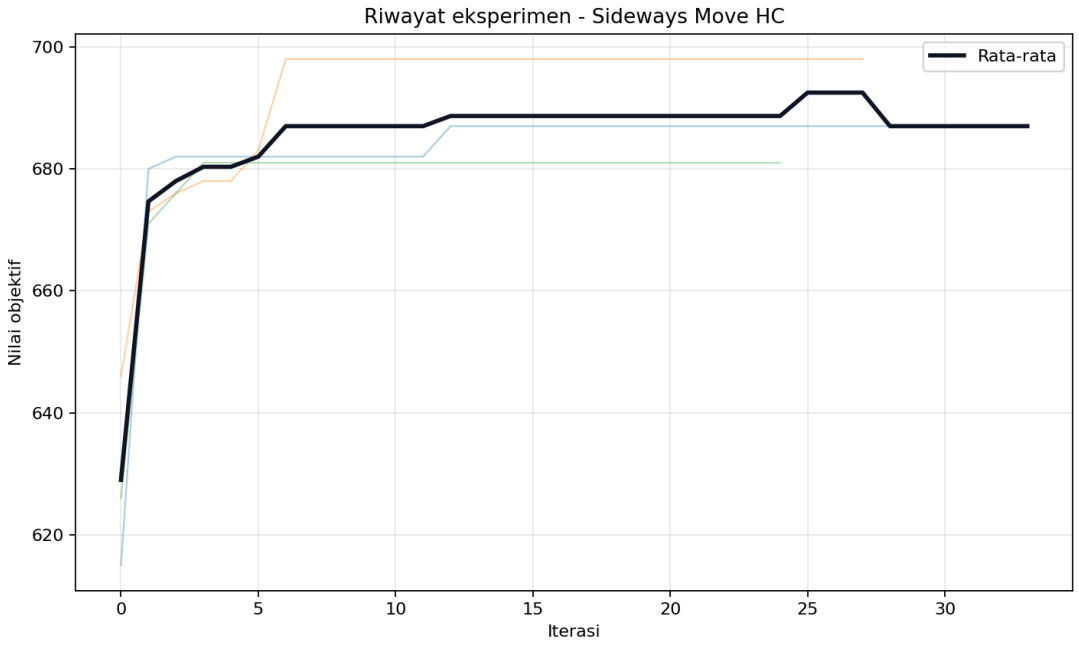
"orientation":
"Horizontal"
          }
        },
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<tr><td></td><td colspan="2">Objective Function Score: 681</td></tr>
<tr><td colspan="3">Visualisasi Initial State:</td></tr>
</table>



Visualisasi Final State:


Sideways Move HC run 3 - Final State | objective armada 681.00
IntimLine-Deck-01
10 kendaraan | fee 681 | berat 500/500
V10
V08
V09
V07
V06
V11
V17
V13
V02
V15
Panjang geladak (y)
Lebar geladak (x)
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20

Plot:



Highest Objective Function Score: 681
Iterations: 24
Duration: 0.6859

2.5.3 *Stochastic*

Percobaan 1		
Initial	"initial_state": { "ship": { "id": "IntimLine-Deck-01", "w": 10, "l": 20, "max_capacity": 500 }, "vehicles": { "V01": { "id":	"placements": { "V01": { "vehicle_id": "V01", "in_ship": true, "x": 5, "y": 5, "orientation": "Horizontal" },

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"V01",
            "w": 3,
            "l": 4,

"shipping_fee": 60,
            "weight":
50,
            "eta": 3
          },
          "V02": {
            "id":
"V02",
            "w": 3,
            "l": 5,

"shipping_fee": 85,
            "weight":
65,
            "eta": 2
          },
          "V03": {
            "id":
"V03",
            "w": 2,
            "l": 4,

"shipping_fee": 45,
            "weight":
35,
            "eta": 4
          },
          "V04": {
            "id":
"V04",
            "w": 4,
            "l": 6,

"shipping_fee": 130,
            "weight":
```

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```
          "V02": {

"vehicle_id": "V02",
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true,
            "x": 3,
            "y": 9,

"orientation":
"Vertical"
          },
          "V03": {

"vehicle_id": "V03",
            "in_ship":
true,
            "x": 5,
            "y": 14,

"orientation":
"Horizontal"
          },
          "V04": {

"vehicle_id": "V04",
            "in_ship":
true,
            "x": 0,
            "y": 3,

"orientation":
"Horizontal"
          },
          "V05": {

"vehicle_id": "V05",
            "in_ship":
true,
            "x": 8,
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95,
            "eta": 1
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          "V05": {
            "id":
"V05",
            "w": 2,
            "l": 3,

"shipping_fee": 35,
            "weight":
25,
            "eta": 5
          },
          "V06": {
            "id":
"V06",
            "w": 3,
            "l": 4,

"shipping_fee": 65,
            "weight":
45,
            "eta": 2
          },
          "V07": {
            "id":
"V07",
            "w": 4,
            "l": 5,

"shipping_fee": 110,
            "weight":
80,
            "eta": 3
          },
          "V08": {
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"V08",
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<td>
<pre>
            "y": 14,

"orientation":
"Horizontal"
          },
          "V06": {

"vehicle_id": "V06",
            "in_ship":
true,
            "x": 1,
            "y": 14,

"orientation":
"Vertical"
          },
          "V07": {

"vehicle_id": "V07",
            "in_ship":
false,
            "x": -1,
            "y": -1,

"orientation":
"Horizontal"
          },
          "V08": {

"vehicle_id": "V08",
            "in_ship":
true,
            "x": 1,
            "y": 18,

"orientation":
"Vertical"
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          "V09": {
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"shipping_fee": 55,
            "weight":
40,
            "eta": 2
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          "V09": {
            "id":
"V09",
            "w": 3,
            "l": 6,

"shipping_fee": 95,
            "weight":
70,
            "eta": 1
          },
          "V10": {
            "id":
"V10",
            "w": 2,
            "l": 3,

"shipping_fee": 40,
            "weight":
30,
            "eta": 4
          },
          "V11": {
            "id":
"V11",
            "w": 3,
            "l": 3,

"shipping_fee": 50,
            "weight":
40,
```

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<td>

```

"vehicle_id": "V09",
            "in_ship":
false,
            "x": -1,
            "y": -1,

"orientation":
"Horizontal"
          },
          "V10": {

"vehicle_id": "V10",
            "in_ship":
true,
            "x": 7,
            "y": 12,

"orientation":
"Vertical"
          },
          "V11": {

"vehicle_id": "V11",
            "in_ship":
false,
            "x": -1,
            "y": -1,

"orientation":
"Horizontal"
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          "V12": {

"vehicle_id": "V12",
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false,
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            "y": -1,
```

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<td>
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            "eta": 3
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          "V12": {
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"V12",
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"shipping_fee": 90,
            "weight":
60,
            "eta": 2
          },
          "V13": {
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"V13",
            "w": 2,
            "l": 4,

"shipping_fee": 48,
            "weight":
35,
            "eta": 5
          },
          "V14": {
            "id":
"V14",
            "w": 3,
            "l": 5,

"shipping_fee": 82,
            "weight":
60,
            "eta": 1
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          "V15": {
            "id":
"V15",
            "w": 2,
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<pre>

"orientation":
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          "V13": {

"vehicle_id": "V13",
            "in_ship":
false,
            "x": -1,
            "y": -1,

"orientation":
"Horizontal"
          },
          "V14": {

"vehicle_id": "V14",
            "in_ship":
true,
            "x": 0,
            "y": 0,

"orientation":
"Vertical"
          },
          "V15": {

"vehicle_id": "V15",
            "in_ship":
false,
            "x": -1,
            "y": -1,

"orientation":
"Horizontal"
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          "V16": {
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            "weight": 20,
            "eta": 3
          },
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            "w": 3,
            "l": 4,
            "shipping_fee": 62,
            "weight": 48,
            "eta": 2
          },
          "V17": {
            "id": "V17",
            "w": 4,
            "l": 5,
            "shipping_fee": 105,
            "weight": 75,
            "eta": 4
          },
          "V18": {
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            "w": 2,
            "l": 3,
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            "weight": 28,
            "eta": 2
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          },
          "V17": {
            "vehicle_id": "V17",
            "in_ship": false,
            "x": -1,
            "y": -1,
            "orientation": "Horizontal"
          },
          "V18": {
            "vehicle_id": "V18",
            "in_ship": true,
            "x": 6,
            "y": 3,
            "orientation": "Vertical"
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        },
</pre>
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</table>

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<pre>
        }
      },
</pre>

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<tr><td>Final</td><td>

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        "w": 10,
        "l": 20,

"max_capacity": 500
      },
      "vehicles": {
        "V01": {
          "id":
"V01",
          "w": 3,
          "l": 4,

"shipping_fee": 60,
          "weight":
50,
          "eta": 3
        },
        "V02": {
          "id":
"V02",
          "w": 3,
          "l": 5,

"shipping_fee": 85,
          "weight":
65,
          "eta": 2
        },
        "V03": {
          "id":
"V03",
</pre>

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<pre>
"placements": {
        "V01": {

"vehicle_id": "V01",
          "in_ship":
false,
          "x": -1,
          "y": -1,

"orientation":
"Horizontal"
        },
        "V02": {

"vehicle_id": "V02",
          "in_ship":
true,
          "x": 3,
          "y": 9,

"orientation":
"Vertical"
        },
        "V03": {

"vehicle_id": "V03",
          "in_ship":
false,
          "x": -1,
          "y": -1,

"orientation":
"Horizontal"
        },
        "V04": {
</pre>

</td></tr>
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<pre>
            "w": 2,
            "l": 4,

"shipping_fee": 45,
            "weight":
35,
            "eta": 4
          },
          "V04": {
            "id":
"V04",
            "w": 4,
            "l": 6,

"shipping_fee": 130,
            "weight":
95,
            "eta": 1
          },
          "V05": {
            "id":
"V05",
            "w": 2,
            "l": 3,

"shipping_fee": 35,
            "weight":
25,
            "eta": 5
          },
          "V06": {
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"V06",
            "w": 3,
            "l": 4,

"shipping_fee": 65,
            "weight":
45,
</pre>
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<pre>

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true,
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            "y": 3,

"orientation":
"Horizontal"
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            "in_ship":
true,
            "x": 8,
            "y": 14,

"orientation":
"Horizontal"
          },
          "V06": {

"vehicle_id": "V06",
            "in_ship":
true,
            "x": 1,
            "y": 14,

"orientation":
"Vertical"
          },
          "V07": {

"vehicle_id": "V07",
            "in_ship":
false,
            "x": -1,
            "y": -1,
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```
            "eta": 2
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          "V07": {
            "id":
"V07",
            "w": 4,
            "l": 5,

"shipping_fee": 110,
            "weight":
80,
            "eta": 3
          },
          "V08": {
            "id":
"V08",
            "w": 2,
            "l": 5,

"shipping_fee": 55,
            "weight":
40,
            "eta": 2
          },
          "V09": {
            "id":
"V09",
            "w": 3,
            "l": 6,

"shipping_fee": 95,
            "weight":
70,
            "eta": 1
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          "V10": {
            "id":
"V10",
            "w": 2,
```

```
"orientation":
"Horizontal"
          },
          "V08": {

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            "in_ship":
true,
            "x": 1,
            "y": 18,

"orientation":
"Vertical"
          },
          "V09": {

"vehicle_id": "V09",
            "in_ship":
false,
            "x": -1,
            "y": -1,

"orientation":
"Horizontal"
          },
          "V10": {

"vehicle_id": "V10",
            "in_ship":
true,
            "x": 7,
            "y": 12,

"orientation":
"Vertical"
          },
          "V11": {
```

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"l": 3,
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"eta": 4
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"V11": {
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"w": 3,
"l": 3,
"shipping_fee": 50,
"weight": 40,
"eta": 3
},
"V12": {
"id": "V12",
"w": 4,
"l": 4,
"shipping_fee": 90,
"weight": 60,
"eta": 2
},
"V13": {
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"w": 2,
"l": 4,
"shipping_fee": 48,
"weight": 35,
"eta": 5
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<td>
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"in_ship": true,
"x": 5,
"y": 14,
"orientation": "Horizontal"
},
"V12": {
"vehicle_id": "V12",
"in_ship": false,
"x": -1,
"y": -1,
"orientation": "Horizontal"
},
"V13": {
"vehicle_id": "V13",
"in_ship": false,
"x": -1,
"y": -1,
"orientation": "Horizontal"
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"vehicle_id": "V14",
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<td></td>
<td>
<pre>
          },
          "V14": {
            "id":
"V14",
            "w": 3,
            "l": 5,

"shipping_fee": 82,
            "weight":
60,
            "eta": 1
          },
          "V15": {
            "id":
"V15",
            "w": 2,
            "l": 2,

"shipping_fee": 28,
            "weight":
20,
            "eta": 3
          },
          "V16": {
            "id":
"V16",
            "w": 3,
            "l": 4,

"shipping_fee": 62,
            "weight":
48,
            "eta": 2
          },
          "V17": {
            "id":
"V17",
            "w": 4,
            "l": 5,
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</td>
<td>
<pre>
"orientation":
"Vertical"
          },
          "V15": {

"vehicle_id": "V15",
            "in_ship":
true,
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            "y": 10,

"orientation":
"Horizontal"
          },
          "V16": {

"vehicle_id": "V16",
            "in_ship":
true,
            "x": 5,
            "y": 5,

"orientation":
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false,
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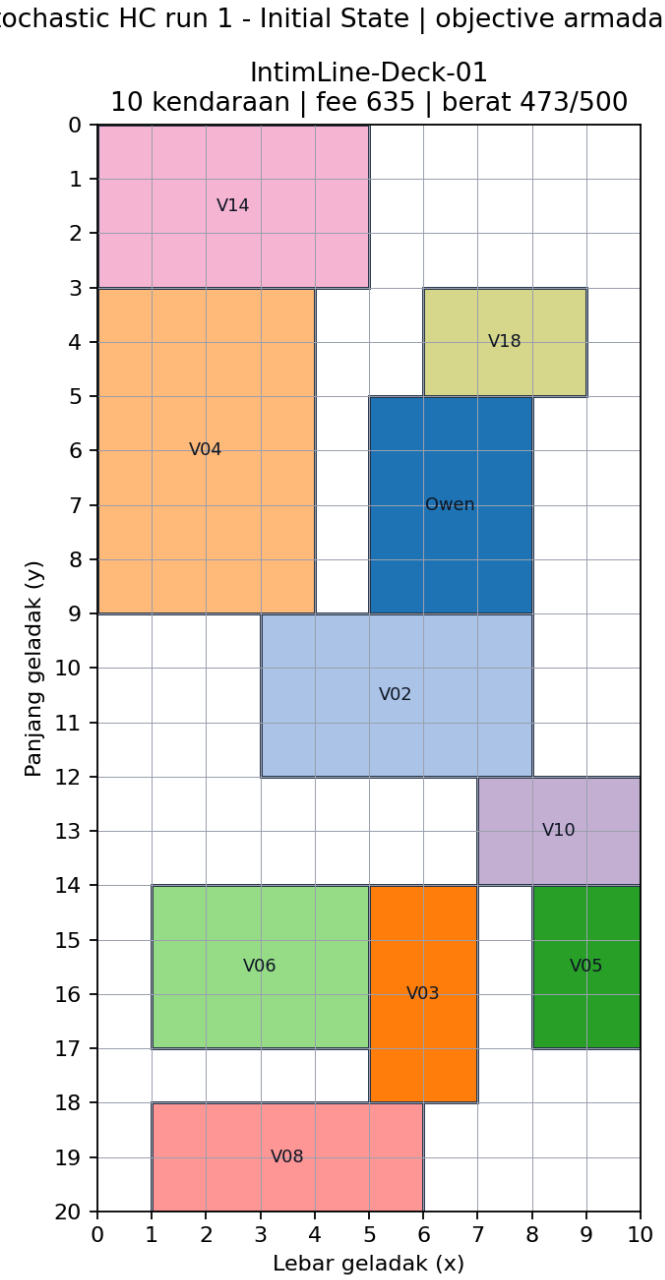
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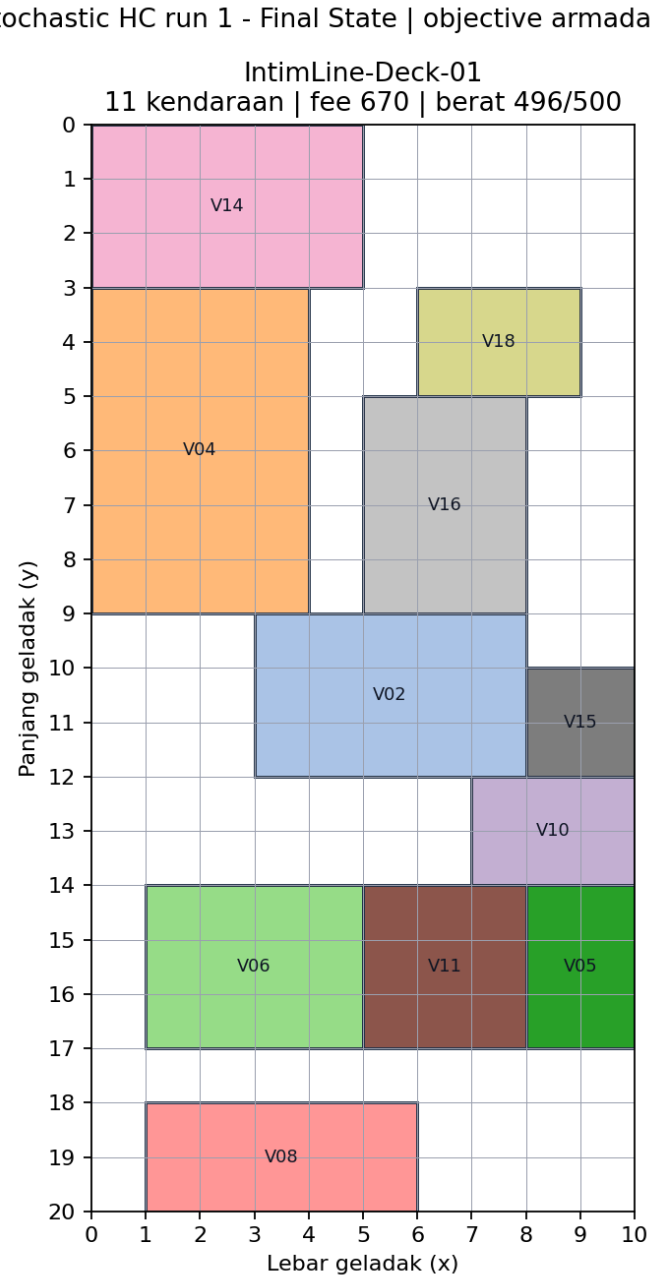
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75,
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          },
          "V18": {
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"V18",
            "w": 2,
            "l": 3,

"shipping_fee": 38,
            "weight":
28,
            "eta": 2
          }
        },
</pre>
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<td>
<pre>
            "in_ship":
true,
            "x": 6,
            "y": 3,

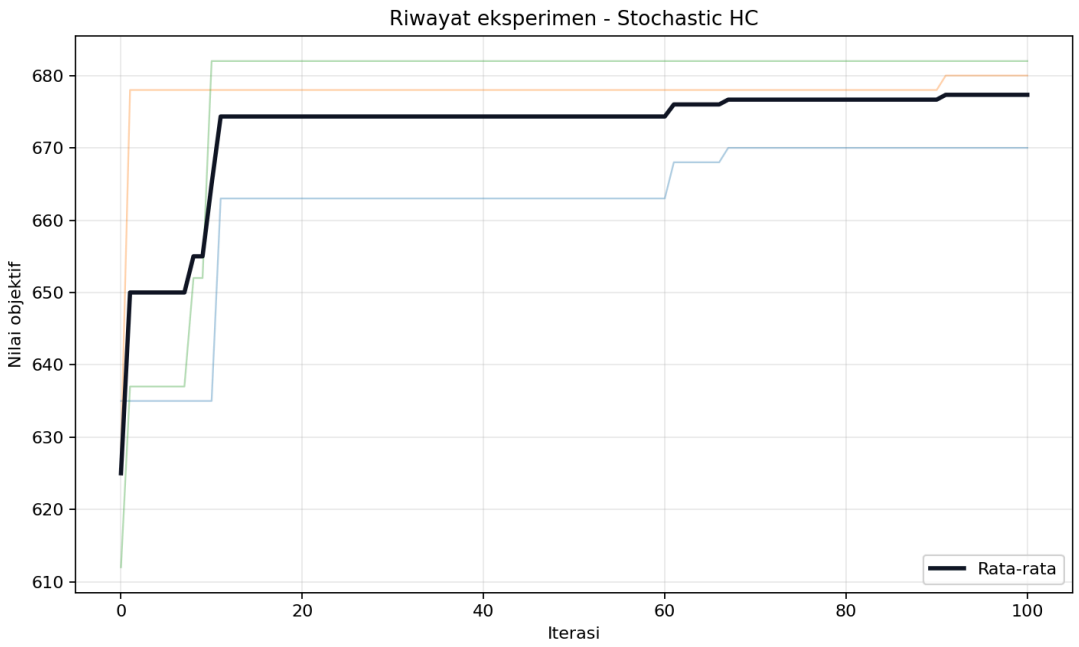
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"Vertical"
          }
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<td colspan="2">Objective Function Score: 670</td>
</tr>
<tr>
<td colspan="3">Visualisasi Initial State:</td>
</tr>
</table>



Visualisasi Final State:


Stochastic HC run 1 - Final State | objective armada 670.00
IntimLine-Deck-01
11 kendaraan | fee 670 | berat 496/500
V14
V18
V04
V16
V02
V15
V10
V06
V11
V05
V08
Panjang geladak (y)
Lebar geladak (x)
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20

Plot:



Highest Objective Function Score: 670

Iterations: 100

Duration: 0.0280

Percobaan 2		
Initial	"initial_state": { "ship": { "id": "IntimLine-Deck-01", "w": 10, "l": 20, "max_capacity": 500 }, "vehicles": { "V01": { "id": "V01", "w": 3,	"placements": { "V01": { "vehicle_id": "V01", "in_ship": true, "x": 0, "y": 3, "orientation": "Horizontal" }, "V02": {

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            "eta": 3
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            "w": 3,
            "l": 5,
            "shipping_fee": 85,
            "weight": 65,
            "eta": 2
          },
          "V03": {
            "id": "V03",
            "w": 2,
            "l": 4,
            "shipping_fee": 45,
            "weight": 35,
            "eta": 4
          },
          "V04": {
            "id": "V04",
            "w": 4,
            "l": 6,
            "shipping_fee": 130,
            "weight": 95,
            "eta": 1
</pre>
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<td>
<pre>
"vehicle_id": "V02",
            "in_ship": false,
            "x": -1,
            "y": -1,
            "orientation": "Horizontal"
          },
          "V03": {
            "vehicle_id": "V03",
            "in_ship": true,
            "x": 7,
            "y": 16,
            "orientation": "Horizontal"
          },
          "V04": {
            "vehicle_id": "V04",
            "in_ship": false,
            "x": -1,
            "y": -1,
            "orientation": "Horizontal"
          },
          "V05": {
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"V05",
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25,
            "eta": 5
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"V06",
            "w": 3,
            "l": 4,

"shipping_fee": 65,
            "weight":
45,
            "eta": 2
          },
          "V07": {
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            "w": 4,
            "l": 5,

"shipping_fee": 110,
            "weight":
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            "eta": 3
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"V08",
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"Vertical"
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false,
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"Horizontal"
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true,
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"orientation":
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"eta": 4
},
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"orientation":
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"V12",
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            "l": 4,

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            "weight":
60,
            "eta": 2
          },
          "V13": {
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"V13",
            "w": 2,
            "l": 4,

"shipping_fee": 48,
            "weight":
35,
            "eta": 5
          },
          "V14": {
            "id":
"V14",
            "w": 3,
            "l": 5,

"shipping_fee": 82,
            "weight":
60,
            "eta": 1
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"V15",
            "w": 2,
            "l": 2,
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          "V13": {

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true,
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            "y": 8,

"orientation":
"Horizontal"
          },
          "V14": {

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            "in_ship":
true,
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            "y": 15,

"orientation":
"Horizontal"
          },
          "V15": {

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false,
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            "y": -1,

"orientation":
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            "in_ship":
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"w": 3,
"l": 4,

"shipping_fee": 62,
"weight": 48,
"eta": 2
},
"V17": {
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"w": 4,
"l": 5,

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"weight": 75,
"eta": 4
},
"V18": {
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"w": 2,
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"shipping_fee": 38,
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},
"V17": {

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"x": 5,
"y": 1,

"orientation": "Horizontal"
},
"V18": {

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"in_ship": true,
"x": 2,
"y": 9,

"orientation": "Horizontal"
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},
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	Objective Function Score: 628	
Final	"final_state": { "ship": { "id": "IntimLine-Deck-01", "w": 10, "l": 20, "max_capacity": 500 }, "vehicles": { "V01": { "id": "V01", "w": 3, "l": 4, "shipping_fee": 60, "weight": 50, "eta": 3 }, "V02": { "id": "V02", "w": 3, "l": 5, "shipping_fee": 85, "weight": 65, "eta": 2 }, "V03": { "id": "V03", "w": 2, "l": 4,	"placements": { "V01": { "vehicle_id": "V01", "in_ship": false, "x": -1, "y": -1, "orientation": "Horizontal" }, "V02": { "vehicle_id": "V02", "in_ship": false, "x": -1, "y": -1, "orientation": "Horizontal" }, "V03": { "vehicle_id": "V03", "in_ship": true, "x": 7, "y": 16, "orientation": "Horizontal" }, "V04": { "vehicle_id": "V04",

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"weight": 95,
"eta": 1
},
"V05": {
"id": "V05",
"w": 2,
"l": 3,
"shipping_fee": 35,
"weight": 25,
"eta": 5
},
"V06": {
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"eta": 2
},
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"x": -1,
"y": -1,
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},
"V06": {
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"in_ship": true,
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"w": 2,
"l": 5,
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"weight": 40,
"eta": 2
},
"V09": {
"id": "V09",
"w": 3,
"l": 6,
"shipping_fee": 95,
"weight": 70,
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},
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"l": 3,
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<pre>
"Horizontal"
},
"V08": {
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"in_ship": true,
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"y": 6,
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"V09": {
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	"weight": 40,
	"eta": 3
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	"w": 4,
	"l": 4,
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	"weight": 60,
	"eta": 2
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	"V13": {
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	"w": 2,
	"l": 4,
	"shipping_fee": 48,
	"weight": 35,
	"eta": 5
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"V15",
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20,
            "eta": 3
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"shipping_fee": 62,
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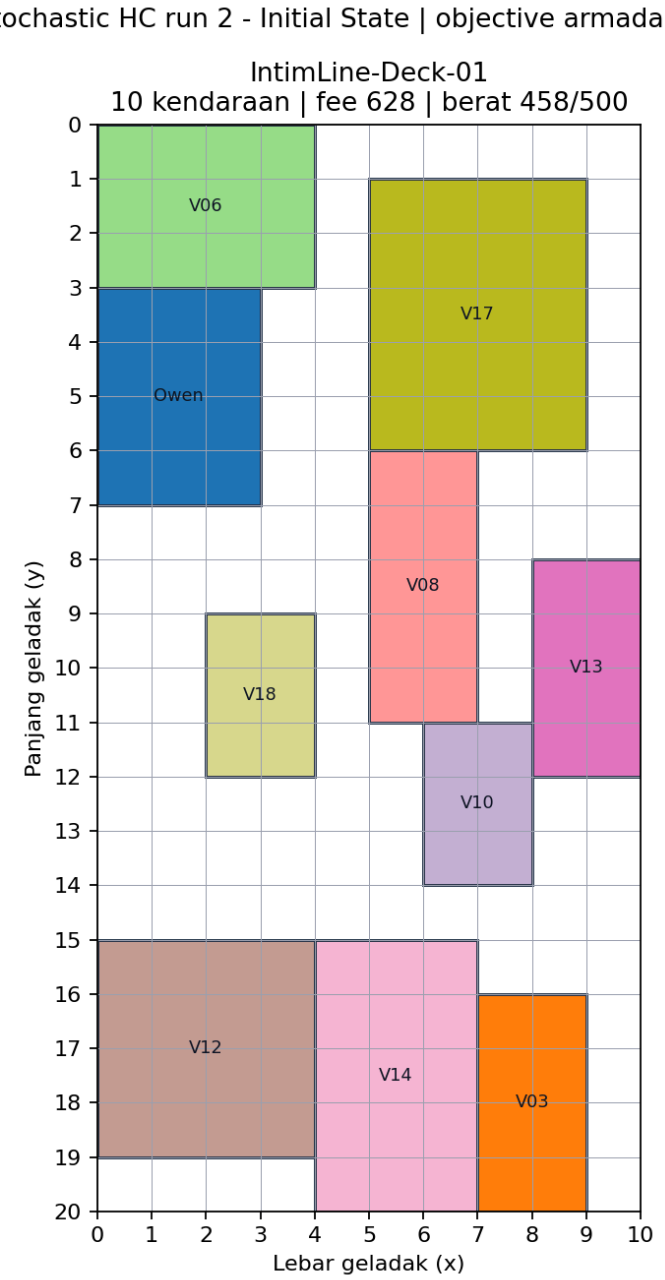
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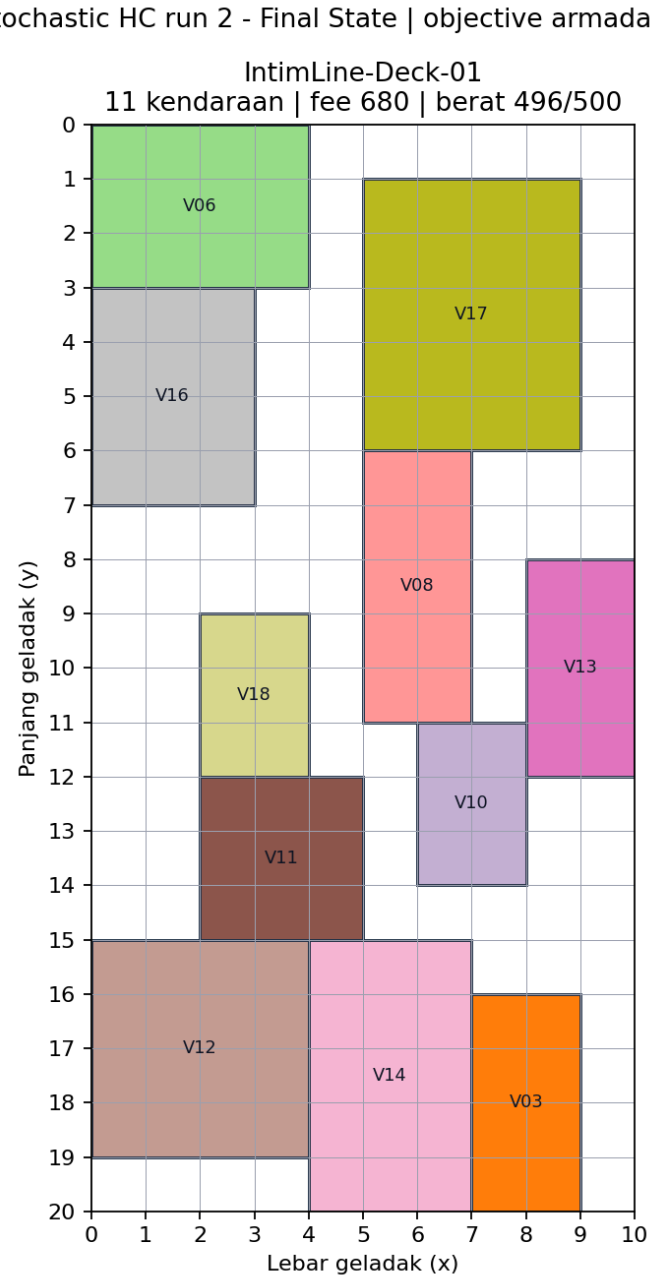
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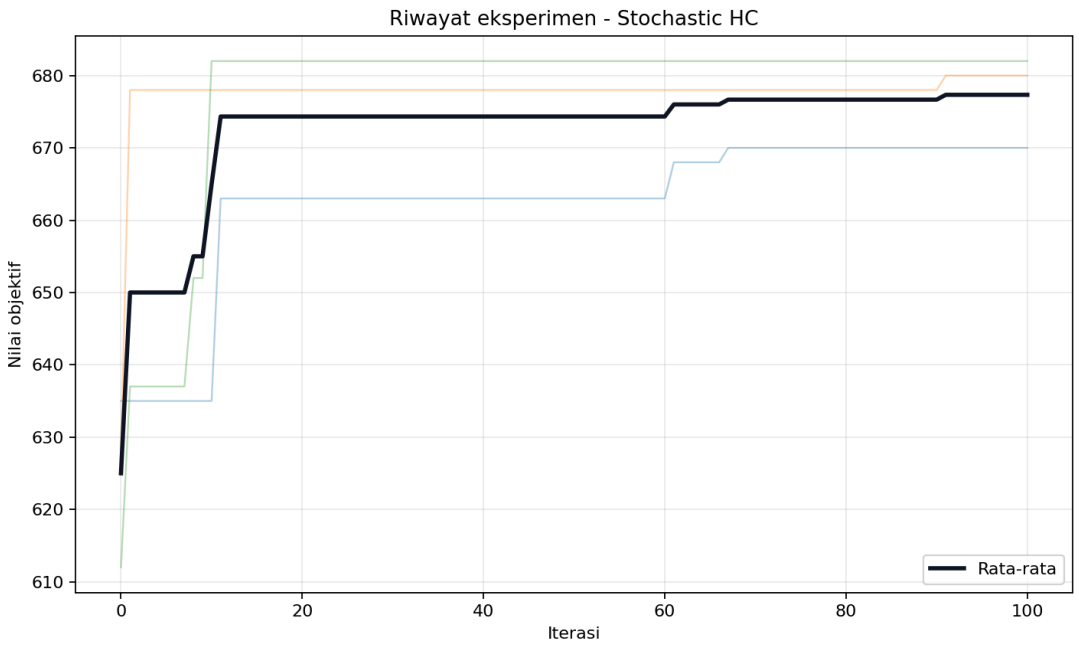
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<td colspan="2">Objective Function Score: 680</td>
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<tr>
<td colspan="3">Visualisasi Initial State:</td>
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Visualisasi Final State:


Stochastic HC run 2 - Final State | objective armada 680.00
IntimLine-Deck-01
11 kendaraan | fee 680 | berat 496/500
V06
V17
V16
V08
V18
V13
V10
V11
V12
V14
V03
Panjang geladak (y)
Lebar geladak (x)
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20

Plot:



Highest Objective Function Score: 680

Iterations: 100

Duration: 0.02803

Percobaan 3		
Initial	"initial_state": { "ship": { "id": "IntimLine-Deck-01", "w": 10, "l": 20, "max_capacity": 500 }, "vehicles": { "V01": { "id": "V01", "w": 3,	"placements": { "V01": { "vehicle_id": "V01", "in_ship": true, "x": 3, "y": 13, "orientation": "Horizontal" }, "V02": {

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            "eta": 3
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            "w": 3,
            "l": 5,
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            "eta": 2
          },
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            "w": 2,
            "l": 4,
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            "weight":
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            "eta": 4
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"V04",
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95,
            "eta": 1
```

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"orientation":
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            "in_ship":
false,
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            "y": -1,
"orientation":
"Horizontal"
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          "V05": {
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          "eta": 5
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"shipping_fee": 65,
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          "eta": 2
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"V09": {
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"shipping_fee": 95,
"weight": 70,
"eta": 1
},
"V10": {
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"w": 2,
"l": 3,
"shipping_fee": 40,
"weight": 30,
"eta": 4
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"V11": {
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"weight": 40,
"eta": 3
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"in_ship": true,
"x": 5,
"y": 17,
"orientation": "Vertical"
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"vehicle_id": "V11",
"in_ship": true,
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"y": 1,
"orientation": "Horizontal"
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"V12": {
"vehicle_id": "V12",
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"orientation":
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<pre>
"V12": {
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"eta": 2
},
"V13": {
"id": "V13",
"w": 2,
"l": 4,
"shipping_fee": 48,
"weight": 35,
"eta": 5
},
"V14": {
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"y": 5,
"orientation": "Horizontal"
},
"V14": {
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"x": 7,
"y": 12,
"orientation": "Horizontal"
},
"V15": {
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"in_ship": false,
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"V16": {
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            "eta": 3
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"V16",
            "w": 3,
            "l": 4,

"shipping_fee": 62,
            "weight":
48,
            "eta": 2
          },
          "V17": {
            "id":
"V17",
            "w": 4,
            "l": 5,

"shipping_fee": 105,
            "weight":
75,
            "eta": 4
          },
          "V18": {
            "id":
"V18",
            "w": 2,
            "l": 3,

"shipping_fee": 38,
            "weight":
28,
            "eta": 2
          }
        },
```

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true,
            "x": 1,
            "y": 10,

"orientation":
"Vertical"
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            "in_ship":
true,
            "x": 1,
            "y": 4,

"orientation":
"Vertical"
          },
          "V18": {

"vehicle_id": "V18",
            "in_ship":
false,
            "x": -1,
            "y": -1,

"orientation":
"Horizontal"
          }
        },
```

	Objective Function Score: 612	
Final	"final_state": { "ship": { "id": "IntimLine-Deck-01", "w": 10, "l": 20, "max_capacity": 500 }, "vehicles": { "V01": { "id": "V01", "w": 3, "l": 4, "shipping_fee": 60, "weight": 50, "eta": 3 }, "V02": { "id": "V02", "w": 3, "l": 5, "shipping_fee": 85, "weight": 65, "eta": 2 }, "V03": { "id": "V03", "w": 2, "l": 4,	"placements": { "V01": { "vehicle_id": "V01", "in_ship": false, "x": -1, "y": -1, "orientation": "Horizontal" }, "V02": { "vehicle_id": "V02", "in_ship": true, "x": 0, "y": 17, "orientation": "Vertical" }, "V03": { "vehicle_id": "V03", "in_ship": true, "x": 8, "y": 1, "orientation": "Horizontal" }, "V04": { "vehicle_id": "V04",

```
"shipping_fee": 45,
            "weight": 35,
            "eta": 4
          },
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            "w": 4,
            "l": 6,
"shipping_fee": 130,
            "weight": 95,
            "eta": 1
          },
          "V05": {
            "id": "V05",
            "w": 2,
            "l": 3,
"shipping_fee": 35,
            "weight": 25,
            "eta": 5
          },
          "V06": {
            "id": "V06",
            "w": 3,
            "l": 4,
"shipping_fee": 65,
            "weight": 45,
            "eta": 2
          },
```

```
            "in_ship": true,
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            "y": 4,
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            "in_ship": true,
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          "V06": {
"vehicle_id": "V06",
            "in_ship": true,
            "x": 3,
            "y": 1,
"orientation": "Vertical"
          },
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"orientation":
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"shipping_fee": 110,
            "weight":
80,
            "eta": 3
          },
          "V08": {
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"V08",
            "w": 2,
            "l": 5,

"shipping_fee": 55,
            "weight":
40,
            "eta": 2
          },
          "V09": {
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"V09",
            "w": 3,
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"shipping_fee": 95,
            "weight":
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            "eta": 1
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"V10",
            "w": 2,
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          "V08": {

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"orientation":
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"Horizontal"
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            "in_ship":
true,
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            "y": 17,

"orientation":
"Vertical"
          },
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"vehicle_id": "V11",
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            "w": 4,
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"shipping_fee": 90,
            "weight": 60,
            "eta": 2
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          "V13": {
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            "w": 2,
            "l": 4,

"shipping_fee": 48,
            "weight": 35,
            "eta": 5
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"orientation": "Horizontal"
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"orientation": "Horizontal"
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"orientation": "Horizontal"
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            "l": 5,

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          "V15": {
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"V15",
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            "l": 2,

"shipping_fee": 28,
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            "eta": 3
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false,
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"orientation":
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true,
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            "y": 10,

"orientation":
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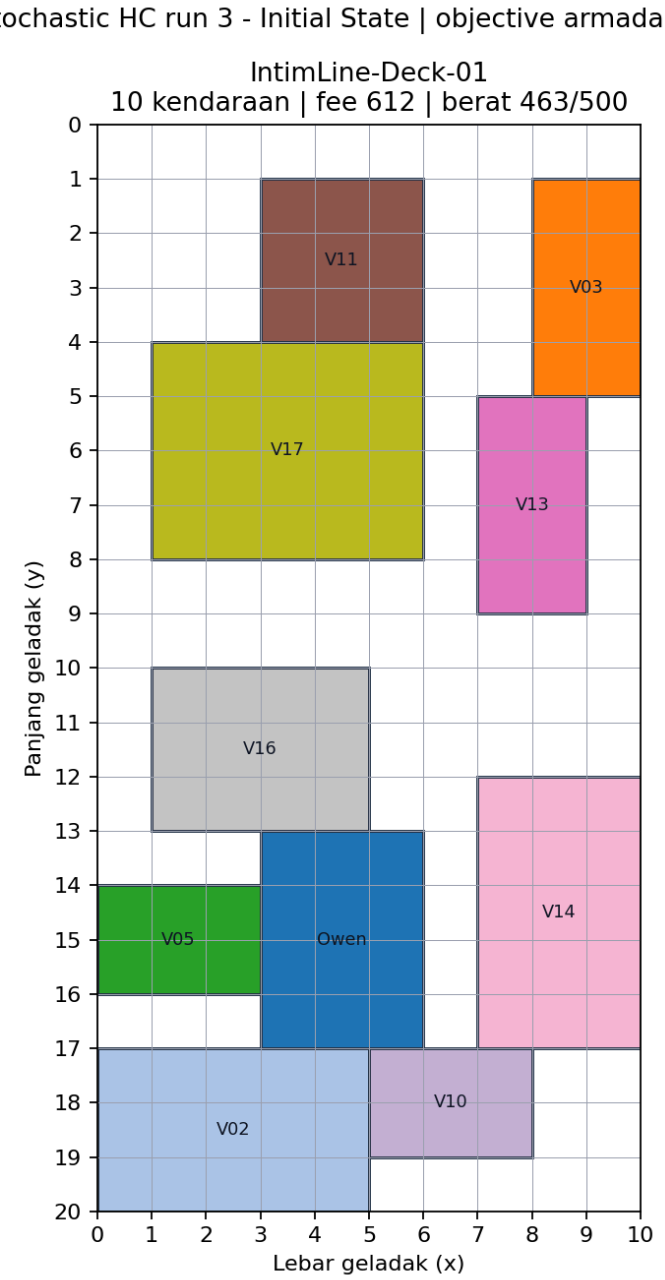
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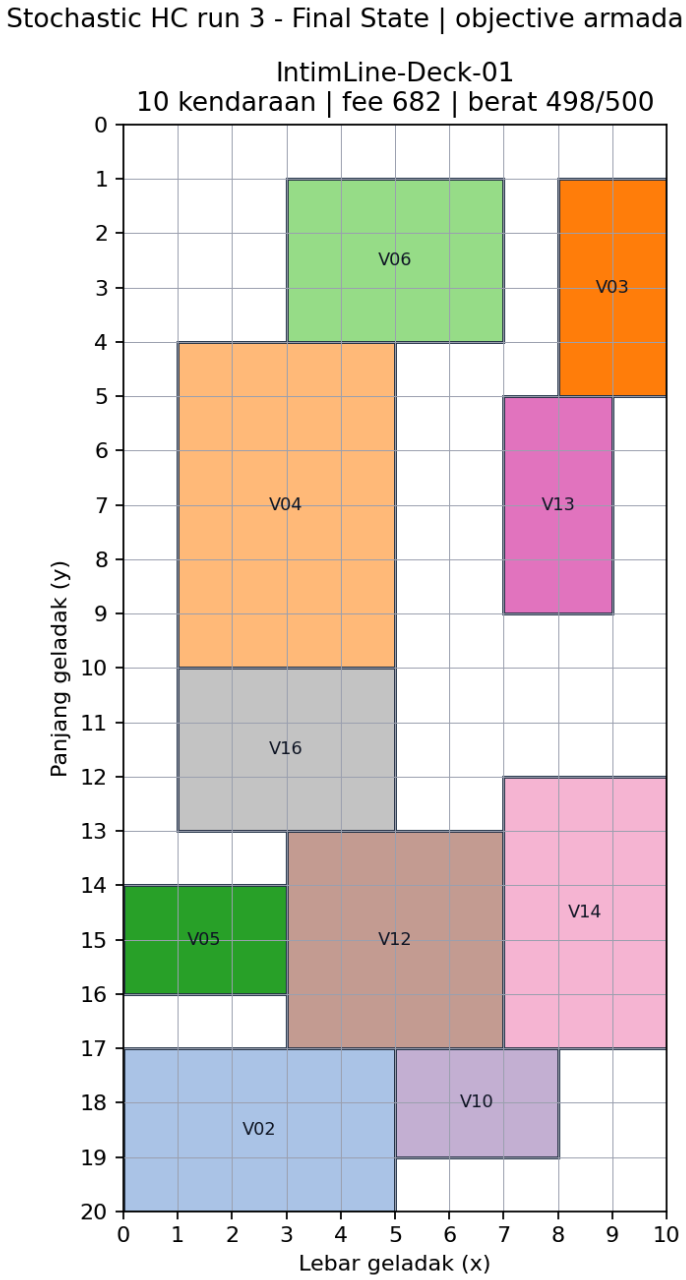
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            "l": 3,

"shipping_fee": 38,
            "weight":
28,
            "eta": 2
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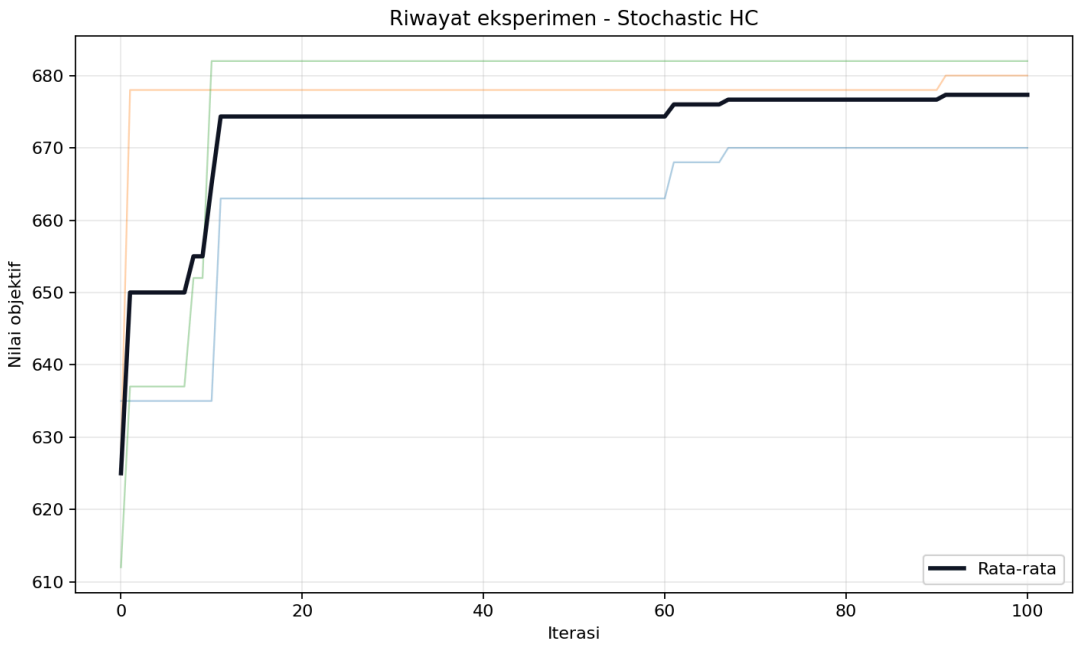
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<td colspan="2">Objective Function Score: 682</td>
</tr>
<tr>
<td colspan="3">Visualisasi Initial State:</td>
</tr>
</table>



Visualisasi Final State:


Stochastic HC run 3 - Final State | objective armada 682.00
IntimLine-Deck-01
10 kendaraan | fee 682 | berat 498/500
V06
V03
V04
V13
V16
V14
V05
V12
V10
V02
Panjang geladak (y)
Lebar geladak (x)

Plot:



Highest Objective Function Score: 682
Iterations: 100
Duration: 0.02604

2.5.4 *Random Restart*

Percobaan 1		
Initial	"initial_state": { "ship": { "id": "IntimLine-Deck-01", "w": 10, "l": 20, "max_capacity": 500 }, "vehicles": { "V01": { "id": "V01",	"placements": { "V01": { "vehicle_id": "V01", "in_ship": false, "x": -1, "y": -1, "orientation": "Horizontal" }, "V02": {

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            "l": 4,

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            "weight":
50,
            "eta": 3
          },
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            "id":
"V02",
            "w": 3,
            "l": 5,

"shipping_fee": 85,
            "weight":
65,
            "eta": 2
          },
          "V03": {
            "id":
"V03",
            "w": 2,
            "l": 4,

"shipping_fee": 45,
            "weight":
35,
            "eta": 4
          },
          "V04": {
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"V04",
            "w": 4,
            "l": 6,

"shipping_fee": 130,
            "weight":
95,
</pre>
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<td>
<pre>

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true,
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          "V03": {

"vehicle_id": "V03",
            "in_ship":
true,
            "x": 6,
            "y": 18,

"orientation":
"Vertical"
          },
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"vehicle_id": "V04",
            "in_ship":
true,
            "x": 2,
            "y": 0,

"orientation":
"Horizontal"
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"vehicle_id": "V05",
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true,
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          "V05": {
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"V05",
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            "l": 3,

"shipping_fee": 35,
            "weight":
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            "eta": 5
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"V06",
            "w": 3,
            "l": 4,

"shipping_fee": 65,
            "weight":
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            "eta": 2
          },
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"V07",
            "w": 4,
            "l": 5,

"shipping_fee": 110,
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"V08",
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```

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```

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"Vertical"
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"vehicle_id": "V06",
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"orientation":
"Horizontal"
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"vehicle_id": "V07",
            "in_ship":
false,
            "x": -1,
            "y": -1,

"orientation":
"Horizontal"
          },
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            "in_ship":
true,
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"orientation":
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40,
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"V09",
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"shipping_fee": 95,
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70,
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            "w": 2,
            "l": 3,
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false,
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            "y": -1,
"orientation":
"Horizontal"
          },
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true,
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            "y": 17,
"orientation":
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          },
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"V13",
            "w": 2,
            "l": 4,

"shipping_fee": 48,
            "weight":
35,
            "eta": 5
          },
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"V14",
            "w": 3,
            "l": 5,

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true,
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"Vertical"
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"shipping_fee": 62,
            "weight": 48,
            "eta": 2
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            "id": "V17",
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            "l": 5,

"shipping_fee": 105,
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            "eta": 4
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          "V18": {
            "id": "V18",
            "w": 2,
            "l": 3,

"shipping_fee": 38,
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            "eta": 2
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```

```
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```

	},	
	Objective Function Score: 669	
Final	"final_state": { "ship": { "id": "IntimLine-Deck-01", "w": 10, "l": 20, "max_capacity": 500 }, "vehicles": { "V01": { "id": "V01", "w": 3, "l": 4, "shipping_fee": 60, "weight": 50, "eta": 3 }, "V02": { "id": "V02", "w": 3, "l": 5, "shipping_fee": 85, "weight": 65, "eta": 2 }, "V03": { "id": "V03", "w": 2,	"placements": { "V01": { "vehicle_id": "V01", "in_ship": false, "x": -1, "y": -1, "orientation": "Horizontal" }, "V02": { "vehicle_id": "V02", "in_ship": false, "x": -1, "y": -1, "orientation": "Vertical" }, "V03": { "vehicle_id": "V03", "in_ship": true, "x": 8, "y": 1, "orientation": "Horizontal" }, "V04": {

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            "l": 6,
"shipping_fee": 130,
            "weight":
95,
            "eta": 1
          },
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"V05",
            "w": 2,
            "l": 3,
"shipping_fee": 35,
            "weight":
25,
            "eta": 5
          },
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"vehicle_id": "V05",
            "in_ship":
false,
            "x": -1,
            "y": -1,
"orientation":
"Horizontal"
          },
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false,
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            "w": 4,
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"shipping_fee": 110,
            "weight":
80,
            "eta": 3
          },
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"V08",
            "w": 2,
            "l": 5,

"shipping_fee": 55,
            "weight":
40,
            "eta": 2
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"V09",
            "w": 3,
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"shipping_fee": 95,
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true,
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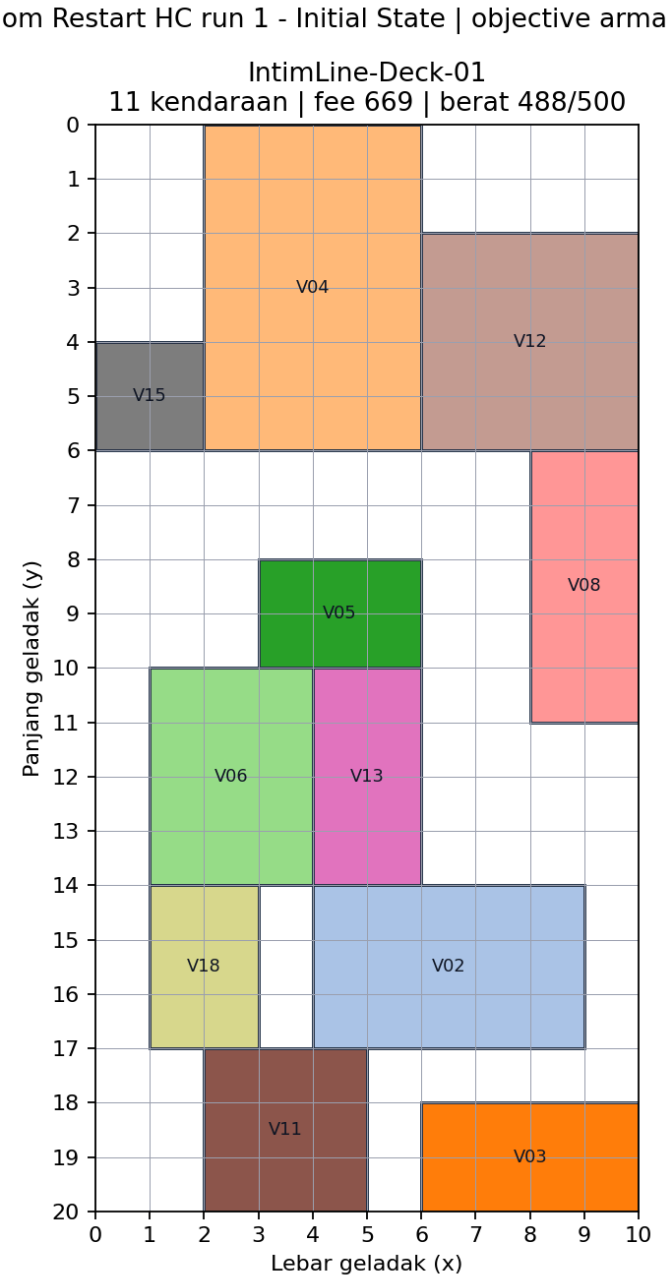
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"eta": 3
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"V16": {
"id": "V16",
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"orientation": "Horizontal"
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"V16": {
"vehicle_id": "V16",
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"x": -1,
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"orientation": "Horizontal"
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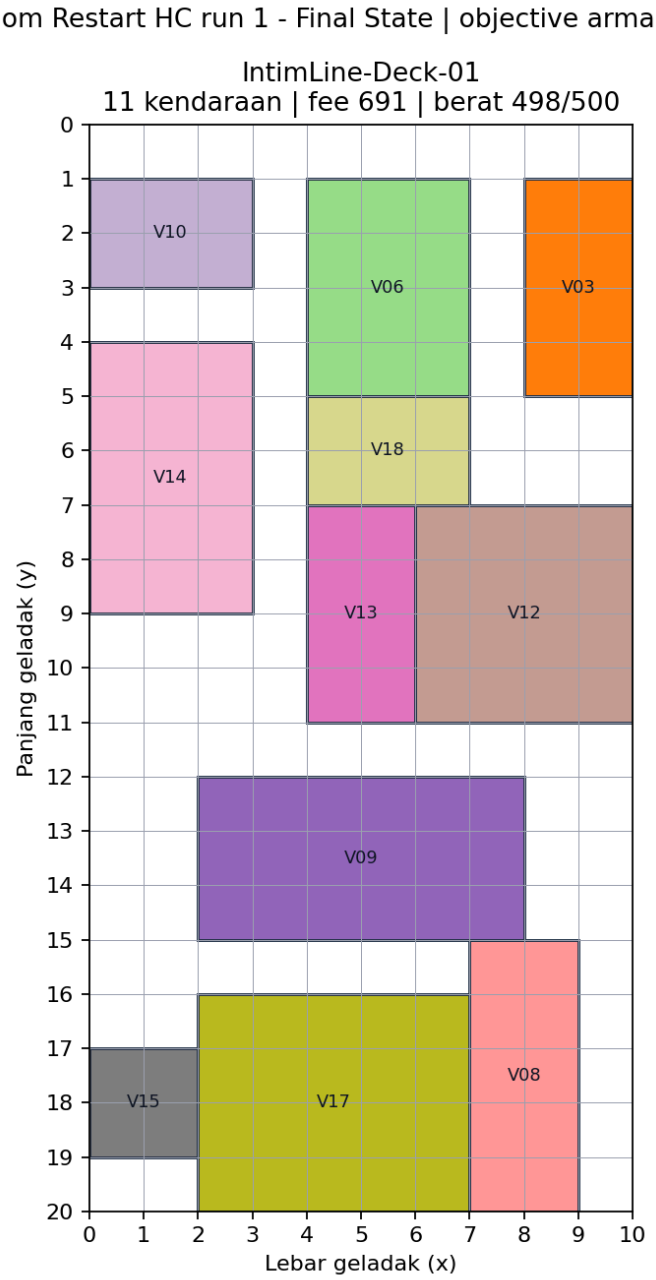
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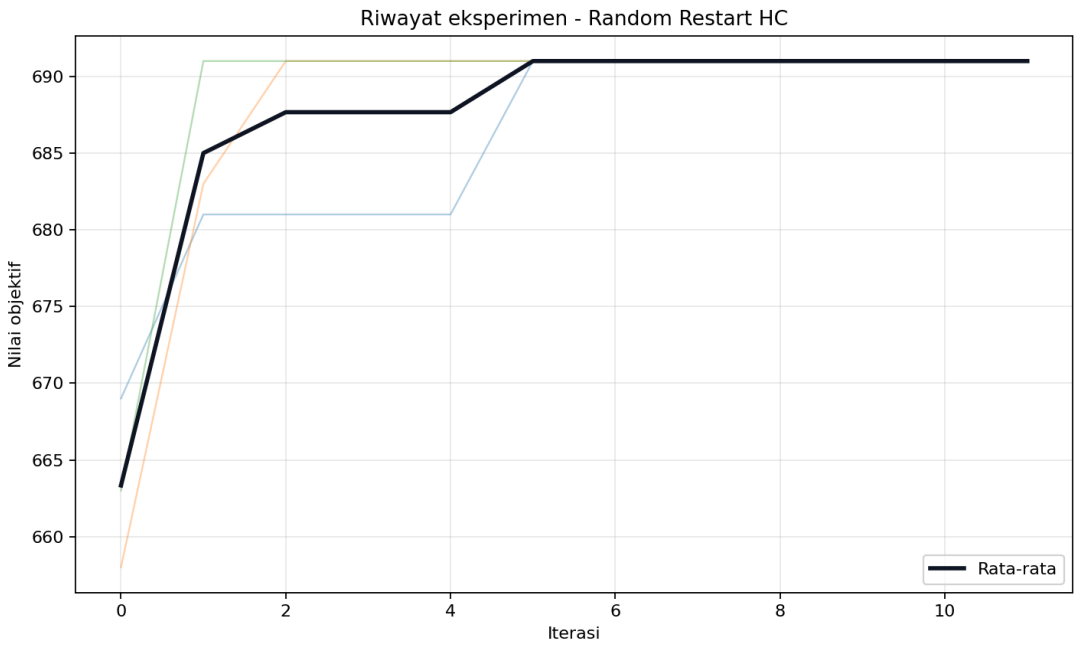
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<tr><td colspan="3">Visualisasi Initial State:</td></tr>
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Visualisasi Final State:


Random Restart HC run 1 - Final State | objective armada 691.00
IntimLine-Deck-01
11 kendaraan | fee 691 | berat 498/500
V10
V06
V03
V14
V18
V13
V12
V09
V15
V17
V08
Panjang geladak (y)
Lebar geladak (x)
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20

Plot:



Highest Objective Function Score: 691

Iterations: 9

Duration: 0.2337

Percobaan 2		
Initial	"initial_state": { "ship": { "id": "IntimLine-Deck-01", "w": 10, "l": 20, "max_capacity": 500 }, "vehicles": { "V01": { "id": "V01", "w": 3,	"placements": { "V01": { "vehicle_id": "V01", "in_ship": false, "x": -1, "y": -1, "orientation": "Horizontal" }, "V02": {

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            "weight": 65,
            "eta": 2
          },
          "V03": {
            "id": "V03",
            "w": 2,
            "l": 4,
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            "weight": 35,
            "eta": 4
          },
          "V04": {
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            "x": -1,
            "y": -1,
            "orientation": "Horizontal"
          },
          "V04": {
            "vehicle_id": "V04",
            "in_ship": true,
            "x": 0,
            "y": 5,
            "orientation": "Vertical"
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          "eta": 5
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          "w": 3,
          "l": 4,

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          "eta": 2
        },
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"V07",
          "w": 4,
          "l": 5,

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},
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"l": 4,
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"eta": 5
},
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            "eta": 2
          },
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"V17",
            "w": 4,
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            "eta": 2
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	Objective Function Score: 658	
Final	"final_state": { "ship": { "id": "IntimLine-Deck-01", "w": 10, "l": 20, "max_capacity": 500 }, "vehicles": { "V01": { "id": "V01", "w": 3, "l": 4, "shipping_fee": 60, "weight": 50, "eta": 3 }, "V02": { "id": "V02", "w": 3, "l": 5, "shipping_fee": 85, "weight": 65, "eta": 2 }, "V03": { "id": "V03", "w": 2, "l": 4,	"placements": { "V01": { "vehicle_id": "V01", "in_ship": false, "x": -1, "y": -1, "orientation": "Horizontal" }, "V02": { "vehicle_id": "V02", "in_ship": false, "x": -1, "y": -1, "orientation": "Vertical" }, "V03": { "vehicle_id": "V03", "in_ship": false, "x": -1, "y": -1, "orientation": "Horizontal" }, "V04": { "vehicle_id": "V04",

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            "eta": 4
          },
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"shipping_fee": 130,
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            "eta": 1
          },
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"shipping_fee": 35,
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            "eta": 5
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            "eta": 2
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"orientation": "Vertical"
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            "in_ship": true,
            "x": 2,
            "y": 0,

"orientation": "Vertical"
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          "V06": {

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            "x": 7,
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"orientation": "Horizontal"
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"V08",
            "w": 2,
            "l": 5,

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            "weight":
40,
            "eta": 2
          },
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"V09",
            "w": 3,
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            "eta": 1
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"V10",
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false,
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"V11",
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            "l": 3,

"shipping_fee": 50,
            "weight":
40,
            "eta": 3
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"V12",
            "w": 4,
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60,
            "eta": 2
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"V13",
            "w": 2,
            "l": 4,

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            "weight":
35,
            "eta": 5
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          "V14": {
```

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"orientation":
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60,
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"V16",
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            "weight":
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"V17",
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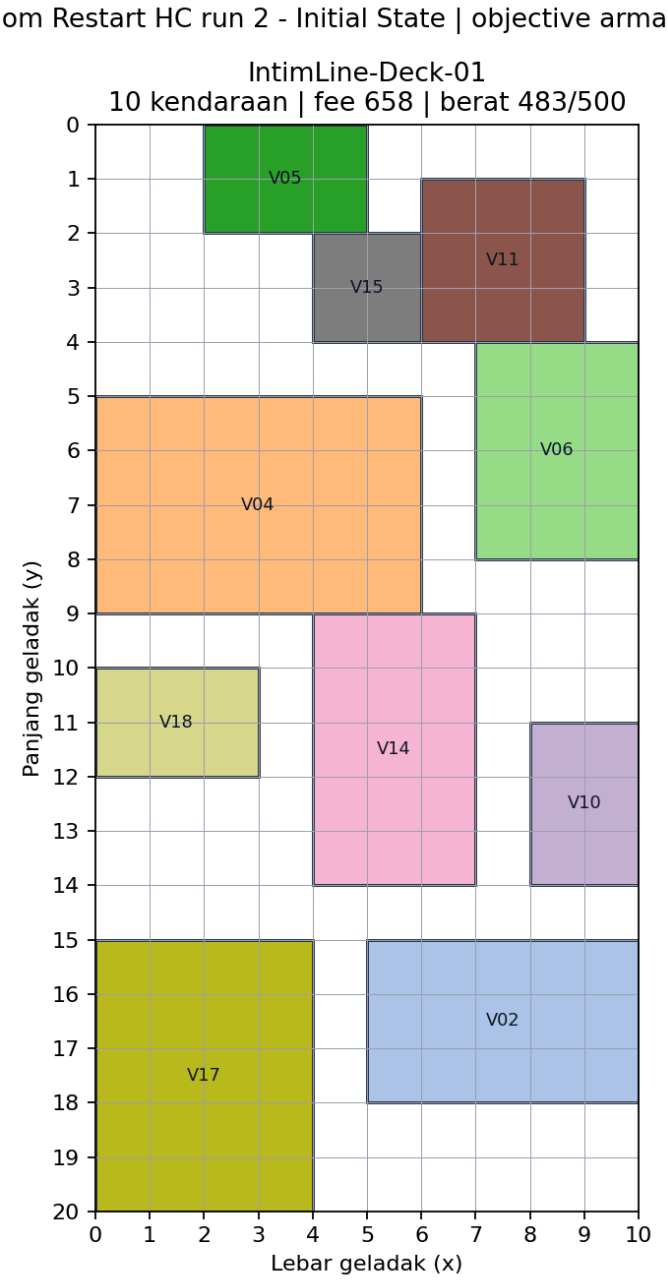
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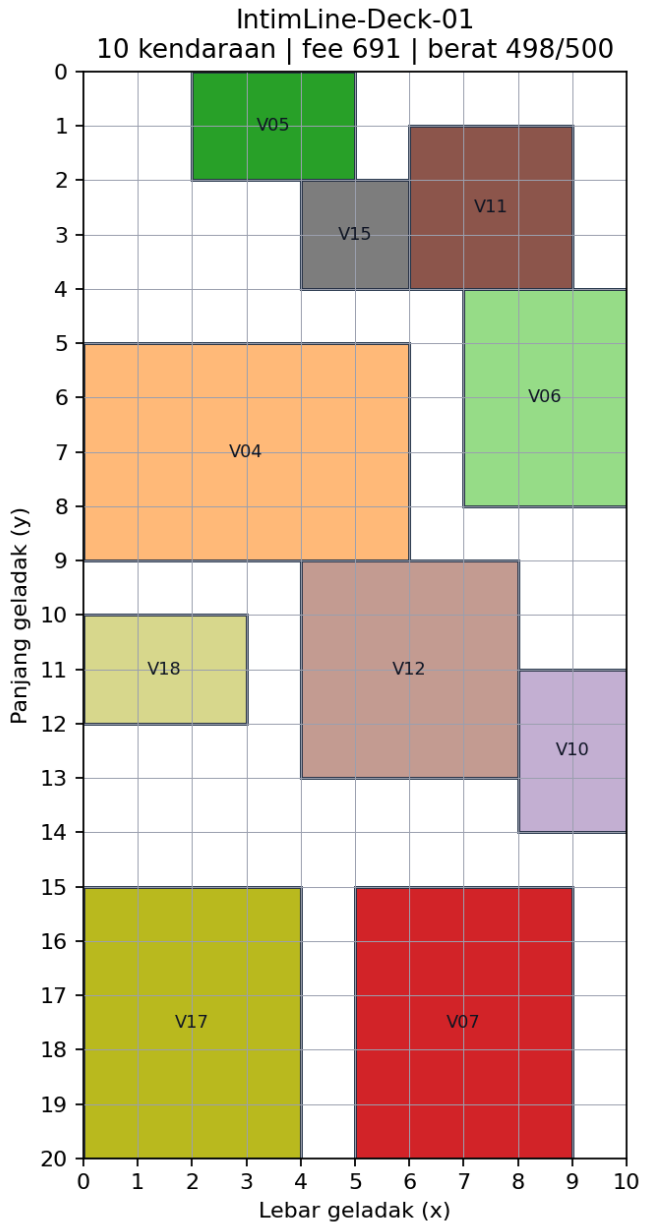
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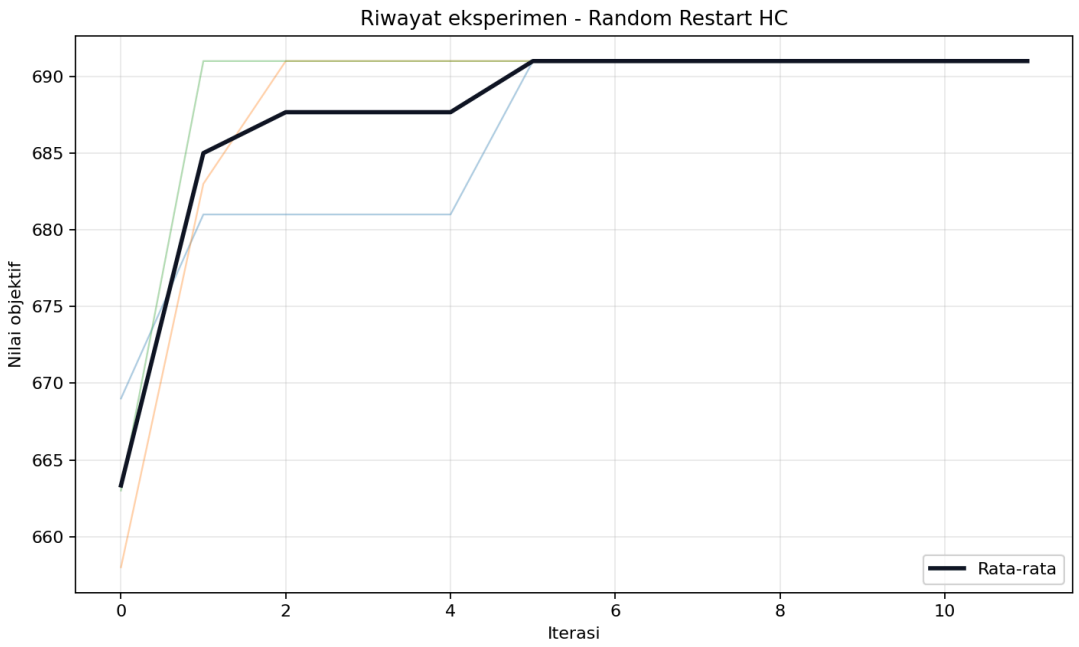
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<tr><td colspan="2">Objective Function Score: 691</td></tr>
<tr><td colspan="3">Visualisasi Initial State:</td></tr>
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Visualisasi Final State:


Random Restart HC run 2 - Final State | objective armada 691.00
IntimLine-Deck-01
10 kendaraan | fee 691 | berat 498/500
V05
V11
V15
V06
V04
V18
V12
V10
V17
V07
Panjang geladak (y)
Lebar geladak (x)
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20

Plot:



Highest Objective Function Score: 691

Iterations: 7

Duration: 0.20854

Percobaan 3		
Initial	"initial_state": { "ship": { "id": "IntimLine-Deck-01", "w": 10, "l": 20, "max_capacity": 500 }, "vehicles": { "V01": { "id": "V01", "w": 3,	"placements": { "V01": { "vehicle_id": "V01", "in_ship": false, "x": -1, "y": -1, "orientation": "Horizontal" }, "V02": {

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            "eta": 5
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            "eta": 2
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          "V09": {

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            "eta": 2
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            "weight": 70,
            "eta": 1
          },
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            "w": 2,
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            "weight": 30,
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            "eta": 5
          },
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"V14",
            "w": 3,
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60,
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```
"Vertical"
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          "V14": {

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            "y": 10,

"orientation":
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          },
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"V17",
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"shipping_fee": 105,
            "weight":
75,
            "eta": 4
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false,
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	Objective Function Score: 663	
Final	"final_state": { "ship": { "id": "IntimLine-Deck-01", "w": 10, "l": 20, "max_capacity": 500 }, "vehicles": { "V01": { "id": "V01", "w": 3, "l": 4, "shipping_fee": 60, "weight": 50, "eta": 3 }, "V02": { "id": "V02", "w": 3, "l": 5, "shipping_fee": 85, "weight": 65, "eta": 2 }, "V03": { "id": "V03", "w": 2, "l": 4,	"placements": { "V01": { "vehicle_id": "V01", "in_ship": false, "x": -1, "y": -1, "orientation": "Horizontal" }, "V02": { "vehicle_id": "V02", "in_ship": false, "x": -1, "y": -1, "orientation": "Horizontal" }, "V03": { "vehicle_id": "V03", "in_ship": false, "x": -1, "y": -1, "orientation": "Horizontal" }, "V04": { "vehicle_id": "V04",

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"weight": 95,
"eta": 1
},
"V05": {
"id": "V05",
"w": 2,
"l": 3,
"shipping_fee": 35,
"weight": 25,
"eta": 5
},
"V06": {
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"w": 3,
"l": 4,
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"orientation": "Vertical"
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"eta": 3
},
"V08": {
"id": "V08",
"w": 2,
"l": 5,
"shipping_fee": 55,
"weight": 40,
"eta": 2
},
"V09": {
"id": "V09",
"w": 3,
"l": 6,
"shipping_fee": 95,
"weight": 70,
"eta": 1
},
"V10": {
"id": "V10",
"w": 2,
"l": 3,
</pre>
</td>
<td>
<pre>
"Horizontal"
},
"V08": {
"vehicle_id": "V08",
"in_ship": true,
"x": 5,
"y": 1,
"orientation": "Vertical"
},
"V09": {
"vehicle_id": "V09",
"in_ship": false,
"x": -1,
"y": -1,
"orientation": "Horizontal"
},
"V10": {
"vehicle_id": "V10",
"in_ship": true,
"x": 8,
"y": 5,
"orientation": "Horizontal"
},
"V11": {
"vehicle_id": "V11",
"in_ship":
</pre>
</td>
</tr>
</table>

```
"shipping_fee": 40,
            "weight":
30,
            "eta": 4
          },
          "V11": {
            "id":
"V11",
            "w": 3,
            "l": 3,

"shipping_fee": 50,
            "weight":
40,
            "eta": 3
          },
          "V12": {
            "id":
"V12",
            "w": 4,
            "l": 4,

"shipping_fee": 90,
            "weight":
60,
            "eta": 2
          },
          "V13": {
            "id":
"V13",
            "w": 2,
            "l": 4,

"shipping_fee": 48,
            "weight":
35,
            "eta": 5
          },
          "V14": {
```

```
true,
            "x": 4,
            "y": 5,

"orientation":
"Vertical"
          },
          "V12": {

"vehicle_id": "V12",
            "in_ship":
true,
            "x": 0,
            "y": 1,

"orientation":
"Vertical"
          },
          "V13": {

"vehicle_id": "V13",
            "in_ship":
true,
            "x": 8,
            "y": 9,

"orientation":
"Horizontal"
          },
          "V14": {

"vehicle_id": "V14",
            "in_ship":
false,
            "x": -1,
            "y": -1,

"orientation":
"Horizontal"
```

```
            "id":
"V14",
            "w": 3,
            "l": 5,

"shipping_fee": 82,
            "weight":
60,
            "eta": 1
          },
          "V15": {
            "id":
"V15",
            "w": 2,
            "l": 2,

"shipping_fee": 28,
            "weight":
20,
            "eta": 3
          },
          "V16": {
            "id":
"V16",
            "w": 3,
            "l": 4,

"shipping_fee": 62,
            "weight":
48,
            "eta": 2
          },
          "V17": {
            "id":
"V17",
            "w": 4,
            "l": 5,

"shipping_fee": 105,
```

```
          },
          "V15": {

"vehicle_id": "V15",
            "in_ship":
true,
            "x": 0,
            "y": 5,

"orientation":
"Vertical"
          },
          "V16": {

"vehicle_id": "V16",
            "in_ship":
false,
            "x": -1,
            "y": -1,

"orientation":
"Horizontal"
          },
          "V17": {

"vehicle_id": "V17",
            "in_ship":
true,
            "x": 4,
            "y": 8,

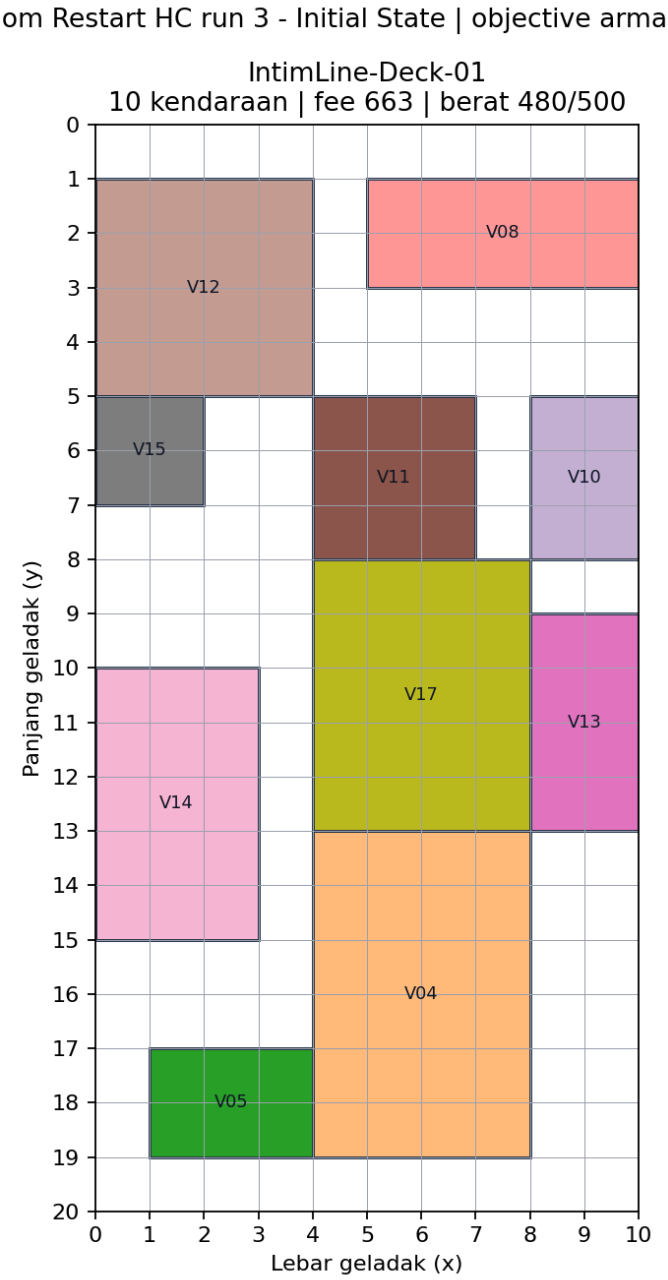
"orientation":
"Horizontal"
          },
          "V18": {

"vehicle_id": "V18",
            "in_ship":
false,
```

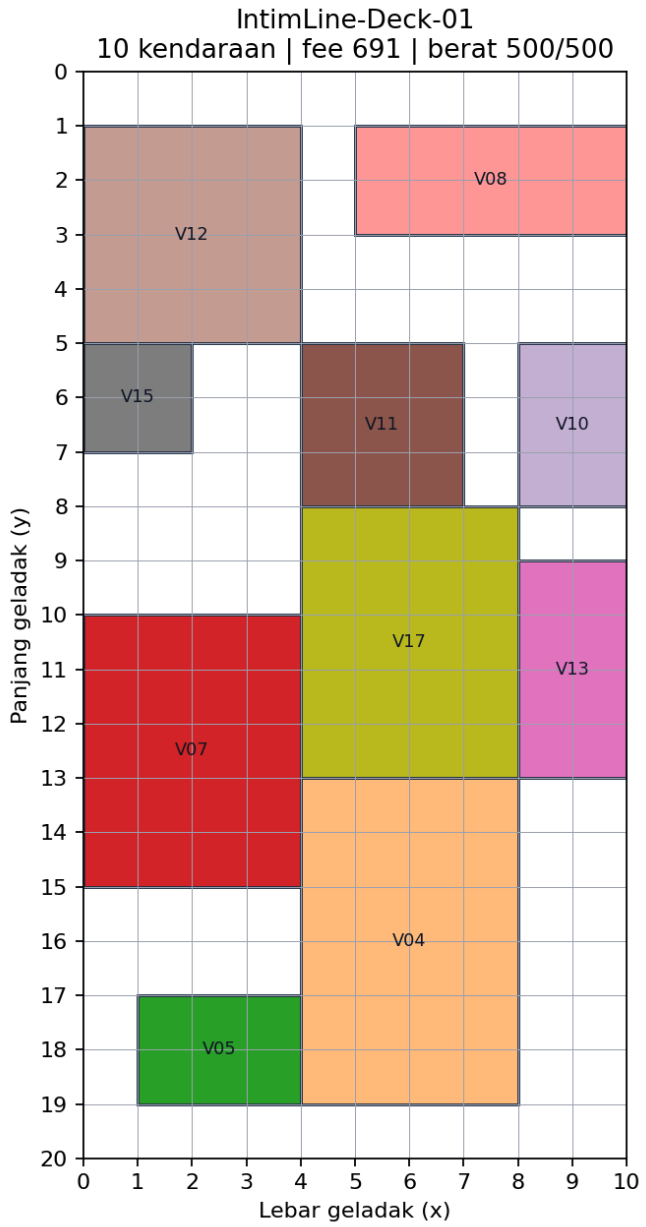
<table>
<tr>
<td rowspan="2"></td>
<td>
<pre>
            "weight":
75,
            "eta": 4
          },
          "V18": {
            "id":
"V18",
            "w": 2,
            "l": 3,

"shipping_fee": 38,
            "weight":
28,
            "eta": 2
          }
        },
</pre>
</td>
<td>
<pre>
            "x": -1,
            "y": -1,

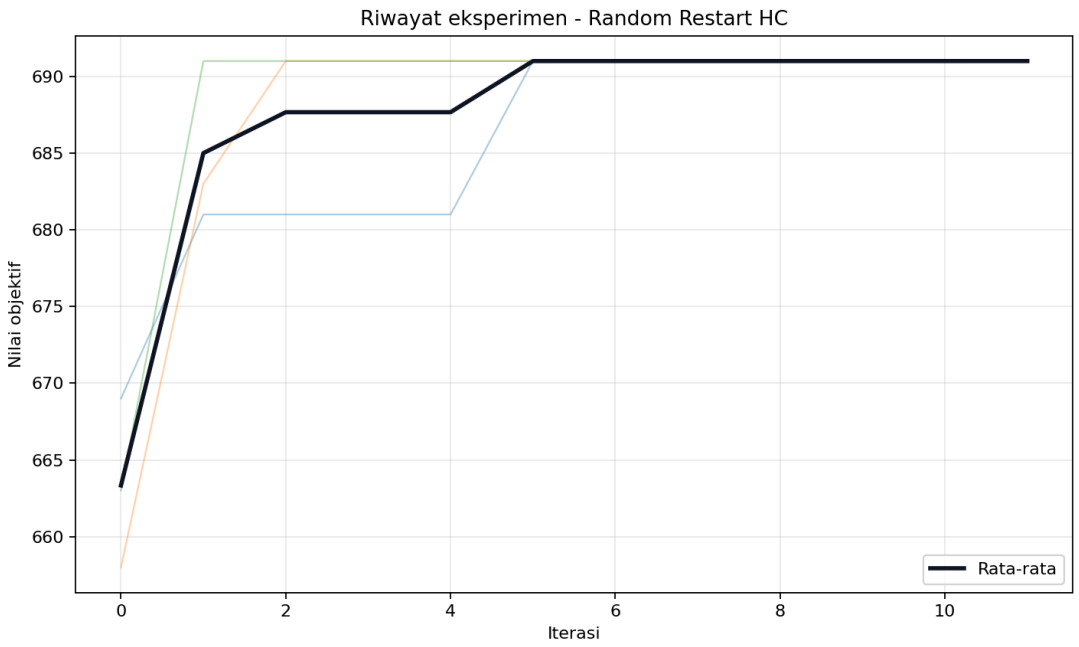
"orientation":
"Horizontal"
          }
        },
</pre>
</td>
</tr>
<tr>
<td colspan="2">Objective Function Score: 691</td>
</tr>
<tr>
<td colspan="3">Visualisasi Initial State:</td>
</tr>
</table>



Visualisasi Final State:


Random Restart HC run 3 - Final State | objective armada 691.00
IntimLine-Deck-01
10 kendaraan | fee 691 | berat 500/500
V12
V08
V15
V11
V10
V17
V13
V07
V04
V05
Panjang geladak (y)
Lebar geladak (x)
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20

Plot:



Highest Objective Function Score: 691
Iterations: 8
Duration: 0.2279

2.5.5 *Simulated Annealing*

Percobaan 1		
Initial	"initial_state": { "ship": { "id": "IntimLine-Deck-01", "w": 10, "l": 20, "max_capacity": 500 }, "vehicles": { "V01": { "id": "V01", "w": 3,	"placements": { "V01": { "vehicle_id": "V01", "in_ship": true, "x": 5, "y": 0, "orientation": "Horizontal" }, "V02": { "vehicle_id": "V02",

```
      "l": 4,
      "shipping_fee": 60,
      "weight": 50,
      "eta": 3
    },
    "V02": {
      "id": "V02",
      "w": 3,
      "l": 5,
      "shipping_fee": 85,
      "weight": 65,
      "eta": 2
    },
    "V03": {
      "id": "V03",
      "w": 2,
      "l": 4,
      "shipping_fee": 45,
      "weight": 35,
      "eta": 4
    },
    "V04": {
      "id": "V04",
      "w": 4,
      "l": 6,
      "shipping_fee": 130,
      "weight": 95,
      "eta": 1
    },
    "V05": {
      "id": "V05",
      "w": 2,
      "l": 3,
      "shipping_fee": 35,
```

```
      "in_ship": false,
      "x": -1,
      "y": -1,
      "orientation": "Horizontal"
    },
    "V03": {
      "vehicle_id": "V03",
      "in_ship": true,
      "x": 1,
      "y": 12,
      "orientation": "Vertical"
    },
    "V04": {
      "vehicle_id": "V04",
      "in_ship": false,
      "x": -1,
      "y": -1,
      "orientation": "Horizontal"
    },
    "V05": {
      "vehicle_id": "V05",
      "in_ship": true,
      "x": 7,
      "y": 8,
      "orientation": "Vertical"
    },
    "V06": {
      "vehicle_id": "V06",
      "in_ship": true,
```

<table>
<tr>
<td></td>
<td>

```
      "weight": 25,
      "eta": 5
    },
    "V06": {
      "id": "V06",
      "w": 3,
      "l": 4,
      "shipping_fee":
65,
      "weight": 45,
      "eta": 2
    },
    "V07": {
      "id": "V07",
      "w": 4,
      "l": 5,
      "shipping_fee":
110,
      "weight": 80,
      "eta": 3
    },
    "V08": {
      "id": "V08",
      "w": 2,
      "l": 5,
      "shipping_fee":
55,
      "weight": 40,
      "eta": 2
    },
    "V09": {
      "id": "V09",
      "w": 3,
      "l": 6,
      "shipping_fee":
95,
      "weight": 70,
      "eta": 1
    },
```

</td>
<td>

```
      "x": 5,
      "y": 5,
      "orientation":
"Vertical"
    },
    "V07": {
      "vehicle_id":
"V07",
      "in_ship":
false,
      "x": -1,
      "y": -1,
      "orientation":
"Horizontal"
    },
    "V08": {
      "vehicle_id":
"V08",
      "in_ship":
false,
      "x": -1,
      "y": -1,
      "orientation":
"Horizontal"
    },
    "V09": {
      "vehicle_id":
"V09",
      "in_ship":
false,
      "x": -1,
      "y": -1,
      "orientation":
"Horizontal"
    },
    "V10": {
      "vehicle_id":
"V10",
      "in_ship":
```

</td>
</tr>
</table>

<table>
<tr>
<td></td>
<td>

```
    "V10": {
      "id": "V10",
      "w": 2,
      "l": 3,
      "shipping_fee":
40,
      "weight": 30,
      "eta": 4
    },
    "V11": {
      "id": "V11",
      "w": 3,
      "l": 3,
      "shipping_fee":
50,
      "weight": 40,
      "eta": 3
    },
    "V12": {
      "id": "V12",
      "w": 4,
      "l": 4,
      "shipping_fee":
90,
      "weight": 60,
      "eta": 2
    },
    "V13": {
      "id": "V13",
      "w": 2,
      "l": 4,
      "shipping_fee":
48,
      "weight": 35,
      "eta": 5
    },
    "V14": {
      "id": "V14",
      "w": 3,
```

</td>
<td>

```
false,
      "x": -1,
      "y": -1,
      "orientation":
"Horizontal"
    },
    "V11": {
      "vehicle_id":
"V11",
      "in_ship": true,
      "x": 0,
      "y": 3,
      "orientation":
"Horizontal"
    },
    "V12": {
      "vehicle_id":
"V12",
      "in_ship": true,
      "x": 0,
      "y": 8,
      "orientation":
"Vertical"
    },
    "V13": {
      "vehicle_id":
"V13",
      "in_ship": true,
      "x": 1,
      "y": 1,
      "orientation":
"Vertical"
    },
    "V14": {
      "vehicle_id":
"V14",
      "in_ship": true,
      "x": 6,
      "y": 11,
```

</td>
</tr>
</table>

<table>
<tr>
<td></td>
<td>

<pre>
      "l": 5,
      "shipping_fee":
82,
      "weight": 60,
      "eta": 1
    },
    "V15": {
      "id": "V15",
      "w": 2,
      "l": 2,
      "shipping_fee":
28,
      "weight": 20,
      "eta": 3
    },
    "V16": {
      "id": "V16",
      "w": 3,
      "l": 4,
      "shipping_fee":
62,
      "weight": 48,
      "eta": 2
    },
    "V17": {
      "id": "V17",
      "w": 4,
      "l": 5,
      "shipping_fee":
105,
      "weight": 75,
      "eta": 4
    },
    "V18": {
      "id": "V18",
      "w": 2,
      "l": 3,
      "shipping_fee":
38,
</pre>

</td>
<td>

<pre>
      "orientation":
"Horizontal"
    },
    "V15": {
      "vehicle_id":
"V15",
      "in_ship": true,
      "x": 1,
      "y": 6,
      "orientation":
"Horizontal"
    },
    "V16": {
      "vehicle_id":
"V16",
      "in_ship":
false,
      "x": -1,
      "y": -1,
      "orientation":
"Horizontal"
    },
    "V17": {
      "vehicle_id":
"V17",
      "in_ship": true,
      "x": 0,
      "y": 15,
      "orientation":
"Vertical"
    },
    "V18": {
      "vehicle_id":
"V18",
      "in_ship": true,
      "x": 6,
      "y": 16,
      "orientation":
"Vertical"
</pre>

</td>
</tr>
</table>

<table>
<tr><td rowspan="2"></td><td><pre>
      "weight": 28,
      "eta": 2
    }
  }
</pre></td><td><pre>
    }
  }
</pre></td></tr>
<tr><td colspan="2">Objective Function Score: 646.0
Vehicle Count: 11
Total Weight: 473</td></tr>
<tr><td>Final</td><td><pre>
"final_state": {
        "ship": {
          "id":
"IntimLine-Deck-01",
          "w": 10,
          "l": 20,

"max_capacity": 500
        },
        "vehicles": {
          "V01": {
            "id":
"V01",
            "w": 3,
            "l": 4,

"shipping_fee": 60,
            "weight":
50,
            "eta": 3
          },
          "V02": {
            "id":
"V02",
            "w": 3,
            "l": 5,

"shipping_fee": 85,
            "weight":
65,
            "eta": 2
</pre></td><td><pre>
"placements": {
          "V01": {

"vehicle_id": "V01",
            "in_ship":
false,
            "x": -1,
            "y": -1,

"orientation":
"Vertical"
          },
          "V02": {

"vehicle_id": "V02",
            "in_ship":
true,
            "x": 0,
            "y": 7,

"orientation":
"Horizontal"
          },
          "V03": {

"vehicle_id": "V03",
            "in_ship":
true,
            "x": 4,
            "y": 17,
</pre></td></tr>
</table>

<table>
<tr>
<td></td>
<td>
<pre>
          },
          "V03": {
            "id":
"V03",
            "w": 2,
            "l": 4,

"shipping_fee": 45,
            "weight":
35,
            "eta": 4
          },
          "V04": {
            "id":
"V04",
            "w": 4,
            "l": 6,

"shipping_fee": 130,
            "weight":
95,
            "eta": 1
          },
          "V05": {
            "id":
"V05",
            "w": 2,
            "l": 3,

"shipping_fee": 35,
            "weight":
25,
            "eta": 5
          },
          "V06": {
            "id":
"V06",
            "w": 3,
            "l": 4,
</pre>
</td>
<td>
<pre>
"orientation":
"Vertical"
          },
          "V04": {

"vehicle_id": "V04",
            "in_ship":
false,
            "x": -1,
            "y": -1,

"orientation":
"Vertical"
          },
          "V05": {

"vehicle_id": "V05",
            "in_ship":
false,
            "x": -1,
            "y": -1,

"orientation":
"Horizontal"
          },
          "V06": {

"vehicle_id": "V06",
            "in_ship":
false,
            "x": -1,
            "y": -1,

"orientation":
"Vertical"
          },
          "V07": {

"vehicle_id": "V07",
</pre>
</td>
</tr>
</table>

```
"shipping_fee": 65,
            "weight": 45,
            "eta": 2
          },
          "V07": {
            "id": "V07",
            "w": 4,
            "l": 5,

"shipping_fee": 110,
            "weight": 80,
            "eta": 3
          },
          "V08": {
            "id": "V08",
            "w": 2,
            "l": 5,

"shipping_fee": 55,
            "weight": 40,
            "eta": 2
          },
          "V09": {
            "id": "V09",
            "w": 3,
            "l": 6,

"shipping_fee": 95,
            "weight": 70,
            "eta": 1
          },
```

```
            "in_ship": false,
            "x": -1,
            "y": -1,

"orientation": "Vertical"
          },
          "V08": {

"vehicle_id": "V08",
            "in_ship": false,
            "x": -1,
            "y": -1,

"orientation": "Vertical"
          },
          "V09": {

"vehicle_id": "V09",
            "in_ship": true,
            "x": 3,
            "y": 5,

"orientation": "Horizontal"
          },
          "V10": {

"vehicle_id": "V10",
            "in_ship": true,
            "x": 6,
            "y": 5,

"orientation":
```

```
          "V10": {
            "id":
"V10",
            "w": 2,
            "l": 3,

"shipping_fee": 40,
            "weight":
30,
            "eta": 4
          },
          "V11": {
            "id":
"V11",
            "w": 3,
            "l": 3,

"shipping_fee": 50,
            "weight":
40,
            "eta": 3
          },
          "V12": {
            "id":
"V12",
            "w": 4,
            "l": 4,

"shipping_fee": 90,
            "weight":
60,
            "eta": 2
          },
          "V13": {
            "id":
"V13",
            "w": 2,
            "l": 4,
```

```
"Vertical"
          },
          "V11": {

"vehicle_id": "V11",
            "in_ship":
true,
            "x": 6,
            "y": 9,

"orientation":
"Vertical"
          },
          "V12": {

"vehicle_id": "V12",
            "in_ship":
true,
            "x": 0,
            "y": 15,

"orientation":
"Vertical"
          },
          "V13": {

"vehicle_id": "V13",
            "in_ship":
true,
            "x": 6,
            "y": 15,

"orientation":
"Vertical"
          },
          "V14": {

"vehicle_id": "V14",
            "in_ship":
```

```
"shipping_fee": 48,
            "weight":
35,
            "eta": 5
          },
          "V14": {
            "id":
"V14",
            "w": 3,
            "l": 5,

"shipping_fee": 82,
            "weight":
60,
            "eta": 1
          },
          "V15": {
            "id":
"V15",
            "w": 2,
            "l": 2,

"shipping_fee": 28,
            "weight":
20,
            "eta": 3
          },
          "V16": {
            "id":
"V16",
            "w": 3,
            "l": 4,

"shipping_fee": 62,
            "weight":
48,
            "eta": 2
          },
          "V17": {
```

```
true,
            "x": 1,
            "y": 0,

"orientation":
"Horizontal"
          },
          "V15": {

"vehicle_id": "V15",
            "in_ship":
true,
            "x": 6,
            "y": 7,

"orientation":
"Vertical"
          },
          "V16": {

"vehicle_id": "V16",
            "in_ship":
true,
            "x": 4,
            "y": 0,

"orientation":
"Vertical"
          },
          "V17": {

"vehicle_id": "V17",
            "in_ship":
false,
            "x": -1,
            "y": -1,

"orientation":
"Horizontal"
```

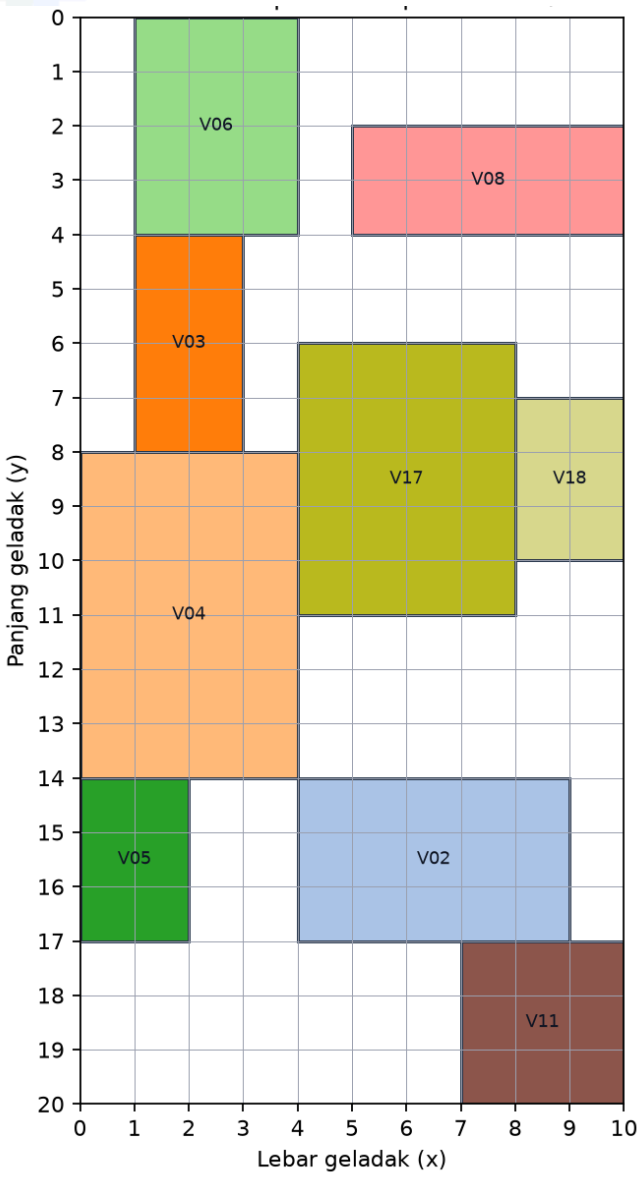
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<td>
<pre>
            "id":
"V17",
            "w": 4,
            "l": 5,

"shipping_fee": 105,
            "weight":
75,
            "eta": 4
          },
          "V18": {
            "id":
"V18",
            "w": 2,
            "l": 3,

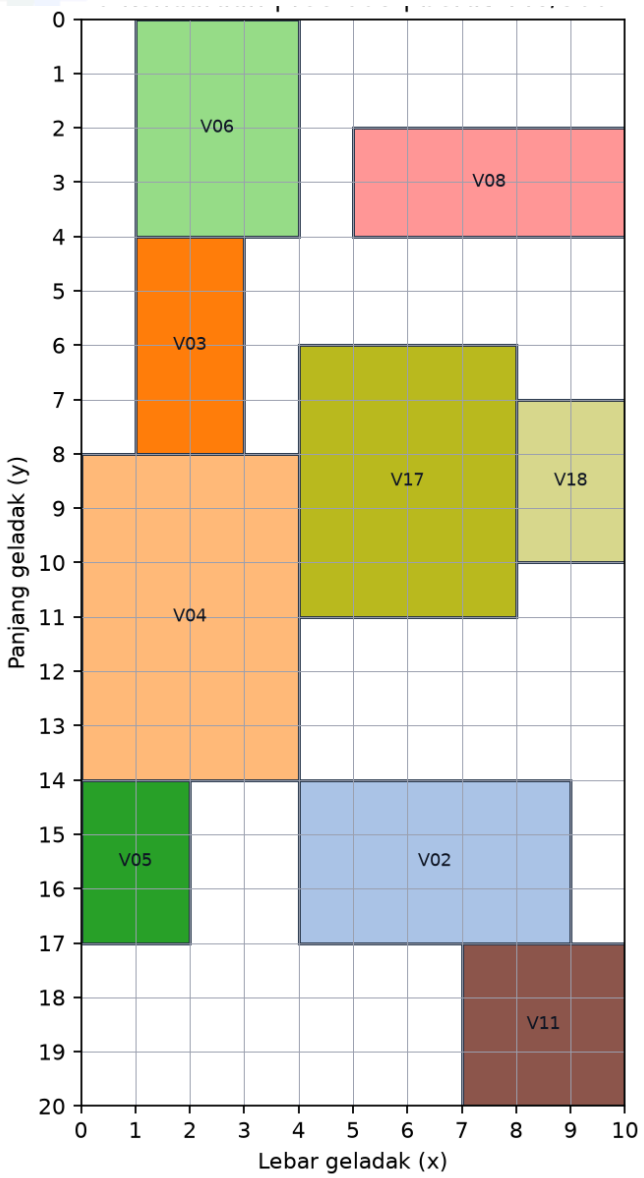
"shipping_fee": 38,
            "weight":
28,
            "eta": 2
          }
        }
</pre>
</td>
<td>
<pre>
          },
          "V18": {

"vehicle_id": "V18",
            "in_ship":
true,
            "x": 2,
            "y": 12,

"orientation":
"Horizontal"
          }
        }
</pre>
</td>
</tr>
<tr>
<td></td>
<td colspan="2">Objective Function Score: 663.0
Vehicle Count: 11
Total Weight: 491</td>
</tr>
<tr>
<td colspan="3">Visualisasi Initial State:</td>
</tr>
</table>



Visualisasi *Final State*:



Highest Objective Function Score: 663.0

Iterations: 100

Duration: 0.03215193748474121

Stuck Frequency: 0

Percobaan 2		
Initial	`"initial_state": {` `"ship": {`	`"placements": {` `"V01": {`

<table>
<tr>
<td></td>
<td>

```
    "id":
"IntimLine-Deck-01",
    "w": 10,
    "l": 20,
    "max_capacity":
500
  },
  "vehicles": {
    "V01": {
      "id": "V01",
      "w": 3,
      "l": 4,
      "shipping_fee":
60,
      "weight": 50,
      "eta": 3
    },
    "V02": {
      "id": "V02",
      "w": 3,
      "l": 5,
      "shipping_fee":
85,
      "weight": 65,
      "eta": 2
    },
    "V03": {
      "id": "V03",
      "w": 2,
      "l": 4,
      "shipping_fee":
45,
      "weight": 35,
      "eta": 4
    },
    "V04": {
      "id": "V04",
      "w": 4,
      "l": 6,
```

</td>
<td>

```
    "vehicle_id":
"V01",
    "in_ship": true,
    "x": 4,
    "y": 10,
    "orientation":
"Vertical"
  },
  "V02": {
    "vehicle_id":
"V02",
    "in_ship": false,
    "x": -1,
    "y": -1,
    "orientation":
"Horizontal"
  },
  "V03": {
    "vehicle_id":
"V03",
    "in_ship": true,
    "x": 5,
    "y": 0,
    "orientation":
"Horizontal"
  },
  "V04": {
    "vehicle_id":
"V04",
    "in_ship": false,
    "x": -1,
    "y": -1,
    "orientation":
"Horizontal"
  },
  "V05": {
    "vehicle_id":
"V05",
    "in_ship": true,
```

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</table>

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```
      "shipping_fee":
130,
      "weight": 95,
      "eta": 1
    },
    "V05": {
      "id": "V05",
      "w": 2,
      "l": 3,
      "shipping_fee":
35,
      "weight": 25,
      "eta": 5
    },
    "V06": {
      "id": "V06",
      "w": 3,
      "l": 4,
      "shipping_fee":
65,
      "weight": 45,
      "eta": 2
    },
    "V07": {
      "id": "V07",
      "w": 4,
      "l": 5,
      "shipping_fee":
110,
      "weight": 80,
      "eta": 3
    },
    "V08": {
      "id": "V08",
      "w": 2,
      "l": 5,
      "shipping_fee":
55,
      "weight": 40,
```

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```
    "x": 6,
    "y": 13,
    "orientation":
"Vertical"
  },
  "V06": {
    "vehicle_id":
"V06",
    "in_ship": true,
    "x": 6,
    "y": 6,
    "orientation":
"Vertical"
  },
  "V07": {
    "vehicle_id":
"V07",
    "in_ship": false,
    "x": -1,
    "y": -1,
    "orientation":
"Horizontal"
  },
  "V08": {
    "vehicle_id":
"V08",
    "in_ship": false,
    "x": -1,
    "y": -1,
    "orientation":
"Horizontal"
  },
  "V09": {
    "vehicle_id":
"V09",
    "in_ship": true,
    "x": 2,
    "y": 0,
    "orientation":
```

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</table>

```
      "eta": 2
    },
    "V09": {
      "id": "V09",
      "w": 3,
      "l": 6,
      "shipping_fee":
95,
      "weight": 70,
      "eta": 1
    },
    "V10": {
      "id": "V10",
      "w": 2,
      "l": 3,
      "shipping_fee":
40,
      "weight": 30,
      "eta": 4
    },
    "V11": {
      "id": "V11",
      "w": 3,
      "l": 3,
      "shipping_fee":
50,
      "weight": 40,
      "eta": 3
    },
    "V12": {
      "id": "V12",
      "w": 4,
      "l": 4,
      "shipping_fee":
90,
      "weight": 60,
      "eta": 2
    },
    "V13": {
```

```
"Horizontal"
  },
  "V10": {
    "vehicle_id":
"V10",
    "in_ship": true,
    "x": 7,
    "y": 18,
    "orientation":
"Vertical"
  },
  "V11": {
    "vehicle_id":
"V11",
    "in_ship": false,
    "x": -1,
    "y": -1,
    "orientation":
"Horizontal"
  },
  "V12": {
    "vehicle_id":
"V12",
    "in_ship": false,
    "x": -1,
    "y": -1,
    "orientation":
"Horizontal"
  },
  "V13": {
    "vehicle_id":
"V13",
    "in_ship": true,
    "x": 1,
    "y": 18,
    "orientation":
"Vertical"
  },
  "V14": {
```

```
      "id": "V13",
      "w": 2,
      "l": 4,
      "shipping_fee": 48,
      "weight": 35,
      "eta": 5
    },
    "V14": {
      "id": "V14",
      "w": 3,
      "l": 5,
      "shipping_fee": 82,
      "weight": 60,
      "eta": 1
    },
    "V15": {
      "id": "V15",
      "w": 2,
      "l": 2,
      "shipping_fee": 28,
      "weight": 20,
      "eta": 3
    },
    "V16": {
      "id": "V16",
      "w": 3,
      "l": 4,
      "shipping_fee": 62,
      "weight": 48,
      "eta": 2
    },
    "V17": {
      "id": "V17",
      "w": 4,
      "l": 5,
```

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    "vehicle_id": "V14",
    "in_ship": true,
    "x": 0,
    "y": 10,
    "orientation": "Horizontal"
  },
  "V15": {
    "vehicle_id": "V15",
    "in_ship": true,
    "x": 1,
    "y": 16,
    "orientation": "Vertical"
  },
  "V16": {
    "vehicle_id": "V16",
    "in_ship": true,
    "x": 3,
    "y": 13,
    "orientation": "Horizontal"
  },
  "V17": {
    "vehicle_id": "V17",
    "in_ship": false,
    "x": -1,
    "y": -1,
    "orientation": "Horizontal"
  },
  "V18": {
    "vehicle_id": "V18",
    "in_ship": true,
```

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<td>
<pre>
      "shipping_fee": 105,
      "weight": 75,
      "eta": 4
    },
    "V18": {
      "id": "V18",
      "w": 2,
      "l": 3,
      "shipping_fee": 38,
      "weight": 28,
      "eta": 2
    }
  }
</pre>
</td>
<td>
<pre>
    "x": 2,
    "y": 7,
    "orientation": "Horizontal"
  }
},
</pre>
</td>
</tr>
<tr>
<td></td>
<td colspan="2">Objective Function Score: 598.0
Vehicle Count: 11
Total Weight: 446/500</td>
</tr>
<tr>
<td>Final</td>
<td>
<pre>
"final_state": {
  "ship": {
    "id": "IntimLine-Deck-01",
    "w": 10,
    "l": 20,
    "max_capacity": 500
  },
  "vehicles": {
    "V01": {
      "id": "V01",
      "w": 3,
      "l": 4,
      "shipping_fee": 60,
      "weight": 50,
      "eta": 3
    },
    "V02": {
</pre>
</td>
<td>
<pre>
"placements": {
    "V01": {
      "vehicle_id": "V01",
      "in_ship": true,
      "x": 6,
      "y": 12,
      "orientation": "Vertical"
    },
    "V02": {
      "vehicle_id": "V02",
      "in_ship": false,
      "x": -1,
      "y": -1,
      "orientation": "Horizontal"
    },
</pre>
</td>
</tr>
</table>

```
      "id": "V02",
      "w": 3,
      "l": 5,
      "shipping_fee": 85,
      "weight": 65,
      "eta": 2
    },
    "V03": {
      "id": "V03",
      "w": 2,
      "l": 4,
      "shipping_fee": 45,
      "weight": 35,
      "eta": 4
    },
    "V04": {
      "id": "V04",
      "w": 4,
      "l": 6,
      "shipping_fee": 130,
      "weight": 95,
      "eta": 1
    },
    "V05": {
      "id": "V05",
      "w": 2,
      "l": 3,
      "shipping_fee": 35,
      "weight": 25,
      "eta": 5
    },
    "V06": {
      "id": "V06",
      "w": 3,
      "l": 4,
```

```
    "V03": {
      "vehicle_id": "V03",
      "in_ship": true,
      "x": 5,
      "y": 6,
      "orientation": "Horizontal"
    },
    "V04": {
      "vehicle_id": "V04",
      "in_ship": false,
      "x": -1,
      "y": -1,
      "orientation": "Vertical"
    },
    "V05": {
      "vehicle_id": "V05",
      "in_ship": false,
      "x": -1,
      "y": -1,
      "orientation": "Horizontal"
    },
    "V06": {
      "vehicle_id": "V06",
      "in_ship": true,
      "x": 5,
      "y": 15,
      "orientation": "Horizontal"
    },
    "V07": {
```

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```
      "shipping_fee":
65,
      "weight": 45,
      "eta": 2
    },
    "V07": {
      "id": "V07",
      "w": 4,
      "l": 5,
      "shipping_fee":
110,
      "weight": 80,
      "eta": 3
    },
    "V08": {
      "id": "V08",
      "w": 2,
      "l": 5,
      "shipping_fee":
55,
      "weight": 40,
      "eta": 2
    },
    "V09": {
      "id": "V09",
      "w": 3,
      "l": 6,
      "shipping_fee":
95,
      "weight": 70,
      "eta": 1
    },
    "V10": {
      "id": "V10",
      "w": 2,
      "l": 3,
      "shipping_fee":
40,
      "weight": 30,
```

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<td>

```
      "vehicle_id":
"V07",
      "in_ship":
false,
      "x": -1,
      "y": -1,
      "orientation":
"Horizontal"
    },
    "V08": {
      "vehicle_id":
"V08",
      "in_ship": true,
      "x": 0,
      "y": 5,
      "orientation":
"Vertical"
    },
    "V09": {
      "vehicle_id":
"V09",
      "in_ship":
false,
      "x": -1,
      "y": -1,
      "orientation":
"Horizontal"
    },
    "V10": {
      "vehicle_id":
"V10",
      "in_ship": true,
      "x": 7,
      "y": 10,
      "orientation":
"Vertical"
    },
    "V11": {
      "vehicle_id":
```

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</table>

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<tr>
<td></td>
<td>

```
      "eta": 4
    },
    "V11": {
      "id": "V11",
      "w": 3,
      "l": 3,
      "shipping_fee":
50,
      "weight": 40,
      "eta": 3
    },
    "V12": {
      "id": "V12",
      "w": 4,
      "l": 4,
      "shipping_fee":
90,
      "weight": 60,
      "eta": 2
    },
    "V13": {
      "id": "V13",
      "w": 2,
      "l": 4,
      "shipping_fee":
48,
      "weight": 35,
      "eta": 5
    },
    "V14": {
      "id": "V14",
      "w": 3,
      "l": 5,
      "shipping_fee":
82,
      "weight": 60,
      "eta": 1
    },
    "V15": {
```

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<td>

```
"V11",
      "in_ship": true,
      "x": 0,
      "y": 15,
      "orientation":
"Vertical"
    },
    "V12": {
      "vehicle_id":
"V12",
      "in_ship": true,
      "x": 6,
      "y": 0,
      "orientation":
"Horizontal"
    },
    "V13": {
      "vehicle_id":
"V13",
      "in_ship": true,
      "x": 0,
      "y": 3,
      "orientation":
"Vertical"
    },
    "V14": {
      "vehicle_id":
"V14",
      "in_ship": true,
      "x": 0,
      "y": 11,
      "orientation":
"Vertical"
    },
    "V15": {
      "vehicle_id":
"V15",
      "in_ship": true,
      "x": 2,
```

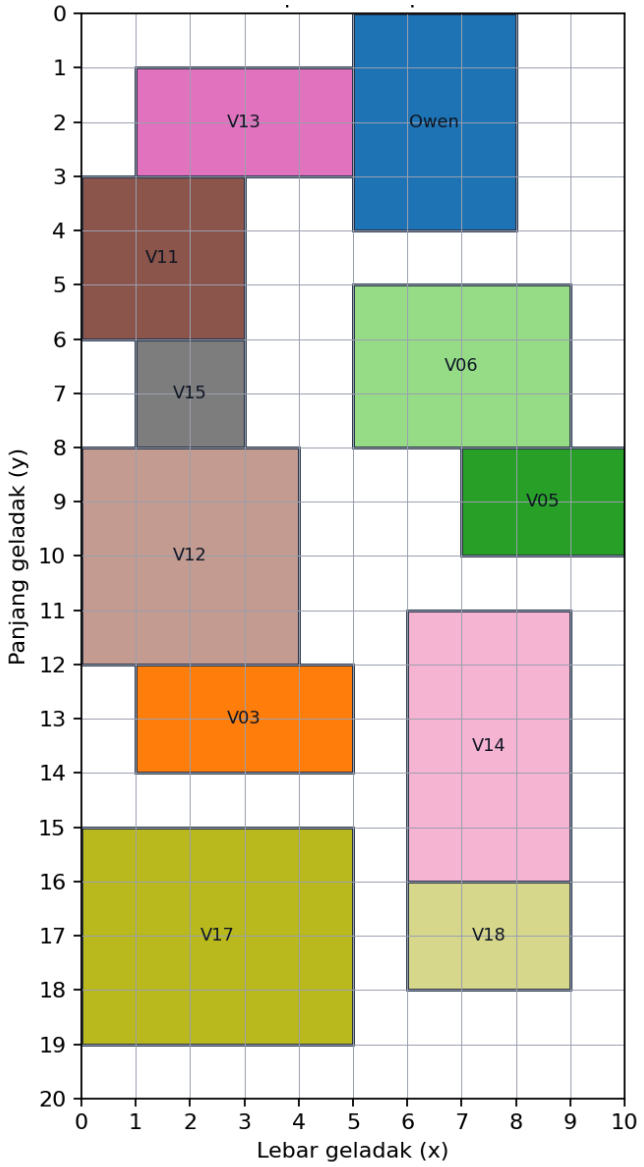
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      "weight": 20,
      "eta": 3
    },
    "V16": {
      "id": "V16",
      "w": 3,
      "l": 4,
      "shipping_fee": 62,
      "weight": 48,
      "eta": 2
    },
    "V17": {
      "id": "V17",
      "w": 4,
      "l": 5,
      "shipping_fee": 105,
      "weight": 75,
      "eta": 4
    },
    "V18": {
      "id": "V18",
      "w": 2,
      "l": 3,
      "shipping_fee": 38,
      "weight": 28,
      "eta": 2
    }
  }
```

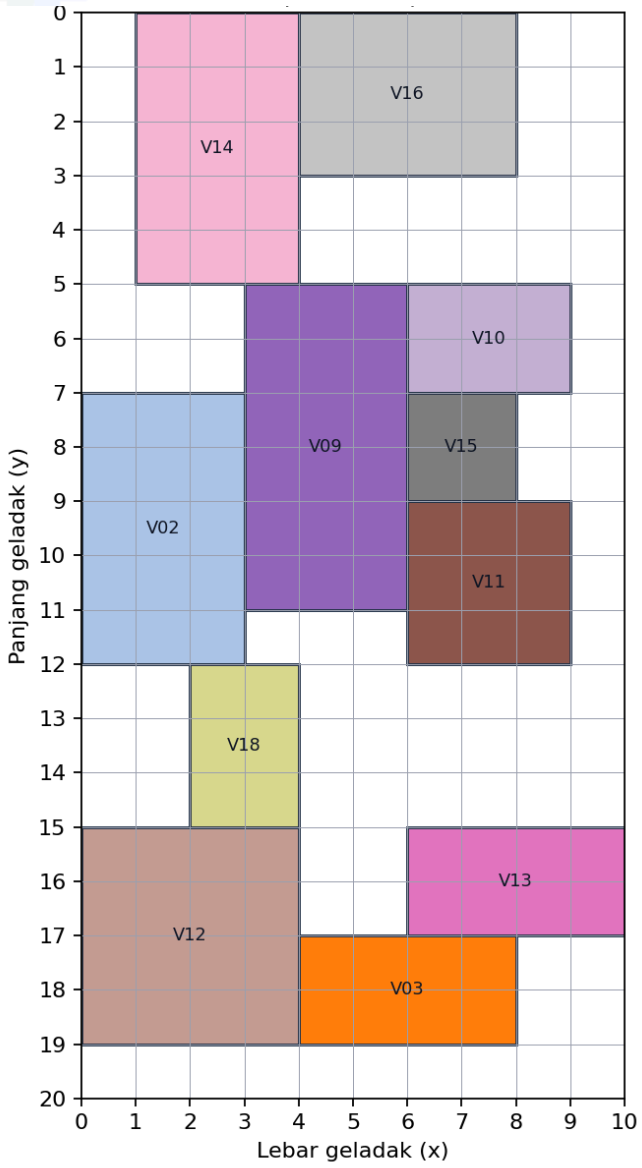
```
      "y": 8,
      "orientation": "Horizontal"
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    "V16": {
      "vehicle_id": "V16",
      "in_ship": true,
      "x": 7,
      "y": 6,
      "orientation": "Horizontal"
    },
    "V17": {
      "vehicle_id": "V17",
      "in_ship": false,
      "x": -1,
      "y": -1,
      "orientation": "Vertical"
    },
    "V18": {
      "vehicle_id": "V18",
      "in_ship": true,
      "x": 2,
      "y": 0,
      "orientation": "Vertical"
    }
  }
```


Objective Function Score: 663.0
Vehicle Count: 12
Total Weight: 491/500

Visualisasi *Initial State*:



Visualisasi *Final State*:



Highest Objective Function Score: 663.0
Iterations: 100
Duration: 0.049959659576416016
Stuck Frequency: 0

Percobaan 3		
Initial	"initial_state": { "ship": {	"placements": { "V01": {

```
    "id":
"IntimLine-Deck-01",
    "w": 10,
    "l": 20,
    "max_capacity":
500
  },
  "vehicles": {
    "V01": {
      "id": "V01",
      "w": 3,
      "l": 4,
      "shipping_fee":
60,
      "weight": 50,
      "eta": 3
    },
    "V02": {
      "id": "V02",
      "w": 3,
      "l": 5,
      "shipping_fee":
85,
      "weight": 65,
      "eta": 2
    },
    "V03": {
      "id": "V03",
      "w": 2,
      "l": 4,
      "shipping_fee":
45,
      "weight": 35,
      "eta": 4
    },
    "V04": {
      "id": "V04",
      "w": 4,
      "l": 6,
```

```
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"V01",
      "in_ship":
false,
      "x": -1,
      "y": -1,
      "orientation":
"Horizontal"
    },
    "V02": {
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"V02",
      "in_ship":
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      "x": -1,
      "y": -1,
      "orientation":
"Horizontal"
    },
    "V03": {
      "vehicle_id":
"V03",
      "in_ship": true,
      "x": 2,
      "y": 3,
      "orientation":
"Vertical"
    },
    "V04": {
      "vehicle_id":
"V04",
      "in_ship": true,
      "x": 3,
      "y": 6,
      "orientation":
"Horizontal"
    },
    "V05": {
      "vehicle_id":
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```
      "shipping_fee":
130,
      "weight": 95,
      "eta": 1
    },
    "V05": {
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      "w": 2,
      "l": 3,
      "shipping_fee":
35,
      "weight": 25,
      "eta": 5
    },
    "V06": {
      "id": "V06",
      "w": 3,
      "l": 4,
      "shipping_fee":
65,
      "weight": 45,
      "eta": 2
    },
    "V07": {
      "id": "V07",
      "w": 4,
      "l": 5,
      "shipping_fee":
110,
      "weight": 80,
      "eta": 3
    },
    "V08": {
      "id": "V08",
      "w": 2,
      "l": 5,
      "shipping_fee":
55,
      "weight": 40,
```

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```
"V05",
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      "x": 8,
      "y": 13,
      "orientation":
"Horizontal"
    },
    "V06": {
      "vehicle_id":
"V06",
      "in_ship": true,
      "x": 1,
      "y": 16,
      "orientation":
"Horizontal"
    },
    "V07": {
      "vehicle_id":
"V07",
      "in_ship":
false,
      "x": -1,
      "y": -1,
      "orientation":
"Horizontal"
    },
    "V08": {
      "vehicle_id":
"V08",
      "in_ship":
false,
      "x": -1,
      "y": -1,
      "orientation":
"Horizontal"
    },
    "V09": {
      "vehicle_id":
"V09",
```

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```
      "eta": 2
    },
    "V09": {
      "id": "V09",
      "w": 3,
      "l": 6,
      "shipping_fee":
95,
      "weight": 70,
      "eta": 1
    },
    "V10": {
      "id": "V10",
      "w": 2,
      "l": 3,
      "shipping_fee":
40,
      "weight": 30,
      "eta": 4
    },
    "V11": {
      "id": "V11",
      "w": 3,
      "l": 3,
      "shipping_fee":
50,
      "weight": 40,
      "eta": 3
    },
    "V12": {
      "id": "V12",
      "w": 4,
      "l": 4,
      "shipping_fee":
90,
      "weight": 60,
      "eta": 2
    },
    "V13": {
```

```
      "in_ship":
false,
      "x": -1,
      "y": -1,
      "orientation":
"Horizontal"
    },
    "V10": {
      "vehicle_id":
"V10",
      "in_ship":
false,
      "x": -1,
      "y": -1,
      "orientation":
"Horizontal"
    },
    "V11": {
      "vehicle_id":
"V11",
      "in_ship": true,
      "x": 5,
      "y": 13,
      "orientation":
"Horizontal"
    },
    "V12": {
      "vehicle_id":
"V12",
      "in_ship": true,
      "x": 6,
      "y": 2,
      "orientation":
"Vertical"
    },
    "V13": {
      "vehicle_id":
"V13",
      "in_ship": true,
```

```
      "id": "V13",
      "w": 2,
      "l": 4,
      "shipping_fee": 48,
      "weight": 35,
      "eta": 5
    },
    "V14": {
      "id": "V14",
      "w": 3,
      "l": 5,
      "shipping_fee": 82,
      "weight": 60,
      "eta": 1
    },
    "V15": {
      "id": "V15",
      "w": 2,
      "l": 2,
      "shipping_fee": 28,
      "weight": 20,
      "eta": 3
    },
    "V16": {
      "id": "V16",
      "w": 3,
      "l": 4,
      "shipping_fee": 62,
      "weight": 48,
      "eta": 2
    },
    "V17": {
      "id": "V17",
      "w": 4,
      "l": 5,
```

```
      "x": 7,
      "y": 9,
      "orientation": "Horizontal"
    },
    "V14": {
      "vehicle_id": "V14",
      "in_ship": true,
      "x": 0,
      "y": 13,
      "orientation": "Vertical"
    },
    "V15": {
      "vehicle_id": "V15",
      "in_ship": true,
      "x": 8,
      "y": 0,
      "orientation": "Vertical"
    },
    "V16": {
      "vehicle_id": "V16",
      "in_ship": true,
      "x": 4,
      "y": 17,
      "orientation": "Vertical"
    },
    "V17": {
      "vehicle_id": "V17",
      "in_ship": false,
      "x": -1,
      "y": -1,
```

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<tr>
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<td>
<pre>
      "shipping_fee": 105,
      "weight": 75,
      "eta": 4
    },
    "V18": {
      "id": "V18",
      "w": 2,
      "l": 3,
      "shipping_fee": 38,
      "weight": 28,
      "eta": 2
    }
  }
</pre>
</td>
<td>
<pre>
      "orientation": "Horizontal"
    },
    "V18": {
      "vehicle_id": "V18",
      "in_ship": true,
      "x": 4,
      "y": 0,
      "orientation": "Horizontal"
    }
  }
</pre>
</td>
</tr>
<tr>
<td colspan="2">Objective Function Score: 673.0
Vehicle Count: 11
Total Weight: 491</td>
</tr>
<tr>
<td>Final</td>
<td>
<pre>
"final_state": {
  "ship": {
    "id": "IntimLine-Deck-01",
    "w": 10,
    "l": 20,
    "max_capacity": 500
  },
  "vehicles": {
    "V01": {
      "id": "V01",
      "w": 3,
      "l": 4,
      "shipping_fee": 60,
      "weight": 50,
      "eta": 3
    },
</pre>
</td>
<td>
<pre>
"placements": {
    "V01": {
      "vehicle_id": "V01",
      "in_ship": false,
      "x": -1,
      "y": -1,
      "orientation": "Vertical"
    },
    "V02": {
      "vehicle_id": "V02",
      "in_ship": true,
      "x": 4,
      "y": 14,
      "orientation": "Vertical"
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</td>
</tr>
</table>

<table>
<tr>
<td></td>
<td>

```
    "V02": {
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      "l": 5,
      "shipping_fee":
85,
      "weight": 65,
      "eta": 2
    },
    "V03": {
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      "w": 2,
      "l": 4,
      "shipping_fee":
45,
      "weight": 35,
      "eta": 4
    },
    "V04": {
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      "w": 4,
      "l": 6,
      "shipping_fee":
130,
      "weight": 95,
      "eta": 1
    },
    "V05": {
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      "w": 2,
      "l": 3,
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35,
      "weight": 25,
      "eta": 5
    },
    "V06": {
      "id": "V06",
      "w": 3,
```

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```
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      "orientation":
"Horizontal"
    },
    "V04": {
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"V04",
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      "y": 8,
      "orientation":
"Horizontal"
    },
    "V05": {
      "vehicle_id":
"V05",
      "in_ship": true,
      "x": 0,
      "y": 14,
      "orientation":
"Horizontal"
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    "V06": {
      "vehicle_id":
"V06",
      "in_ship": true,
      "x": 1,
      "y": 0,
      "orientation":
"Horizontal"
    },
    "V07": {
      "vehicle_id":
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      "w": 4,
      "l": 5,
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110,
      "weight": 80,
      "eta": 3
    },
    "V08": {
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      "w": 2,
      "l": 5,
      "shipping_fee":
55,
      "weight": 40,
      "eta": 2
    },
    "V09": {
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      "w": 3,
      "l": 6,
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      "weight": 70,
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      "w": 2,
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      "shipping_fee":
40,
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"Horizontal"
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    "V08": {
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"V08",
      "in_ship": true,
      "x": 5,
      "y": 2,
      "orientation":
"Vertical"
    },
    "V09": {
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"V09",
      "in_ship":
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      "x": -1,
      "y": -1,
      "orientation":
"Horizontal"
    },
    "V10": {
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      "in_ship":
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      "x": -1,
      "y": -1,
      "orientation":
"Horizontal"
    },
    "V11": {
      "vehicle_id":
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```
      "weight": 30,
      "eta": 4
    },
    "V11": {
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      "w": 3,
      "l": 3,
      "shipping_fee":
50,
      "weight": 40,
      "eta": 3
    },
    "V12": {
      "id": "V12",
      "w": 4,
      "l": 4,
      "shipping_fee":
90,
      "weight": 60,
      "eta": 2
    },
    "V13": {
      "id": "V13",
      "w": 2,
      "l": 4,
      "shipping_fee":
48,
      "weight": 35,
      "eta": 5
    },
    "V14": {
      "id": "V14",
      "w": 3,
      "l": 5,
      "shipping_fee":
82,
      "weight": 60,
      "eta": 1
    },
```

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```
"V11",
      "in_ship": true,
      "x": 7,
      "y": 17,
      "orientation":
"Horizontal"
    },
    "V12": {
      "vehicle_id":
"V12",
      "in_ship":
false,
      "x": -1,
      "y": -1,
      "orientation":
"Vertical"
    },
    "V13": {
      "vehicle_id":
"V13",
      "in_ship":
false,
      "x": -1,
      "y": -1,
      "orientation":
"Horizontal"
    },
    "V14": {
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"V14",
      "in_ship":
false,
      "x": -1,
      "y": -1,
      "orientation":
"Vertical"
    },
    "V15": {
      "vehicle_id":
```

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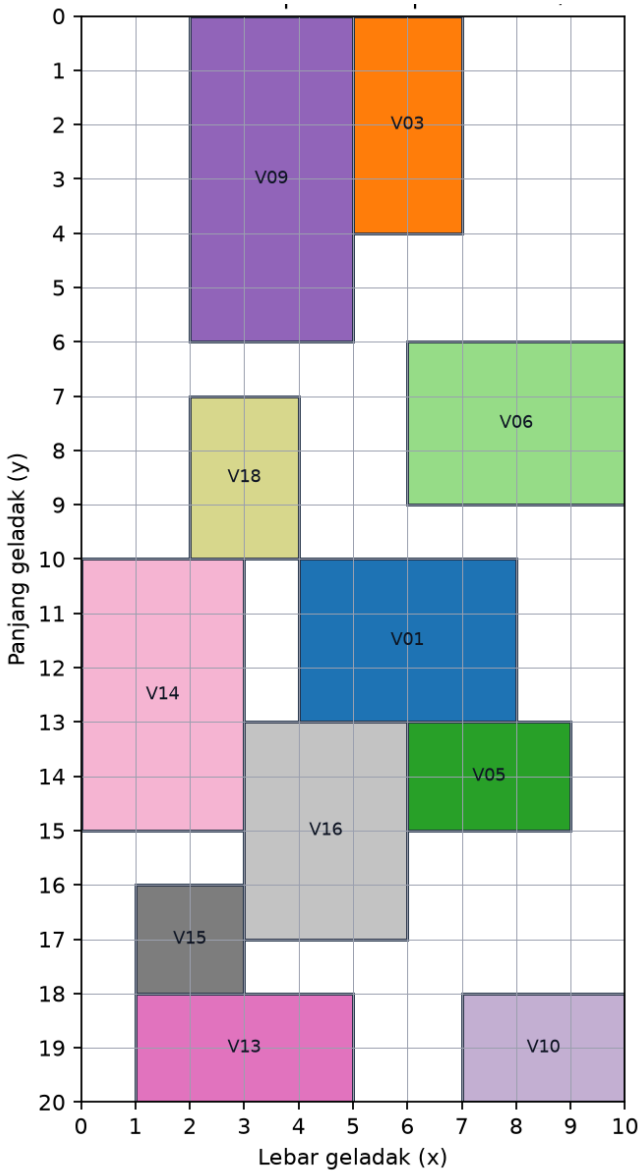
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      "l": 2,
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      "weight": 20,
      "eta": 3
    },
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      "w": 3,
      "l": 4,
      "shipping_fee":
62,
      "weight": 48,
      "eta": 2
    },
    "V17": {
      "id": "V17",
      "w": 4,
      "l": 5,
      "shipping_fee":
105,
      "weight": 75,
      "eta": 4
    },
    "V18": {
      "id": "V18",
      "w": 2,
      "l": 3,
      "shipping_fee":
38,
      "weight": 28,
      "eta": 2
    }
  }
```

```
"V15",
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false,
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      "y": -1,
      "orientation":
"Vertical"
    },
    "V16": {
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"V16",
      "in_ship":
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      "x": -1,
      "y": -1,
      "orientation":
"Horizontal"
    },
    "V17": {
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"V17",
      "in_ship": true,
      "x": 4,
      "y": 6,
      "orientation":
"Horizontal"
    },
    "V18": {
      "vehicle_id":
"V18",
      "in_ship": true,
      "x": 8,
      "y": 7,
      "orientation":
"Horizontal"
    }
  }
```

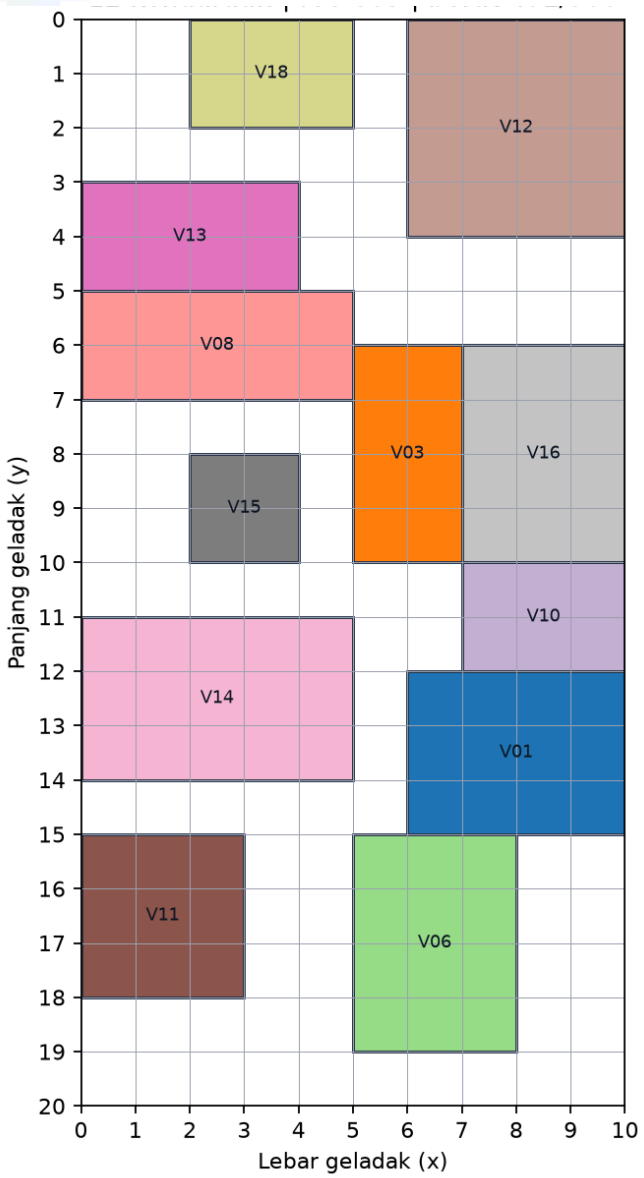

Objective Function Score: 608.0

	Vehicle Count: 9 **Total Weight:** 448

Visualisasi *Initial State*:

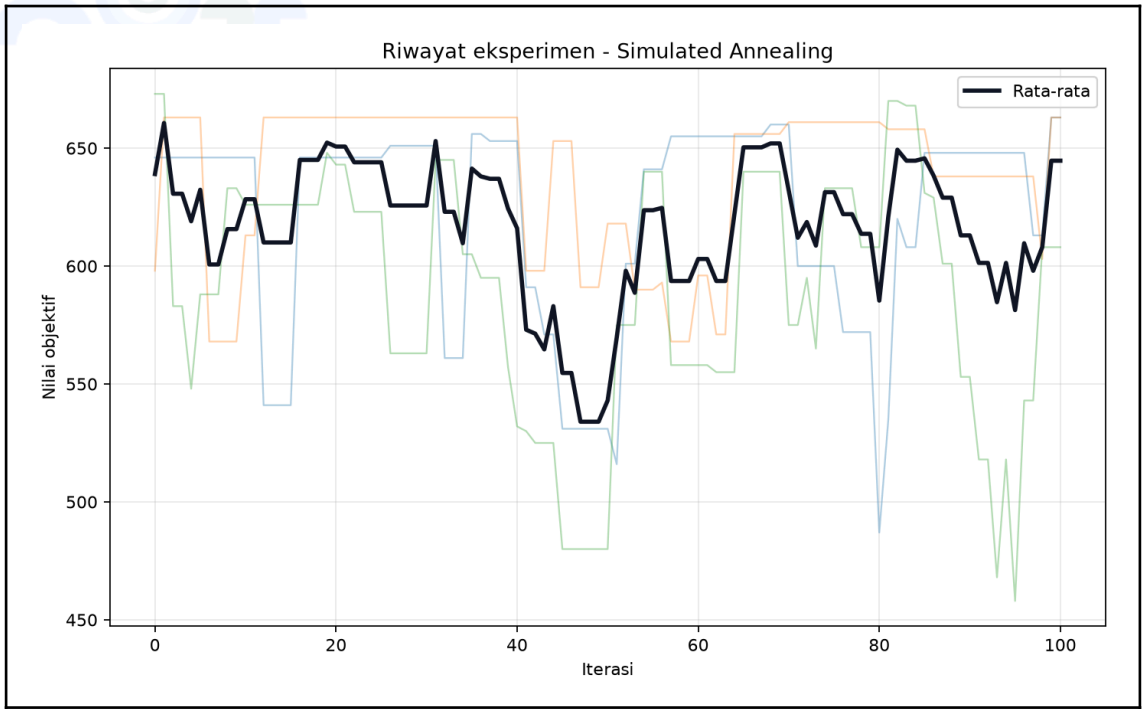


Visualisasi *Final State*:



Highest Objective Function Score: 673.0
Iterations: 100
Duration: 0.017455577850341797
Stuck Frequency: 0

Plot: Perbandingan Riwayat 3 Percobaan Eksperimen *Simulated Annealing*



Analisis Eksperimen *Simulated Annealing*

Berdasarkan hasil dari ketiga percobaan di atas, terdapat beberapa karakteristik utama kinerja *Simulated Annealing* dalam menyelesaikan permasalahan penataan kendaraan pada geladak kapal kargo:

1. Perilaku Eksplorasi vs. Eksploitasi pada Grafik Objective Function
   Pada fase awal iterasi ketika temperatur T masih tinggi, grafik nilai *objective function* memperlihatkan fluktuasi naik-turun yang cukup tajam. Penurunan nilai objektif sementara terjadi karena algoritma mengizinkan langkah *downhill* seperti, mengeluarkan satu kendaraan berukuran besar atau menukar kendaraan berbiaya tinggi dengan kendaraan berbiaya lebih rendah, guna merombak tata letak ruang kosong di geladak kapal. Strategi ini terbukti efektif membebaskan algoritma dari *local optima* awal, sehingga pada pertengahan hingga akhir iterasi, algoritma mampu menyusun ulang kendaraan dengan kepadatan grid yang lebih tinggi dan mencapai nilai `ShippingFee` akhir yang lebih optimal dibandingkan titik awalnya.
2. Analisis Plot Probabilitas Penerimaan $e^{\frac{\Delta E}{T}}$
   Pada grafik plot $e^{\frac{\Delta E}{T}}$ terhadap banyaknya iterasi, terlihat jelas bahwa pada ratusan iterasi pertama, titik-titik nilai probabilitas terkonsentrasi di area atas, mendekati 0.8 hingga 1.0. Seiring menurunnya temperatur secara eksponensial, kurva sebaran

probabilitas penerimaan langkah buruk meluruh mendekati 0.0. Pada fase suhu rendah tersebut, algoritma hampir tidak pernah lagi menerima mutasi yang menurunkan `ShippingFee`, sehingga perilakunya bertransisi mulus menjadi *Stochastic Hill-Climbing* yang menstabilkan solusi di puncak terbaik yang ditemukan.

3. Analisis Frekuensi *Stuck* di *Local Optima* dan Efisiensi Durasi
   Frekuensi stuck yang tercatat pada eksperimen sebagian besar terakumulasi pada paruh akhir iterasi ketika suhu T sudah sangat rendah dan ruang di geladak kapal sudah mendekati kapasitas maksimum. Pada kondisi tersebut, sebagian besar percobaan *move* acak menghasilkan nilai yang lebih buruk atau tidak muat, dan karena probabilitas $e^{\frac{\Delta E}{T}} \approx 0$, langkah-langkah tersebut ditolak secara beruntun. Dari sisi waktu komputasi, *Simulated Annealing* memiliki durasi per iterasi yang sangat ringan karena hanya mengevaluasi satu *random neighbor* di tiap langkahnya, berbeda dengan *Steepest-Ascent Hill-Climbing* yang harus membangkitkan dan mengevaluasi seluruh kandidat tetangga pada setiap iterasi.

2.5.6 *Genetic Algorithm* (Terbaik: *Run* #15)

Percobaan 15	

Initial	"initial_state": {
	"ship": {
	"id": "IntimLine-Deck-01",
	"w": 10,
	"l": 20,
	"max_capacity": 500
	},
	"vehicles": {
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	"id": "V01",
	"w": 3,
	"l": 4,
	"shipping_fee": 60,
	"weight": 50,
	"eta": 3
	},
	"V02": {

```
    "id": "V02",
    "w": 3,
    "l": 5,
    "shipping_fee": 85,
    "weight": 65,
    "eta": 2
  },
  "V03": {
    "id": "V03",
    "w": 2,
    "l": 4,
    "shipping_fee": 45,
    "weight": 35,
    "eta": 4
  },
  "V04": {
    "id": "V04",
    "w": 4,
    "l": 6,
    "shipping_fee": 130,
    "weight": 95,
    "eta": 1
  },
  "V05": {
    "id": "V05",
    "w": 2,
    "l": 3,
    "shipping_fee": 35,
    "weight": 25,
    "eta": 5
  },
  "V06": {
    "id": "V06",
    "w": 3,
    "l": 4,
    "shipping_fee": 65,
    "weight": 45,
    "eta": 2
  },
```

```
  "V07": {
    "id": "V07",
    "w": 4,
    "l": 5,
    "shipping_fee": 110,
    "weight": 80,
    "eta": 3
  },
  "V08": {
    "id": "V08",
    "w": 2,
    "l": 5,
    "shipping_fee": 55,
    "weight": 40,
    "eta": 2
  },
  "V09": {
    "id": "V09",
    "w": 3,
    "l": 6,
    "shipping_fee": 95,
    "weight": 70,
    "eta": 1
  },
  "V10": {
    "id": "V10",
    "w": 2,
    "l": 3,
    "shipping_fee": 40,
    "weight": 30,
    "eta": 4
  },
  "V11": {
    "id": "V11",
    "w": 3,
    "l": 3,
    "shipping_fee": 50,
    "weight": 40,
    "eta": 3
```

```
    },
    "V12": {
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      "w": 4,
      "l": 4,
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      "weight": 60,
      "eta": 2
    },
    "V13": {
      "id": "V13",
      "w": 2,
      "l": 4,
      "shipping_fee": 48,
      "weight": 35,
      "eta": 5
    },
    "V14": {
      "id": "V14",
      "w": 3,
      "l": 5,
      "shipping_fee": 82,
      "weight": 60,
      "eta": 1
    },
    "V15": {
      "id": "V15",
      "w": 2,
      "l": 2,
      "shipping_fee": 28,
      "weight": 20,
      "eta": 3
    },
    "V16": {
      "id": "V16",
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      "l": 4,
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```

```
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    },
    "V17": {
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      "l": 5,
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      "weight": 75,
      "eta": 4
    },
    "V18": {
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      "w": 2,
      "l": 3,
      "shipping_fee": 38,
      "weight": 28,
      "eta": 2
    }
  },
  "placements": {
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      "in_ship": true,
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      "y": 8,
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      "in_ship": false,
      "x": -1,
      "y": -1,
      "orientation": "Horizontal"
    },
    "V03": {
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      "in_ship": false,
      "x": -1,
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```

```
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      "y": 14,
      "orientation": "Horizontal"
    },
    "V05": {
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      "in_ship": true,
      "x": 5,
      "y": 11,
      "orientation": "Horizontal"
    },
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      "vehicle_id": "V06",
      "in_ship": false,
      "x": -1,
      "y": -1,
      "orientation": "Horizontal"
    },
    "V07": {
      "vehicle_id": "V07",
      "in_ship": true,
      "x": 0,
      "y": 6,
      "orientation": "Horizontal"
    },
    "V08": {
      "vehicle_id": "V08",
      "in_ship": false,
      "x": -1,
      "y": -1,
      "orientation": "Horizontal"
    },
    "V09": {
      "vehicle_id": "V09",
```

```
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      "in_ship": false,
      "x": -1,
      "y": -1,
      "orientation": "Horizontal"
    },
    "V11": {
      "vehicle_id": "V11",
      "in_ship": false,
      "x": -1,
      "y": -1,
      "orientation": "Horizontal"
    },
    "V12": {
      "vehicle_id": "V12",
      "in_ship": true,
      "x": 0,
      "y": 0,
      "orientation": "Horizontal"
    },
    "V13": {
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      "in_ship": true,
      "x": 8,
      "y": 13,
      "orientation": "Horizontal"
    },
    "V14": {
      "vehicle_id": "V14",
      "in_ship": true,
      "x": 5,
      "y": 3,
      "orientation": "Horizontal"
```

```
      },
      "V15": {
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        "in_ship": true,
        "x": 8,
        "y": 17,
        "orientation": "Horizontal"
      },
      "V16": {
        "vehicle_id": "V16",
        "in_ship": false,
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        "y": -1,
        "orientation": "Horizontal"
      },
      "V17": {
        "vehicle_id": "V17",
        "in_ship": true,
        "x": 0,
        "y": 11,
        "orientation": "Horizontal"
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      "V18": {
        "vehicle_id": "V18",
        "in_ship": false,
        "x": -1,
        "y": -1,
        "orientation": "Horizontal"
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        "total_weight": 500,
        "shipping_fee": 688
```

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```
        }
      }
    },
```

Objective Function Score: 688.0
Vehicle Count: 9
Total Weight: 500
</td></tr>
<tr><td>Final</td><td>

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"final_state": {
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        "l": 20,
        "max_capacity": 500
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      "vehicles": {
        "V01": {
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          "w": 3,
          "l": 4,
          "shipping_fee": 60,
          "weight": 50,
          "eta": 3
        },
        "V02": {
          "id": "V02",
          "w": 3,
          "l": 5,
          "shipping_fee": 85,
          "weight": 65,
          "eta": 2
        },
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          "id": "V03",
          "w": 2,
          "l": 4,
          "shipping_fee": 45,
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          "eta": 4
```

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```
  },
  "V04": {
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    "l": 6,
    "shipping_fee": 130,
    "weight": 95,
    "eta": 1
  },
  "V05": {
    "id": "V05",
    "w": 2,
    "l": 3,
    "shipping_fee": 35,
    "weight": 25,
    "eta": 5
  },
  "V06": {
    "id": "V06",
    "w": 3,
    "l": 4,
    "shipping_fee": 65,
    "weight": 45,
    "eta": 2
  },
  "V07": {
    "id": "V07",
    "w": 4,
    "l": 5,
    "shipping_fee": 110,
    "weight": 80,
    "eta": 3
  },
  "V08": {
    "id": "V08",
    "w": 2,
    "l": 5,
    "shipping_fee": 55,
    "weight": 40,
```

```
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  },
  "V09": {
    "id": "V09",
    "w": 3,
    "l": 6,
    "shipping_fee": 95,
    "weight": 70,
    "eta": 1
  },
  "V10": {
    "id": "V10",
    "w": 2,
    "l": 3,
    "shipping_fee": 40,
    "weight": 30,
    "eta": 4
  },
  "V11": {
    "id": "V11",
    "w": 3,
    "l": 3,
    "shipping_fee": 50,
    "weight": 40,
    "eta": 3
  },
  "V12": {
    "id": "V12",
    "w": 4,
    "l": 4,
    "shipping_fee": 90,
    "weight": 60,
    "eta": 2
  },
  "V13": {
    "id": "V13",
    "w": 2,
    "l": 4,
    "shipping_fee": 48,
```

```
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    "eta": 5
  },
  "V14": {
    "id": "V14",
    "w": 3,
    "l": 5,
    "shipping_fee": 82,
    "weight": 60,
    "eta": 1
  },
  "V15": {
    "id": "V15",
    "w": 2,
    "l": 2,
    "shipping_fee": 28,
    "weight": 20,
    "eta": 3
  },
  "V16": {
    "id": "V16",
    "w": 3,
    "l": 4,
    "shipping_fee": 62,
    "weight": 48,
    "eta": 2
  },
  "V17": {
    "id": "V17",
    "w": 4,
    "l": 5,
    "shipping_fee": 105,
    "weight": 75,
    "eta": 4
  },
  "V18": {
    "id": "V18",
    "w": 2,
    "l": 3,
```

```
      "shipping_fee": 38,
      "weight": 28,
      "eta": 2
    }
  },
  "placements": {
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      "in_ship": false,
      "x": -1,
      "y": -1,
      "orientation": "Vertical"
    },
    "V02": {
      "vehicle_id": "V02",
      "in_ship": false,
      "x": -1,
      "y": -1,
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    },
    "V03": {
      "vehicle_id": "V03",
      "in_ship": false,
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      "y": -1,
      "orientation": "Horizontal"
    },
    "V04": {
      "vehicle_id": "V04",
      "in_ship": true,
      "x": 4,
      "y": 14,
      "orientation": "Horizontal"
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      "in_ship": false,
      "x": -1,
      "y": -1,
```

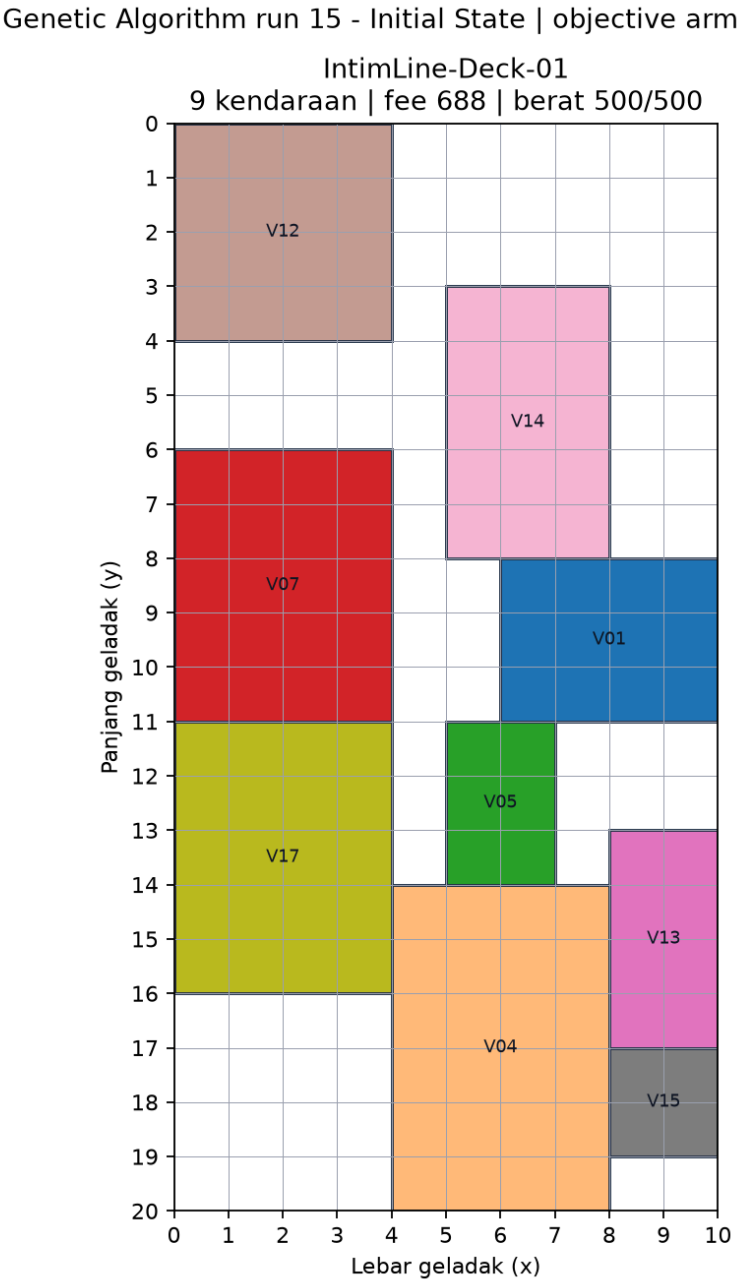
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  },
  "V06": {
    "vehicle_id": "V06",
    "in_ship": true,
    "x": 6,
    "y": 8,
    "orientation": "Vertical"
  },
  "V07": {
    "vehicle_id": "V07",
    "in_ship": true,
    "x": 0,
    "y": 6,
    "orientation": "Horizontal"
  },
  "V08": {
    "vehicle_id": "V08",
    "in_ship": false,
    "x": -1,
    "y": -1,
    "orientation": "Horizontal"
  },
  "V09": {
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    "y": -1,
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  },
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    "in_ship": false,
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```

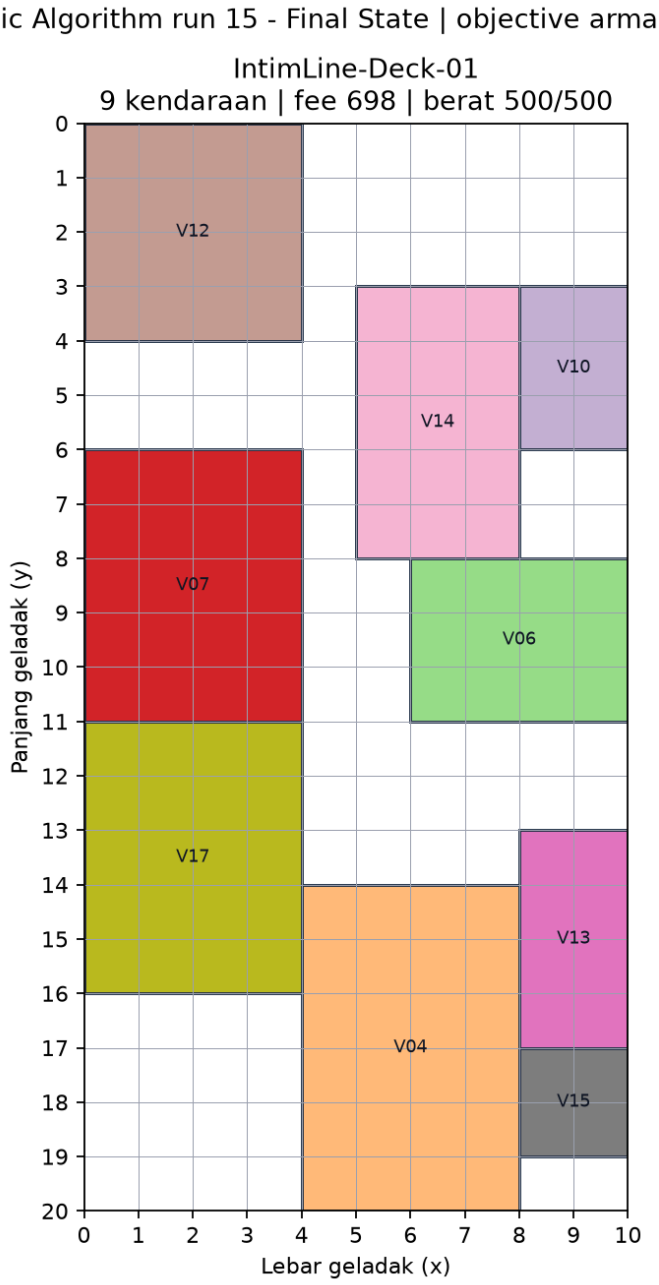
```
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      }
    },
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    "is_valid": true,
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    "total_shipping_fee": 698,
    "ship_totals": {
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      }
    }
  },
```


Objective Function Score: 698.0
Vehicle Count: 9
Total Weight: 500

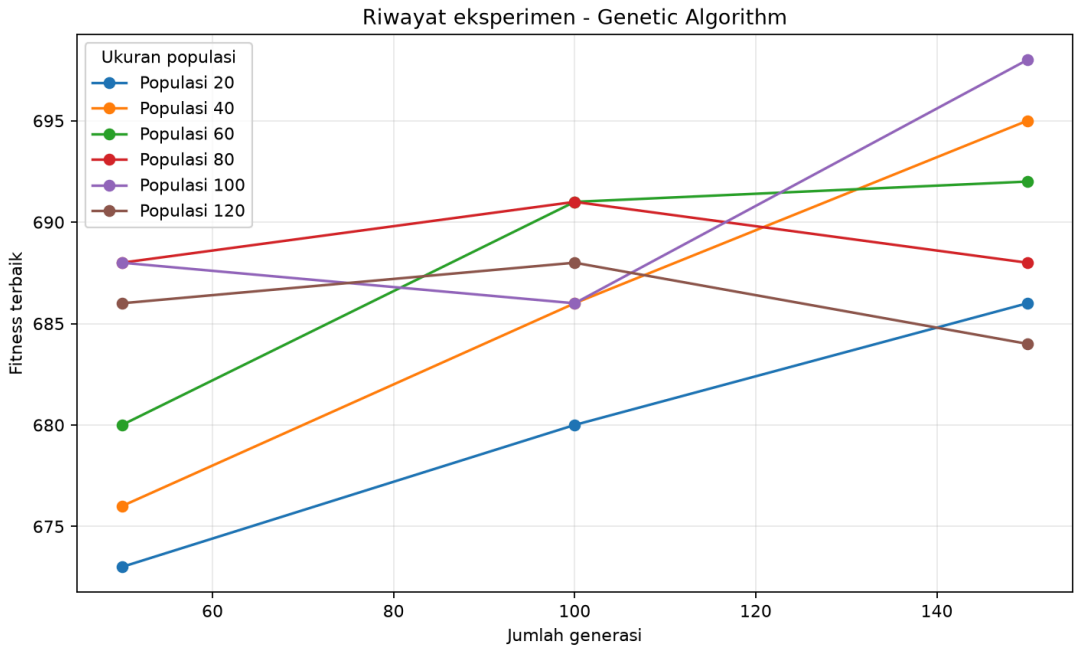
Visualisasi *Initial State*:



Visualisasi *Final State*:


Genetic Algorithm run 15 - Final State | objective armada 698.00
IntimLine-Deck-01
9 kendaraan | fee 698 | berat 500/500
V12
V10
V14
V07
V06
V17
V13
V04
V15
Panjang geladak (y)
Lebar geladak (x)
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20

Plot:



Highest Objective Function Score: 698.0
Iterations: 150
Duration: 31.674 detik

2.5.7 Analisis Perbandingan Algoritma

Eksperimen terdiri dari 33 run:

- 3 run untuk lima algoritma non-GA
- 18 run *Genetic Algorithm*

Solusi feasible terbaik yang ditemukan bernilai **698**, dicapai oleh:

- Hill Climbing Sideways Move, run 2: 646 → 698, durasi 0,681 detik.
- Genetic Algorithm, run 15 (populasi 100, 150 generasi): 688 → 698, durasi 31,674 detik.

Untuk mengukur kedekatan dengan optimum global, dilakukan relaksasi knapsack berdasarkan batas berat 500 tanpa mempertimbangkan kendala geometri geladak. Relaksasi ini memberikan batas atas sebesar **700**. Karena solusi valid bernilai 698 telah ditemukan, maka:

$$698 \leq OPT \leq 700$$

Dengan demikian, skor terbaik 698 telah mencapai minimal:

$$\frac{698}{700} \times 100\% = 99{,}71\%$$

dari optimum global. Selisih terhadap batas atas hanya 2 poin atau maksimal **0,29%**. Nilai 698 dapat disebut solusi terbaik empiris dan sangat dekat dengan optimum global.

Perbandingan kualitas hasil dan durasi

Algoritma	Jumlah run	Objective akhir rata-rata	Min–maks objective	Gap rata-rata dari 698	Durasi rata-rata
Steepest Ascent HC	3	683,33	671–696	2,10%	0,113 s
Sideways Move HC	3	688,67	681–698	1,34%	0,733 s
Stochastic HC	3	677,33	670–682	2,96%	0,027 s
Random Restart HC	3	691,00	691–691	1,00%	0,223 s
Simulated Annealing	3	644,67	608–663	7,64%	0,026 s
Genetic Algorithm	18	686,44	673–698	1,66%	13,756 s

Secara kualitas, Sideways Move HC dan GA adalah dua algoritma yang berhasil mencapai skor tertinggi 698. Namun, Sideways Move HC mencapai skor tersebut sekitar **46 kali lebih cepat** daripada GA terbaik (0,681 s dibanding 31,674 s).

Simulated Annealing memiliki durasi rata-rata paling rendah, tetapi kualitas solusi terendah. Stochastic HC hampir sama cepat dan menghasilkan objective rata-rata yang lebih tinggi daripada Simulated Annealing. Random Restart HC memberikan kualitas rata-rata terbaik di antara algoritma yang tidak mencapai 698, dengan objective konsisten 691.

Konsistensi hasil

Algoritma	*Objective*	Standar deviasi	Interpretasi
Steepest Ascent	671, 696, 683	12,50	Sensitif terhadap state awal dan mudah berhenti pada local optimum.
Sideways Move	687, 698, 681	8,62	Lebih baik daripada Steepest Ascent dan sesekali mencapai solusi terbaik, tetapi belum konsisten.
Stochastic	670, 680, 682	6,43	Cukup konsisten, tetapi berada pada kualitas solusi menengah.
Random Restart	691, 691, 691	**0,00**	Paling konsisten; selalu menghasilkan solusi baik, meskipun belum optimum empiris.
Simulated Annealing	663, 663, 608	31,75	Paling tidak konsisten dan paling sensitif terhadap proses acak.
Genetic Algorithm	673–698	6,27	Variasi terlihat cukup besar antar-konfigurasi; angka ini tidak dapat dianggap konsistensi satu konfigurasi karena populasi, generasi, dan seed berbeda pada tiap run.

Random Restart HC merupakan algoritma paling stabil karena seluruh run menghasilkan objective 691. Sebaliknya, Simulated Annealing mempunyai variasi paling besar. Untuk GA, diperlukan beberapa pengulangan dengan pasangan parameter yang sama agar konsistensi tiap konfigurasi dapat diukur secara lebih adil.

2.6 Bonus

A. Buatlah implementasi untuk seluruh algoritma Hill-climbing yang ada.

Kami mengembangkan algoritma *Hill Climbing* menjadi empat varian berbeda agar performanya bisa dikomparasi secara langsung. Varian pertama adalah *Steepest Ascent* yang akan mengevaluasi semua kemungkinan pergerakan kendaraan lalu selalu mengambil susunan dengan nilai paling tinggi. Varian kedua ditambahkan mekanisme *Sideways Move* yang mengizinkan algoritma bergeser ke konfigurasi tetangga bernilai sama. Mekanisme ini berguna agar pencarian tidak langsung berhenti saat mendapati jalan buntu.

Varian ketiga adalah *Stochastic Hill Climbing* yang bekerja dengan cara memilih satu pergerakan kendaraan secara acak, lalu menerimanya jika pergerakan tersebut memberikan nilai lebih baik daripada susunan sebelumnya. Varian ketiga adalah *Random Restart* yang secara otomatis akan memulai ulang pencarian dari awal ketika algoritma sudah benar benar terjebak di posisi optimal lokal.

B. Multi-kapal. Program dapat menerima jumlah kapal lebih dari satu dan memaksimalkan *value* total yang diangkut armada.

Kami memperluas skala persoalan dari sekadar mengisi satu geladak menjadi pengelolaan sebuah armada yang memiliki banyak kapal. Program tetap mengelola satu daftar kendaraan, namun kini sistem harus mendistribusikannya secara cerdas ke kapal kapal yang tersedia.

Untuk mengakomodasi fitur ini, pemodelan *state* kendaraan diubah agar mampu menyimpan ID kapal tujuannya. Proses validasi aturan seperti

dilarang saling bertumpuk, batas luar geladak, serta batas berat angkut kini dihitung secara terpisah untuk setiap kapal. Sebagai contoh, jika kita memiliki dua kapal yang masing masing berkapasitas lima ratus beban, maka armada tersebut sanggup mengangkut total seribu beban asalkan muatan di tiap kapalnya tidak berlebih.

Pada proses pencarian solusi di semua algoritma, pergerakan acaknya tidak lagi sebatas menggeser koordinat atau orientasi kendaraan di satu tempat. Sistem kini bisa mengeluarkan kendaraan dari armada, memasukkan barang baru, atau bahkan memindahkan satu kendaraan dari kapal pertama ke kapal kedua. Nilai objektif utamanya didapatkan dari total penjumlahan seluruh tarif pengiriman di seluruh armada.

Fitur ini dapat dijalankan dengan mudah melalui terminal "`python -m src.main --num-ships n`"—dengan **'n'** adalah jumlah kapal. Kami juga mendesain modul visualisasi khusus agar geladak kapal-kapal dalam satu armada tersebut bisa digambar secara berdampingan, lengkap dengan ringkasan jumlah beban dan keuntungan di setiap unit kapalnya dengan *syntax* "`python -m src.visualizer`".

Percobaan 1 (*Stochastic Hill Climbing*)	

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	"vehicles": {
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	"eta": 3
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	"V02": {
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	"w": 3,
	"l": 5,
	"shipping_fee": 85,
	"weight": 65,
	"eta": 2
	},
	"V03": {
	"id": "V03",
	"w": 2,
	"l": 4,
	"shipping_fee": 45,
	"weight": 35,
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	"l": 6,

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    "weight": 95,
    "eta": 1
  },
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    "l": 3,
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    "w": 3,
    "l": 4,
    "shipping_fee": 65,
    "weight": 45,
    "eta": 2
  },
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    "id": "V07",
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    "l": 5,
    "shipping_fee": 110,
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    "eta": 3
  },
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```

```
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    "eta": 5
  },
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    },
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      "l": 5,
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  },
```

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    "in_ship": true,
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    "y": 10,
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      "orientation": "Horizontal",
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      "in_ship": true,
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    },
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      "in_ship": true,
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      "in_ship": true,
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      "y": 1,
      "orientation": "Vertical",
      "ship_id": "IntimLine-Deck-01-2"
    },
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      "vehicle_id": "V10",
      "in_ship": true,
      "x": 5,
      "y": 5,
      "orientation": "Vertical",
      "ship_id": "IntimLine-Deck-01-2"
```

```
    },
    "V11": {
      "vehicle_id": "V11",
      "in_ship": true,
      "x": 2,
      "y": 14,
      "orientation": "Horizontal",
      "ship_id": "IntimLine-Deck-01-2"
    },
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      "vehicle_id": "V12",
      "in_ship": true,
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    },
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      "in_ship": true,
      "x": 2,
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      "orientation": "Vertical",
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      },
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        "in_ship": true,
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        "y": 17,
        "orientation": "Vertical",
        "ship_id": "IntimLine-Deck-01-1"
      },
      "V17": {
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        "in_ship": true,
        "x": 2,
        "y": 10,
        "orientation": "Vertical",
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        "in_ship": true,
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        "y": 10,
        "orientation": "Horizontal",
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    },
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    "is_valid": true,
    "objective": 1078.0,
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    "ship_totals": {
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          }
        },
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            "l": 20,
            "max_capacity": 500
          },
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            "l": 20,
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          }
        ]
      },
</code>
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Objective Function Score: 833.0
</td>
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    "l": 4,
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    "weight": 35,
    "eta": 4
  },
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    "l": 6,
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    "eta": 1
  },
  "V05": {
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    "w": 2,
    "l": 3,
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```

```
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    "w": 3,
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    "weight": 45,
    "eta": 2
  },
  "V07": {
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    "w": 4,
    "l": 5,
    "shipping_fee": 110,
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    "eta": 3
  },
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    "l": 5,
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    "weight": 40,
    "eta": 2
  },
  "V09": {
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```

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  },
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        "weight": 48,
        "eta": 2
      },
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        "l": 5,
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        "eta": 4
      },
      "V18": {
        "id": "V18",
        "w": 2,
        "l": 3,
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        "weight": 28,
        "eta": 2
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      "x": 0,
      "y": 14,
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      "ship_id": "IntimLine-Deck-01-3"
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      "y": 10,
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    },
    "V06": {
      "vehicle_id": "V06",
      "in_ship": true,
      "x": 5,
      "y": 7,
      "orientation": "Horizontal",
      "ship_id": "IntimLine-Deck-01-2"
    },
    "V07": {
```

```
    "vehicle_id": "V07",
    "in_ship": true,
    "x": 2,
    "y": 0,
    "orientation": "Horizontal",
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  },
  "V08": {
    "vehicle_id": "V08",
    "in_ship": true,
    "x": 4,
    "y": 5,
    "orientation": "Horizontal",
    "ship_id": "IntimLine-Deck-01-3"
  },
  "V09": {
    "vehicle_id": "V09",
    "in_ship": true,
    "x": 2,
    "y": 1,
    "orientation": "Vertical",
    "ship_id": "IntimLine-Deck-01-2"
  },
  "V10": {
    "vehicle_id": "V10",
    "in_ship": true,
    "x": 5,
    "y": 5,
    "orientation": "Vertical",
    "ship_id": "IntimLine-Deck-01-2"
  },
  "V11": {
    "vehicle_id": "V11",
    "in_ship": true,
    "x": 2,
    "y": 14,
    "orientation": "Horizontal",
    "ship_id": "IntimLine-Deck-01-2"
  },
```

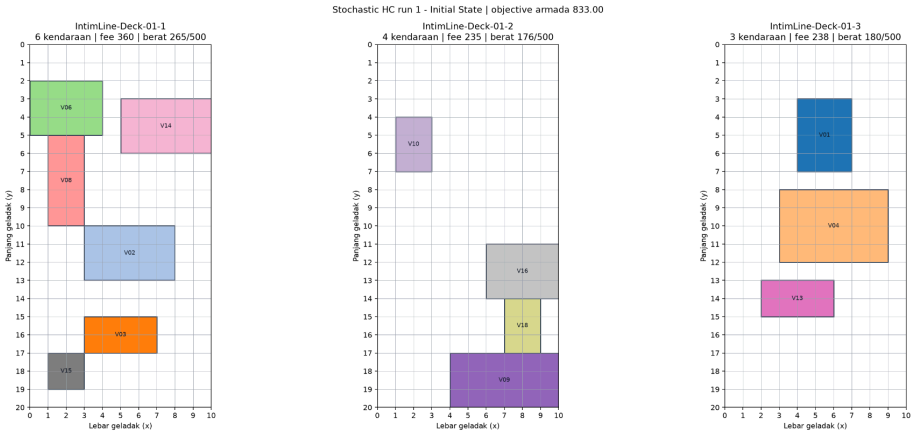
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    "V12": {
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    },
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      "x": 5,
      "y": 13,
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    },
    "V14": {
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      "in_ship": true,
      "x": 6,
      "y": 15,
      "orientation": "Horizontal",
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    },
    "V15": {
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      "in_ship": true,
      "x": 2,
      "y": 10,
      "orientation": "Vertical",
      "ship_id": "IntimLine-Deck-01-2"
    },
    "V16": {
      "vehicle_id": "V16",
      "in_ship": true,
      "x": 5,
      "y": 17,
      "orientation": "Vertical",
      "ship_id": "IntimLine-Deck-01-1"
```

```
      },
      "V17": {
        "vehicle_id": "V17",
        "in_ship": true,
        "x": 2,
        "y": 10,
        "orientation": "Vertical",
        "ship_id": "IntimLine-Deck-01-3"
      },
      "V18": {
        "vehicle_id": "V18",
        "in_ship": true,
        "x": 8,
        "y": 10,
        "orientation": "Horizontal",
        "ship_id": "IntimLine-Deck-01-2"
      }
    },
    "use_bonus_objective": false,
    "is_valid": true,
    "objective": 1223.0,
    "total_weight": 901,
    "total_shipping_fee": 1223,
    "ship_totals": {
      "IntimLine-Deck-01-1": {
        "total_weight": 248,
        "shipping_fee": 332
      },
      "IntimLine-Deck-01-2": {
        "total_weight": 268,
        "shipping_fee": 364
      },
      "IntimLine-Deck-01-3": {
        "total_weight": 385,
        "shipping_fee": 527
      }
    },
    "ships": [
      {
```

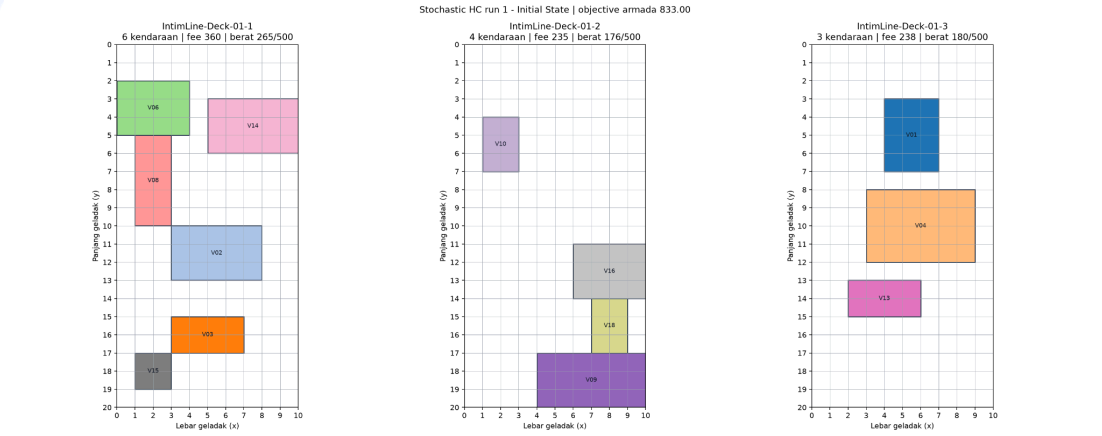
```
        "id": "IntimLine-Deck-01-1",
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        "l": 20,
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      {
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        "w": 10,
        "l": 20,
        "max_capacity": 500
      },
      {
        "id": "IntimLine-Deck-01-3",
        "w": 10,
        "l": 20,
        "max_capacity": 500
      }
    ]
  },
```


Objective Function Score: 1223.0

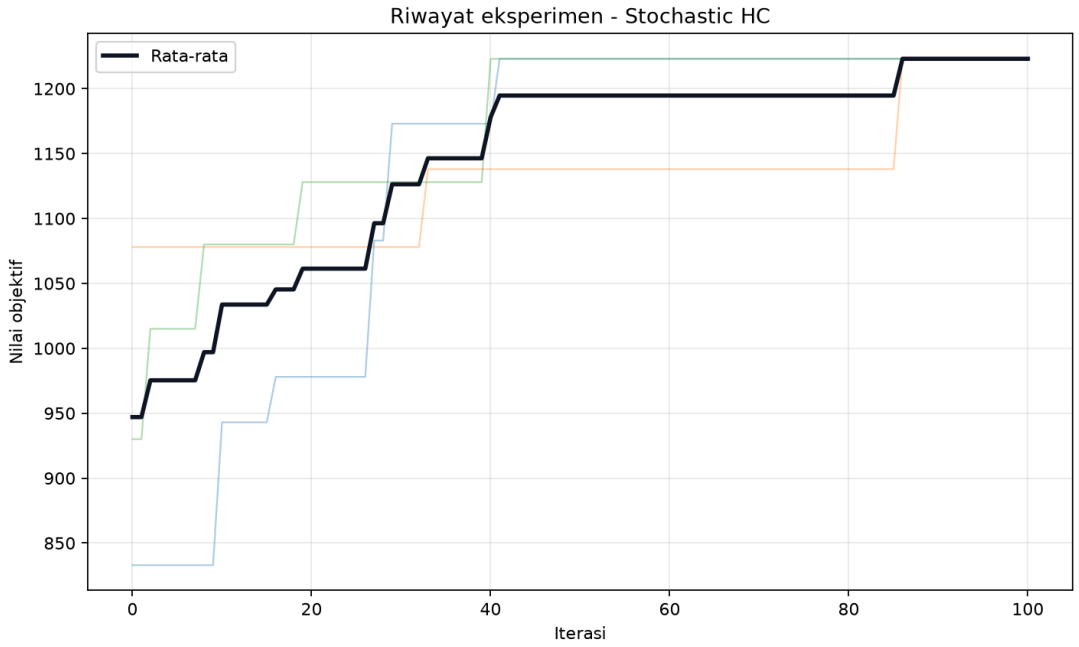
Visualisasi *Initial State*:



Visualisasi *Final State*:



Plot:



Highest Objective Function Score: 1223.0

Iterations: 100

Duration: 0,0103 detik

III KESIMPULAN DAN SARAN

3.1 Kesimpulan & Saran

Sideways Move adalah pilihan paling efisien untuk eksperimen ini: ia memperoleh solusi terbaik empiris 698 dalam waktu singkat. GA juga dapat mencapai skor yang sama, tetapi membutuhkan konfigurasi lebih besar dan waktu jauh lebih lama. Random Restartmmerupakan pilihan paling konsisten ketika stabilitas hasil lebih penting daripada mengejar skor maksimum. Penambahan generasi pada GA cenderung meningkatkan kualitas solusi, sedangkan penambahan populasi hanya efektif hingga sekitar 100 individu.

IV PEMBAGIAN TUGAS

Nama	Abdullah Ahmad Yusuf
NIM	18224073

No.	Tugas
1	Mengerjakan bagian *simulated annealing*

Nama	Abhinaya Rajendra Fargaz
NIM	18224087

No.	Tugas
1	Mengerjakan bagian *genetic algorithm*

Nama	M. Rafif Azis Syahlevi
NIM	18224123

No.	Tugas
1	Mengerjakan bagian *hill climbing*

REFERENSI

S. Russell and P. Norvig, Artificial Intelligence: A Modern Approach, 4th ed. Hoboken, NJ: Pearson, 2020.

LAMPIRAN

FORMAT DEKLARASI PENGGUNAAN AI
SURAT PERNYATAAN PENGGUNAAN AI

Dengan ini, kami:
Nama/NIM : Abdullah Ahmad Yusuf/18224073
Abhinaya Rajendra Fargaz/18224087
M. Rafif Azis Syahlevi
Fakultas/Sekolah : STEI-K/Sistem dan Teknologi Informasi
Kode Mata Kuliah : IF3070
Nama Mata Kuliah : Dasar Inteligensi Artifisial
Judul Tugas/Proposal : Optimasi Pemuatan Kendaraan pada Geladak Kapal Menggunakan Algoritma Hill-Climbing, Simulated Annealing, dan Genetic Algorithm

Menyatakan bahwa kami menggunakan AI dalam pengerjaan maupun penyusunan luaran tugas pada mata kuliah yang tertulis di atas.

Jika Ya, maka alat AI/kecerdasan buatan yang digunakan adalah:

Lingkup Pekerjaan	Digunakan?	Tingkat Penggunaan
Pemeriksaan Ejaan: menggunakan tools seperti Grammarly, Paperpal, Deepl, Quillbot, Grammarbot, LanguageTool, ProWritingAid, ChatGPT, Google Gemini, atau sejenisnya.	Ya	Level 4/5
Pembuatan Teks: menggunakan tools seperti ChatGPT, Gemini, Paperpal, Copilot, GrammarlyGO, WordAI, WriteSonic, Jasper, Jenni AI, atau sejenisnya.	Ya	Level 3/5
Bantuan Diskusi dalam Penyusunan Konten: menggunakan tools seperti ChatGPT, Gemini, Copilot, Perplexity, Jenni AI, Zoom-Companion, atau sejenisnya.	Ya	Level 4/5
Pembuatan Gambar, Video, dan/atau Grafik: menggunakan tools seperti Craiyon, DALL-E, Midjourney, Stable Diffusion, Microsoft Designer, Gemini, Canva AI, atau sejenisnya.	Ya	Level 5/5
Lainnya, Jelaskan: Debugging	Ya	Level 5/5

Kami juga menyatakan bahwa setiap penggunaan AI/kecerdasan buatan yang dilakukan pada mata kuliah tersebut telah dinyatakan di dalam surat ini.

Bandung, 22 Oktober 2026



Abdullah Ahmad Yusuf

18224073